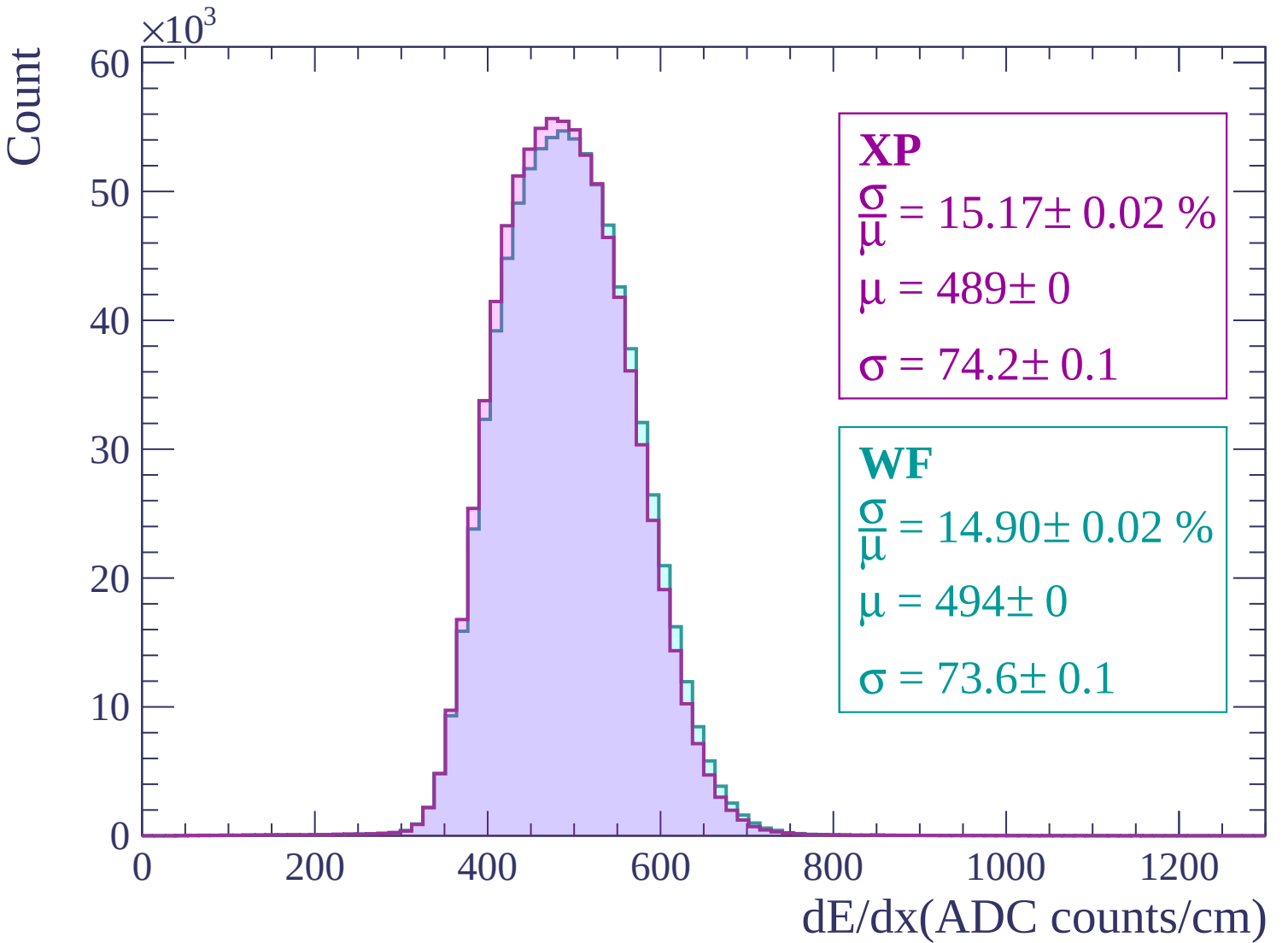
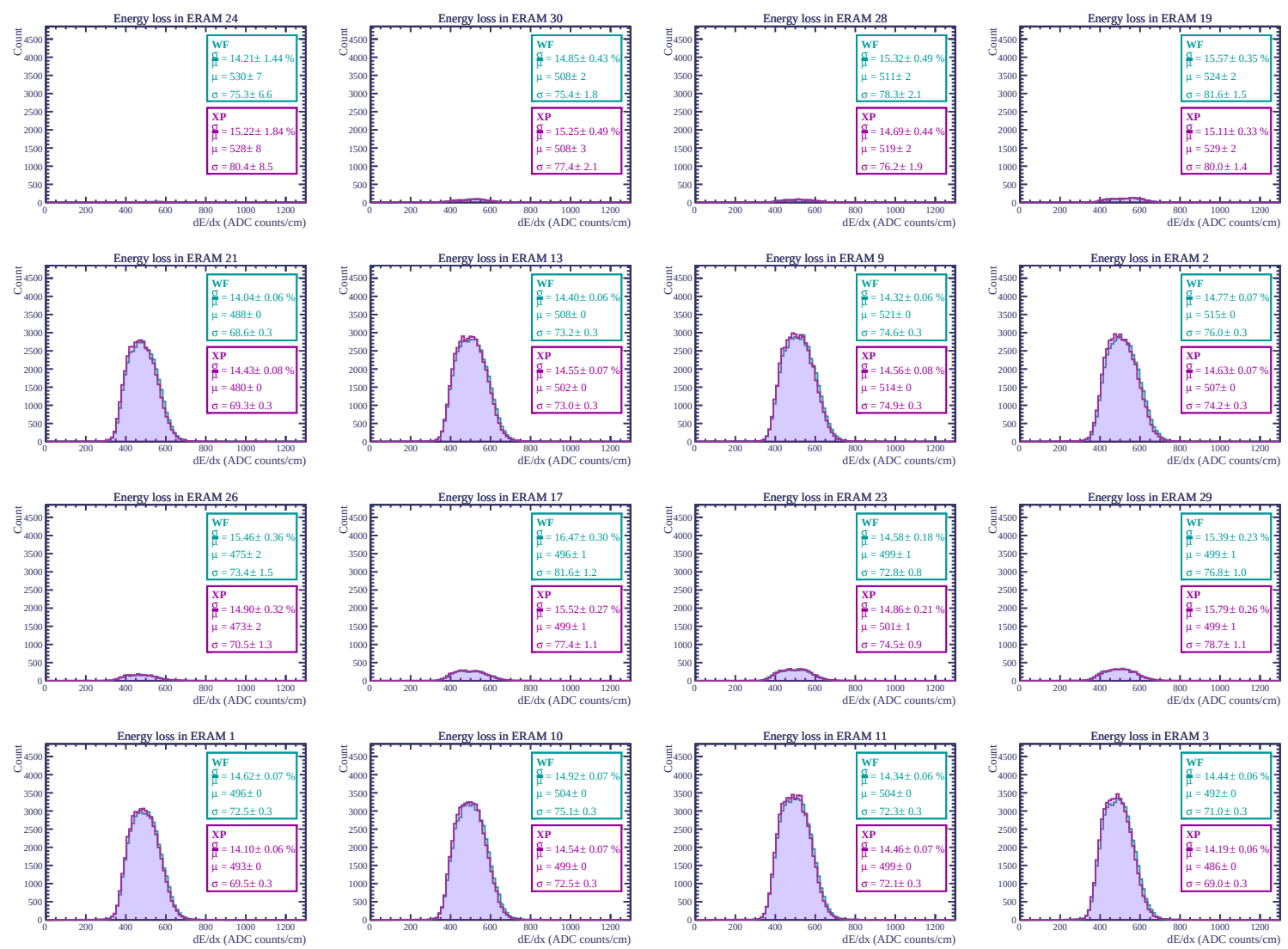
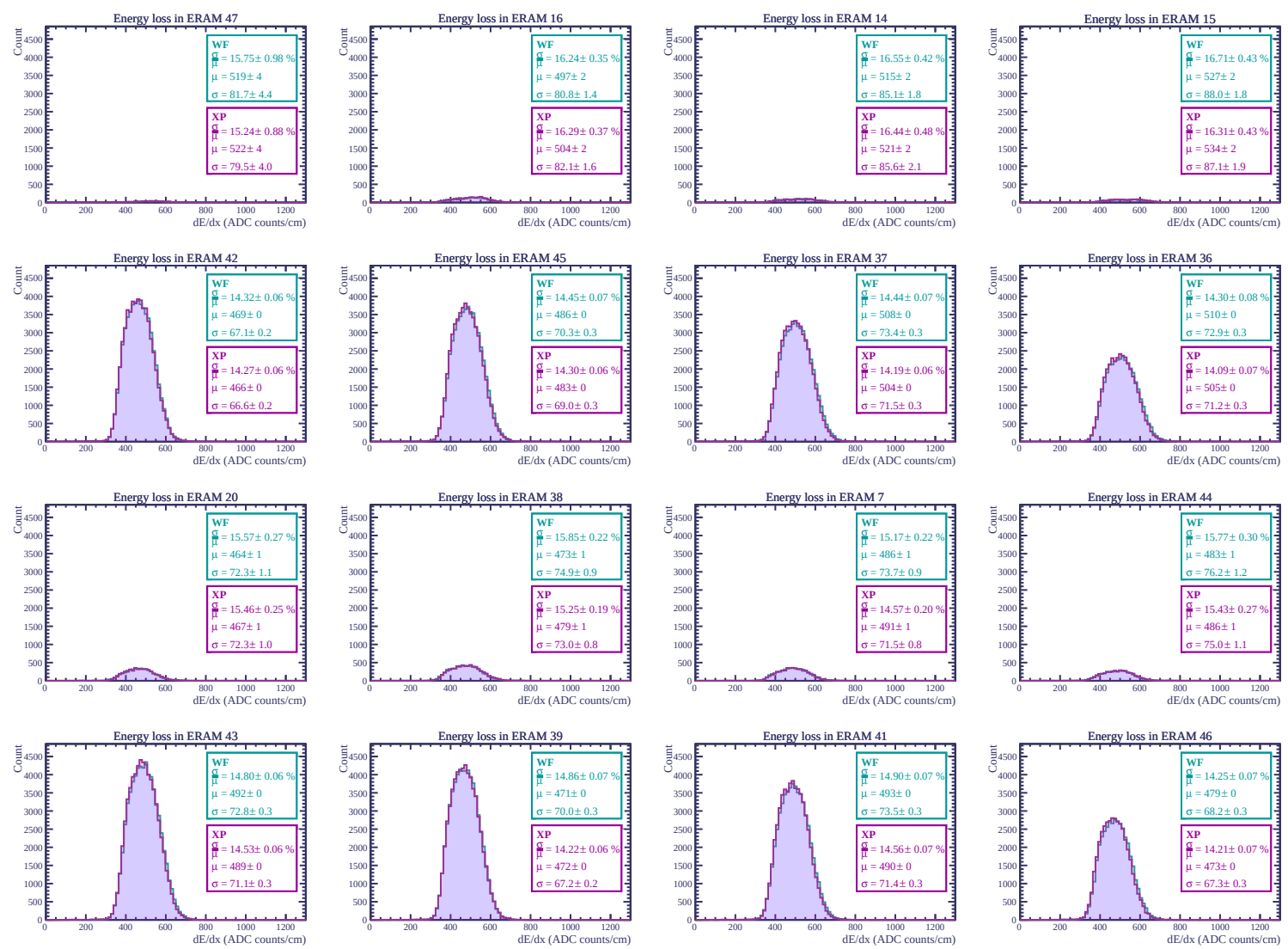
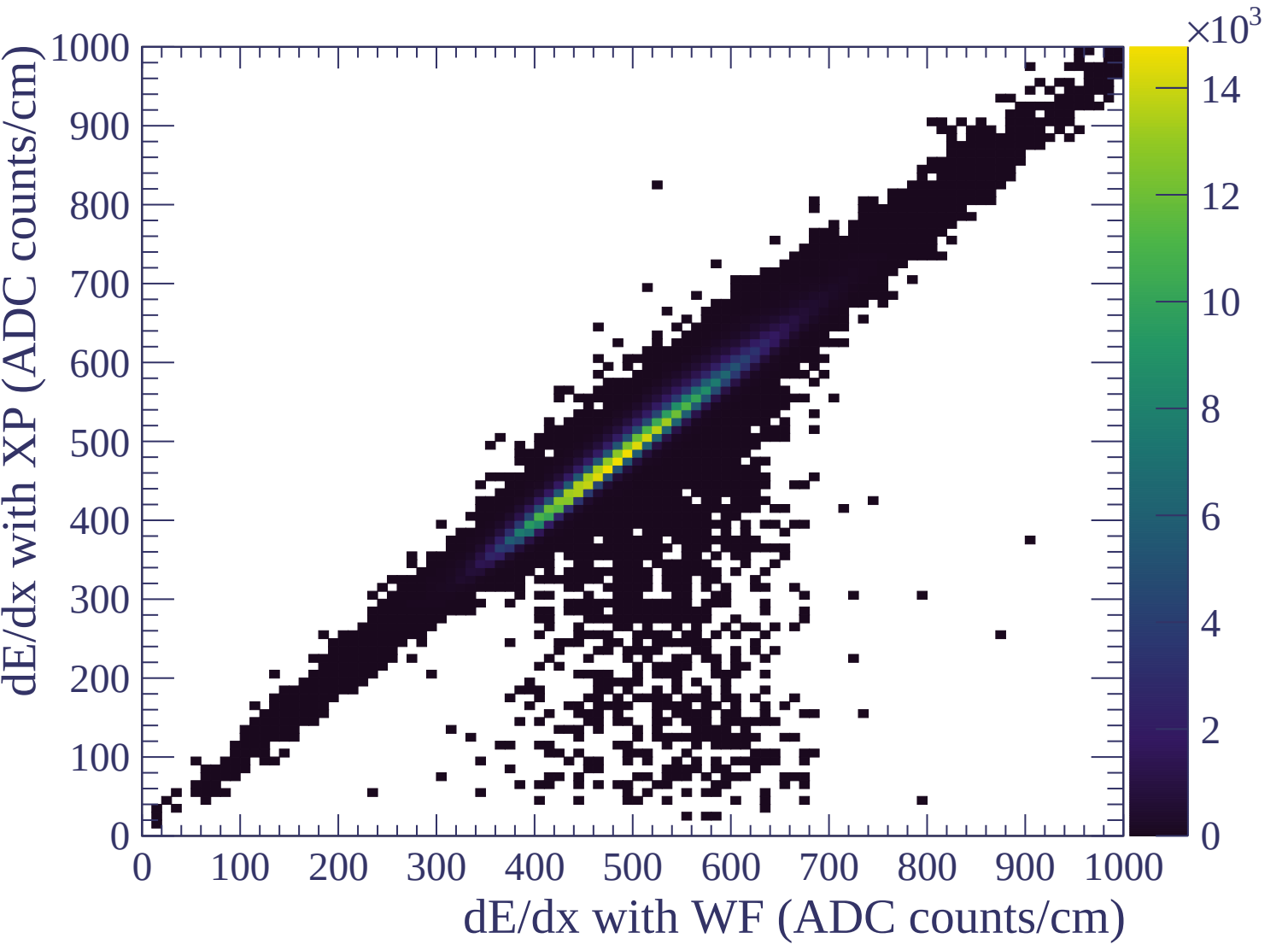
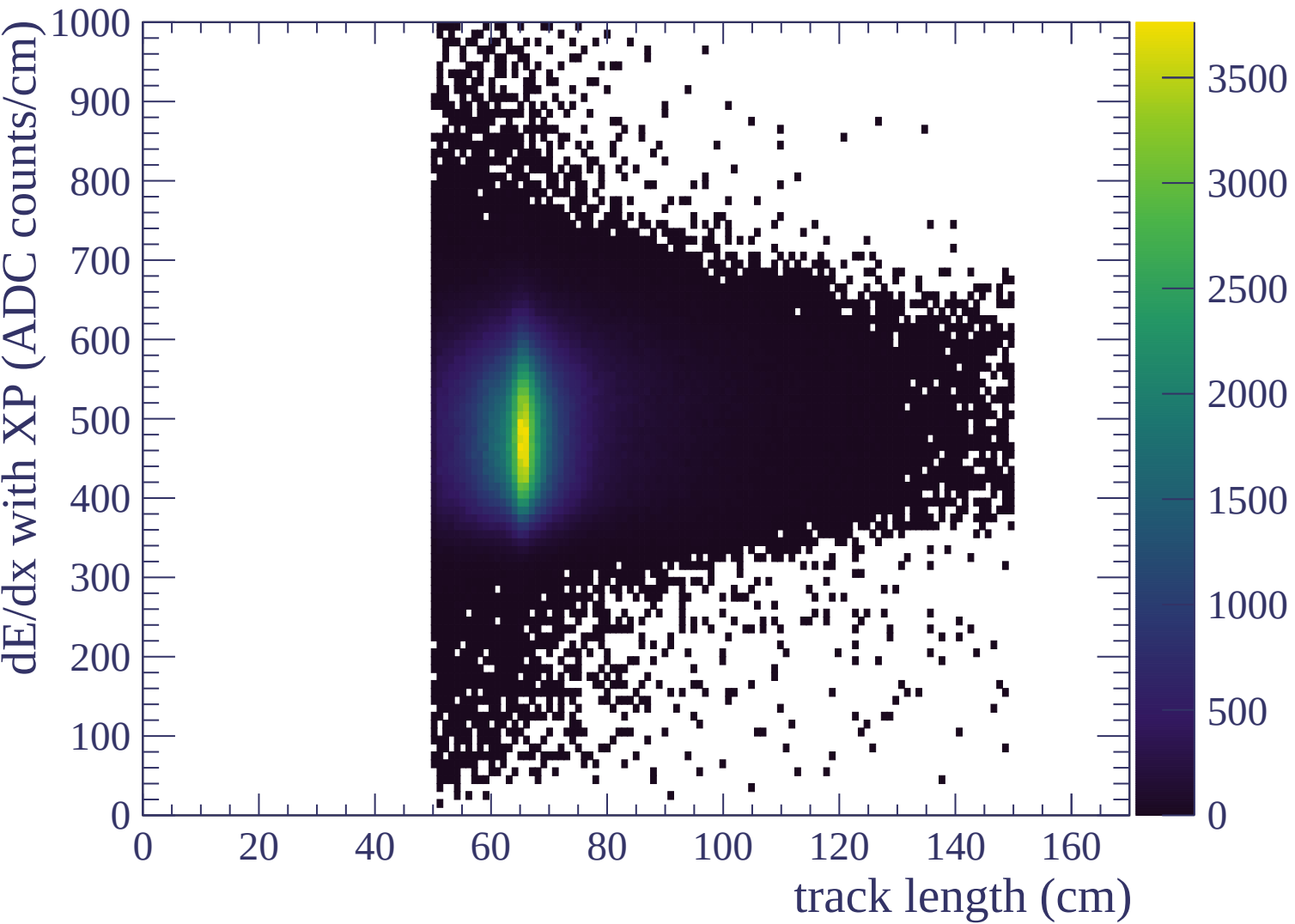


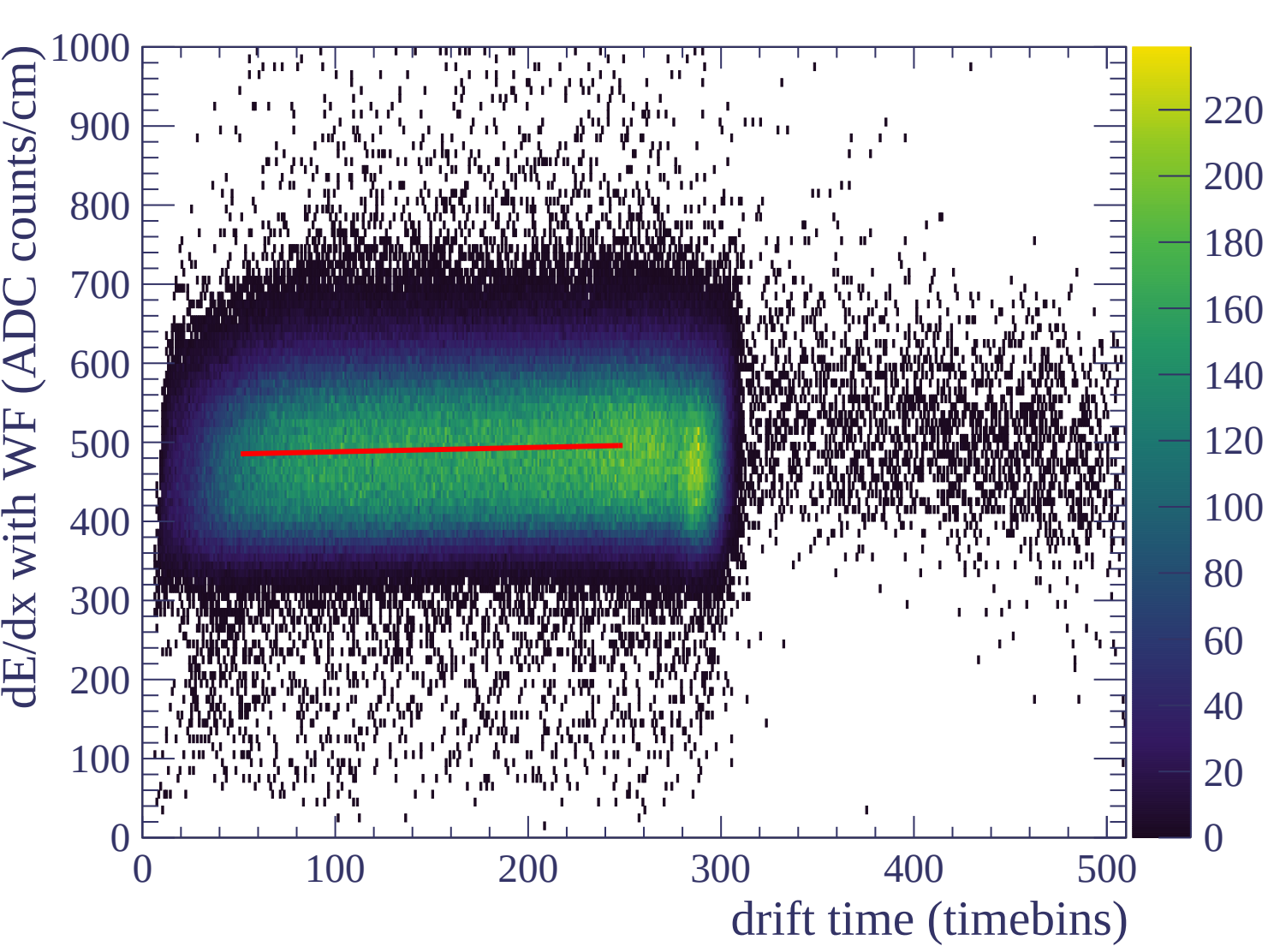


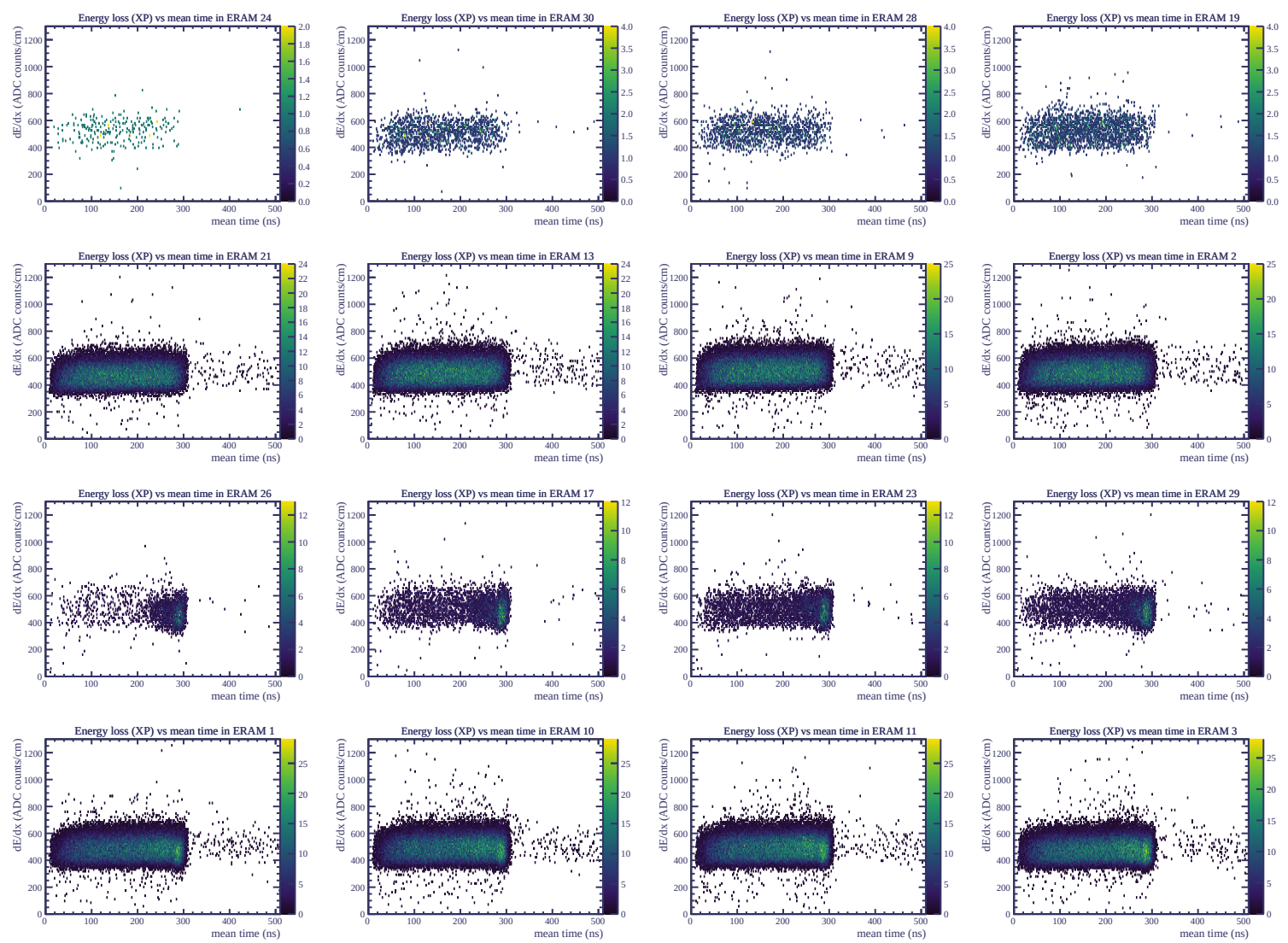


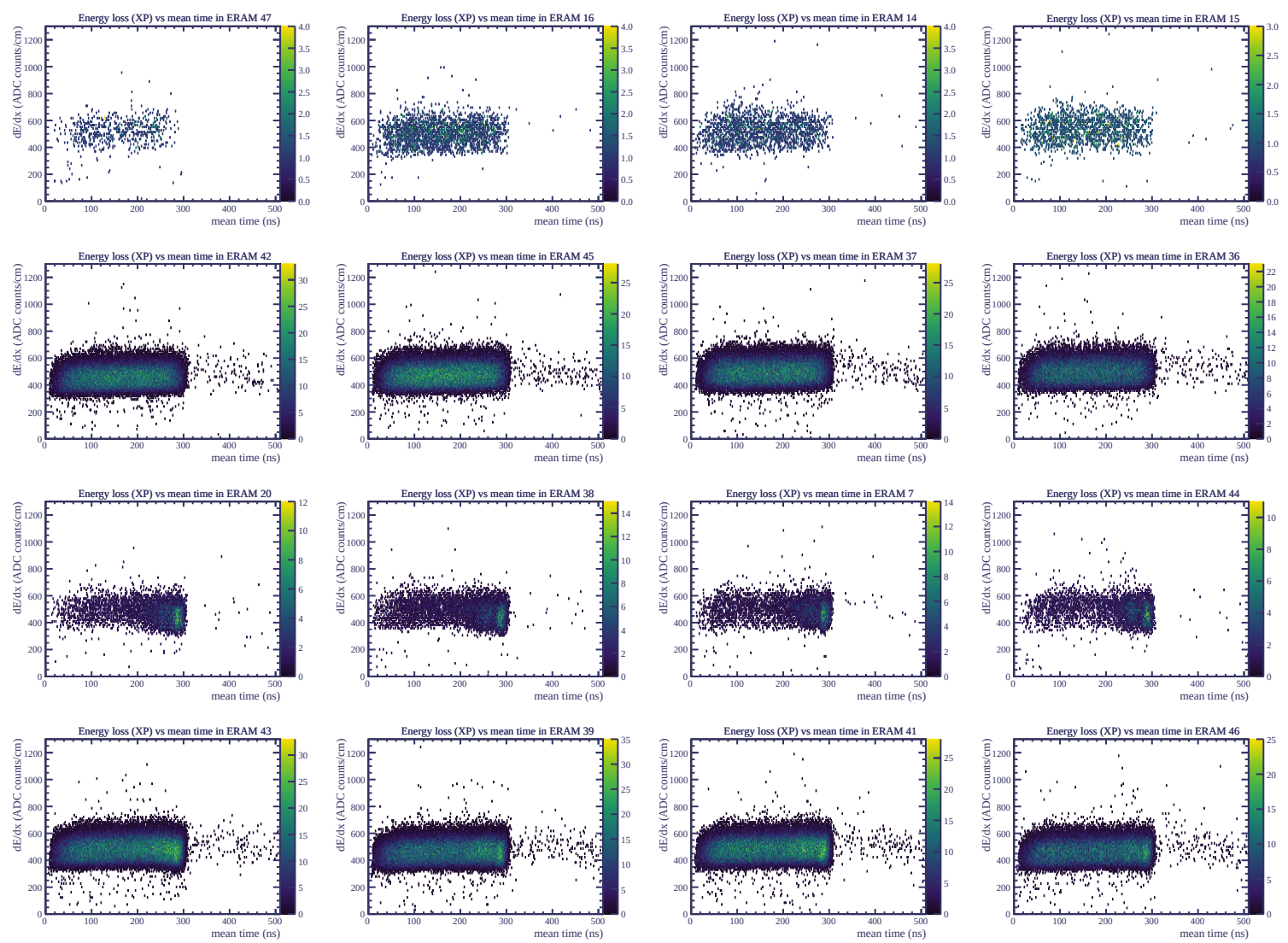




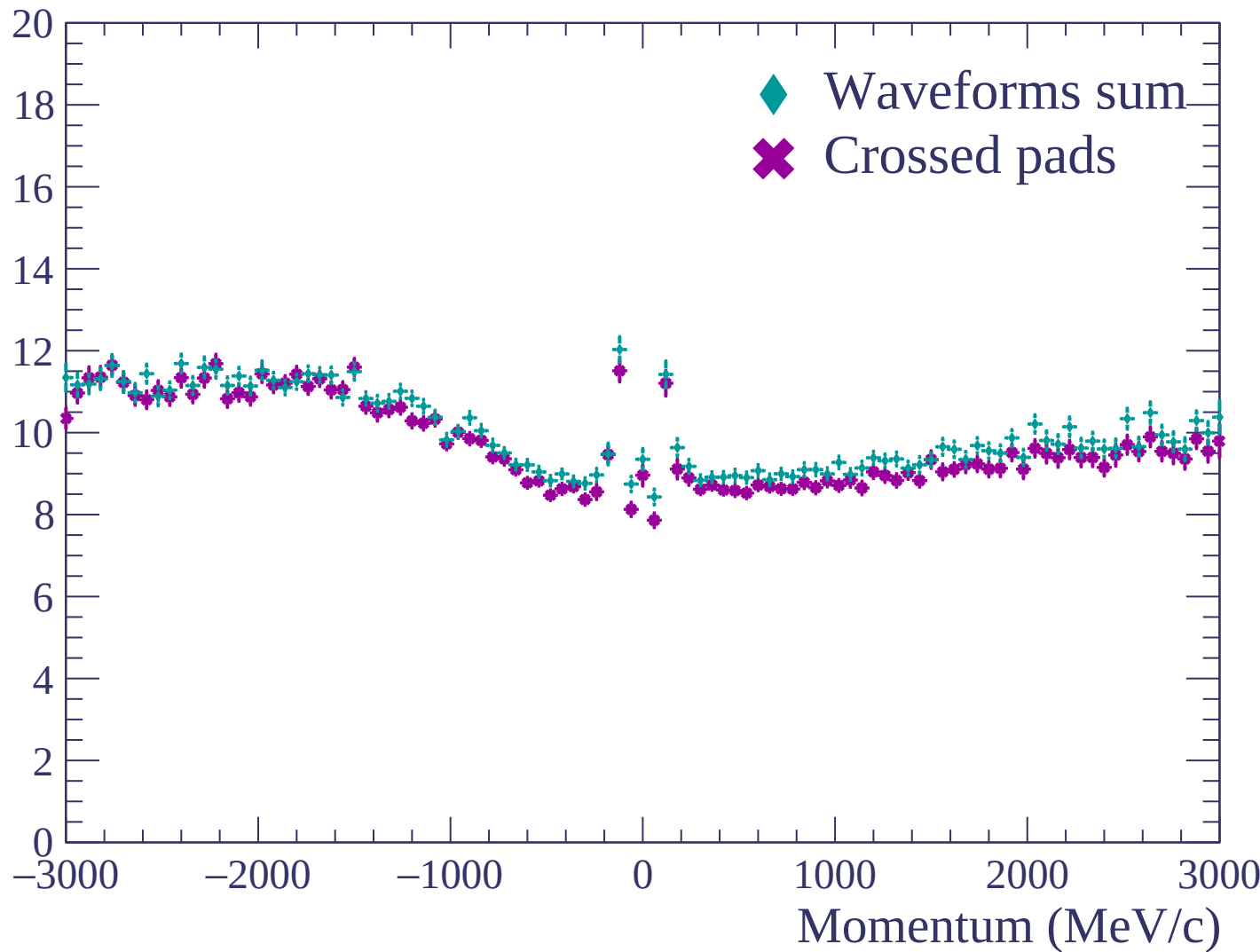


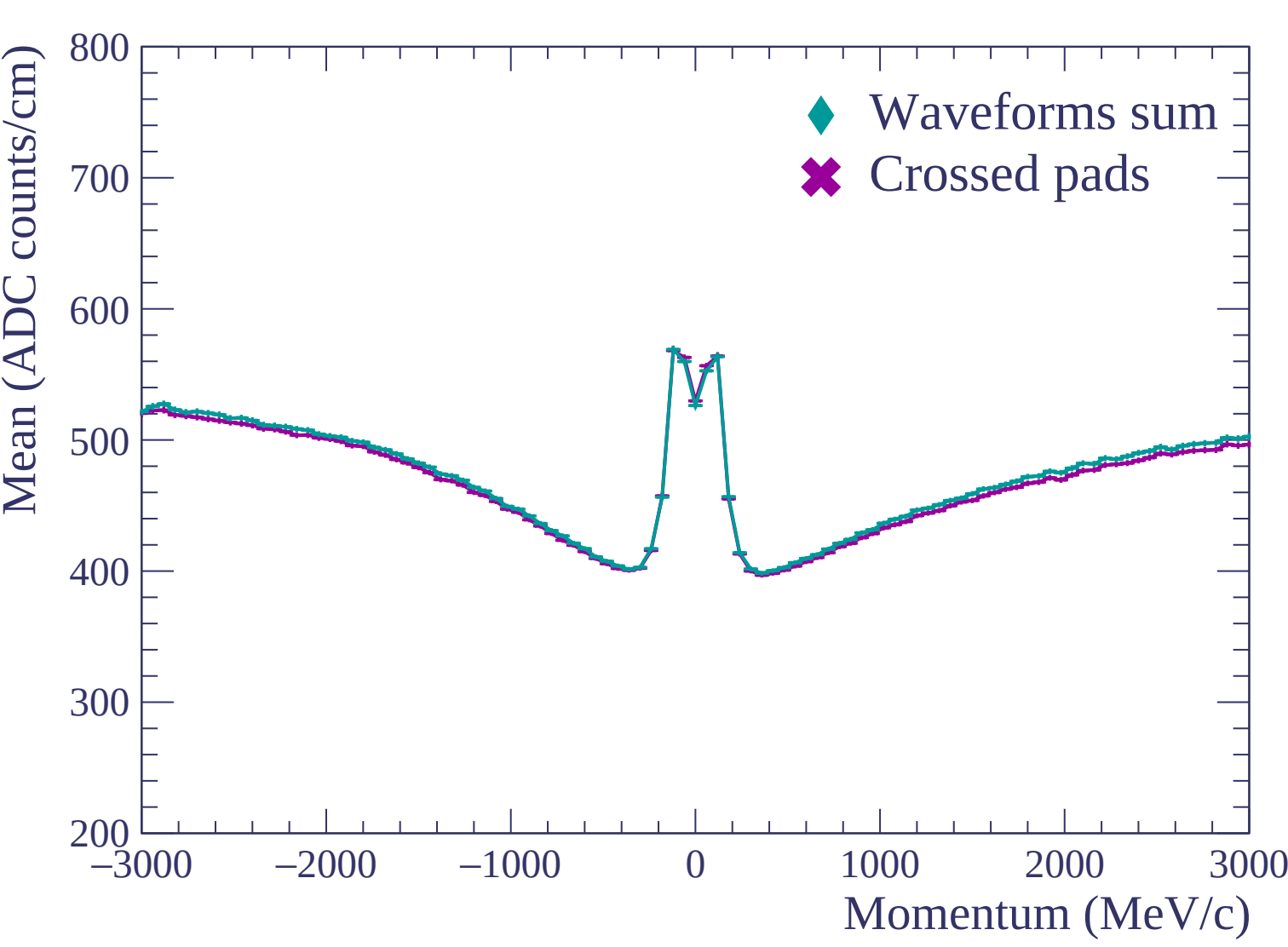


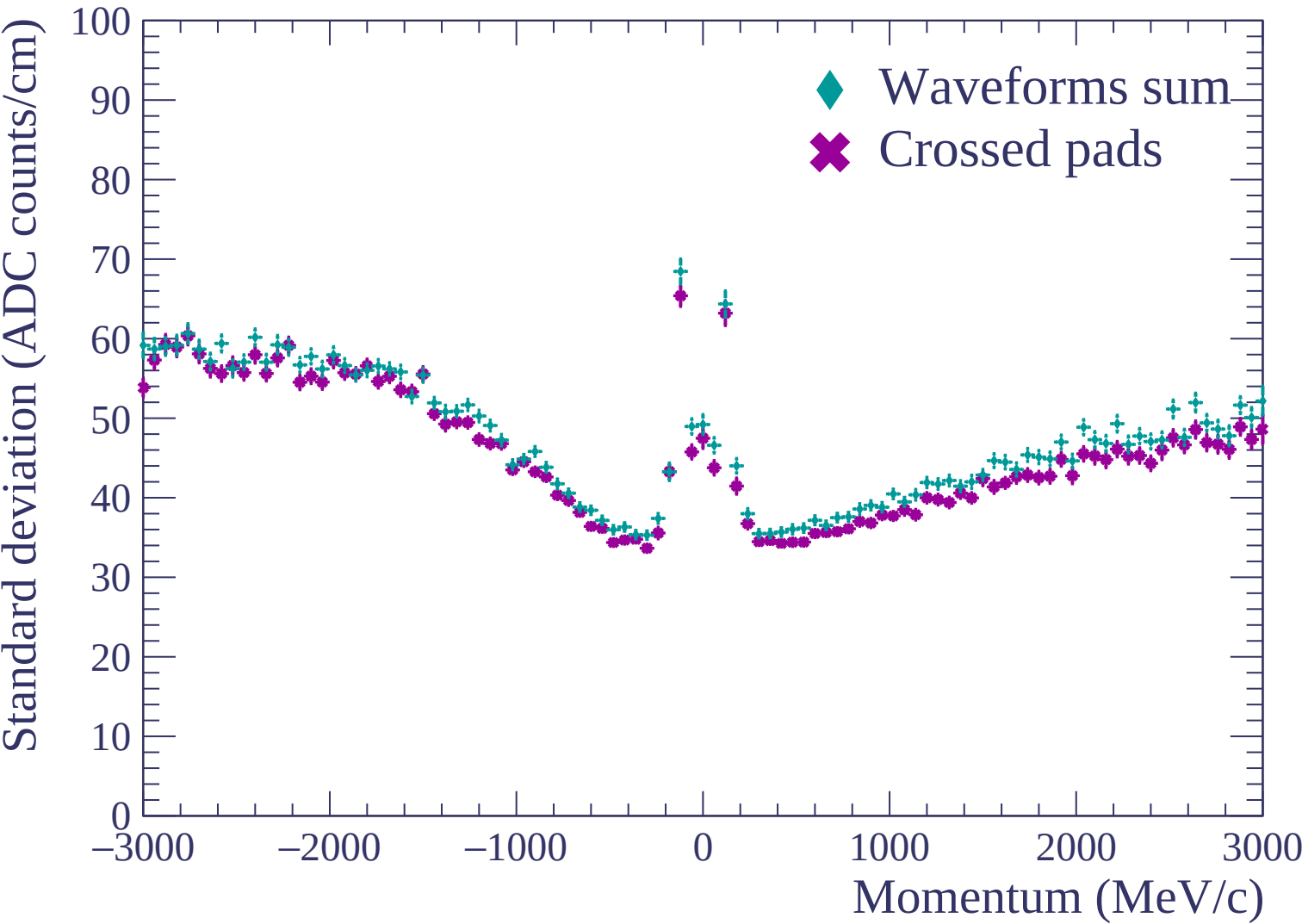


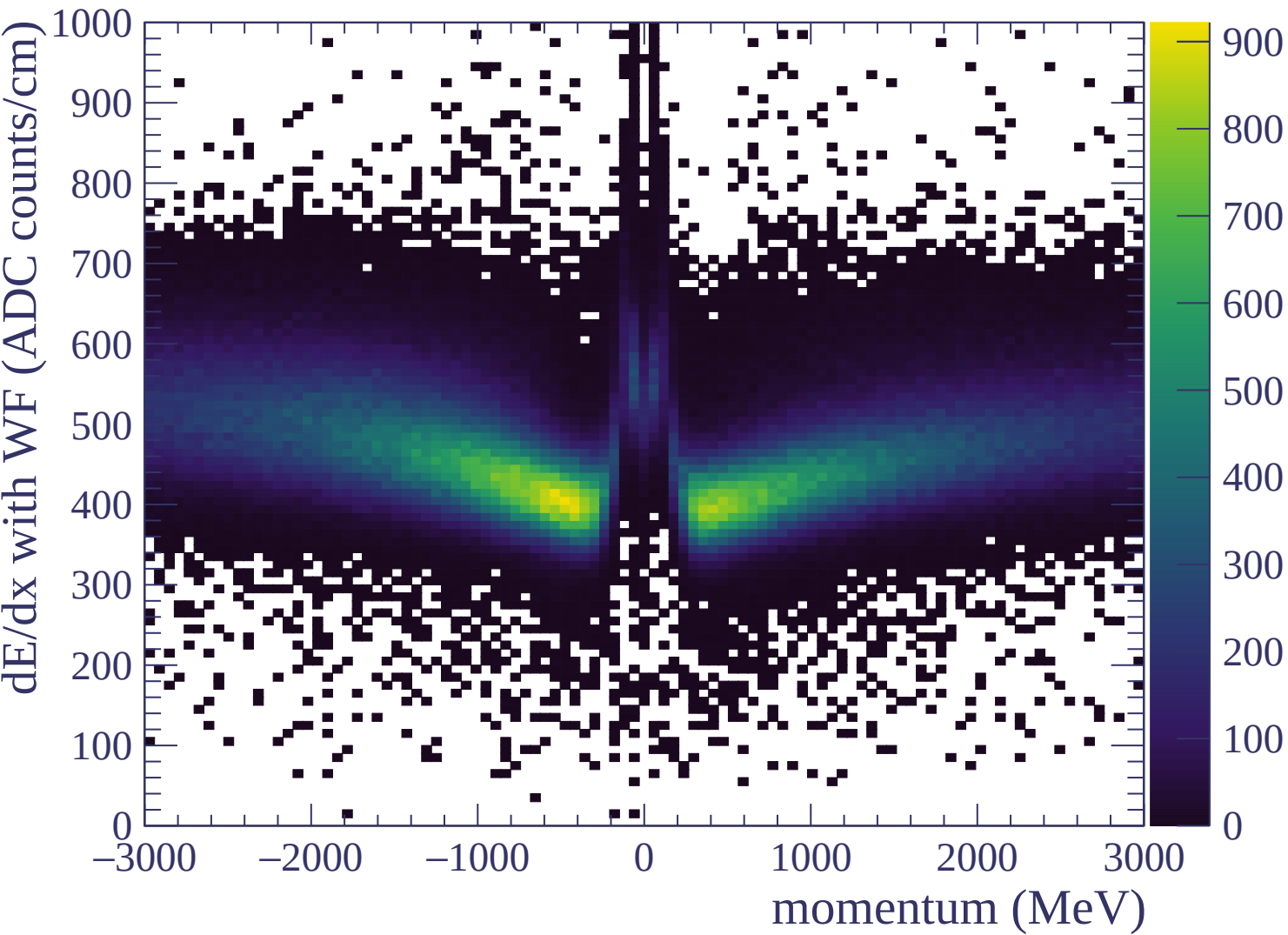


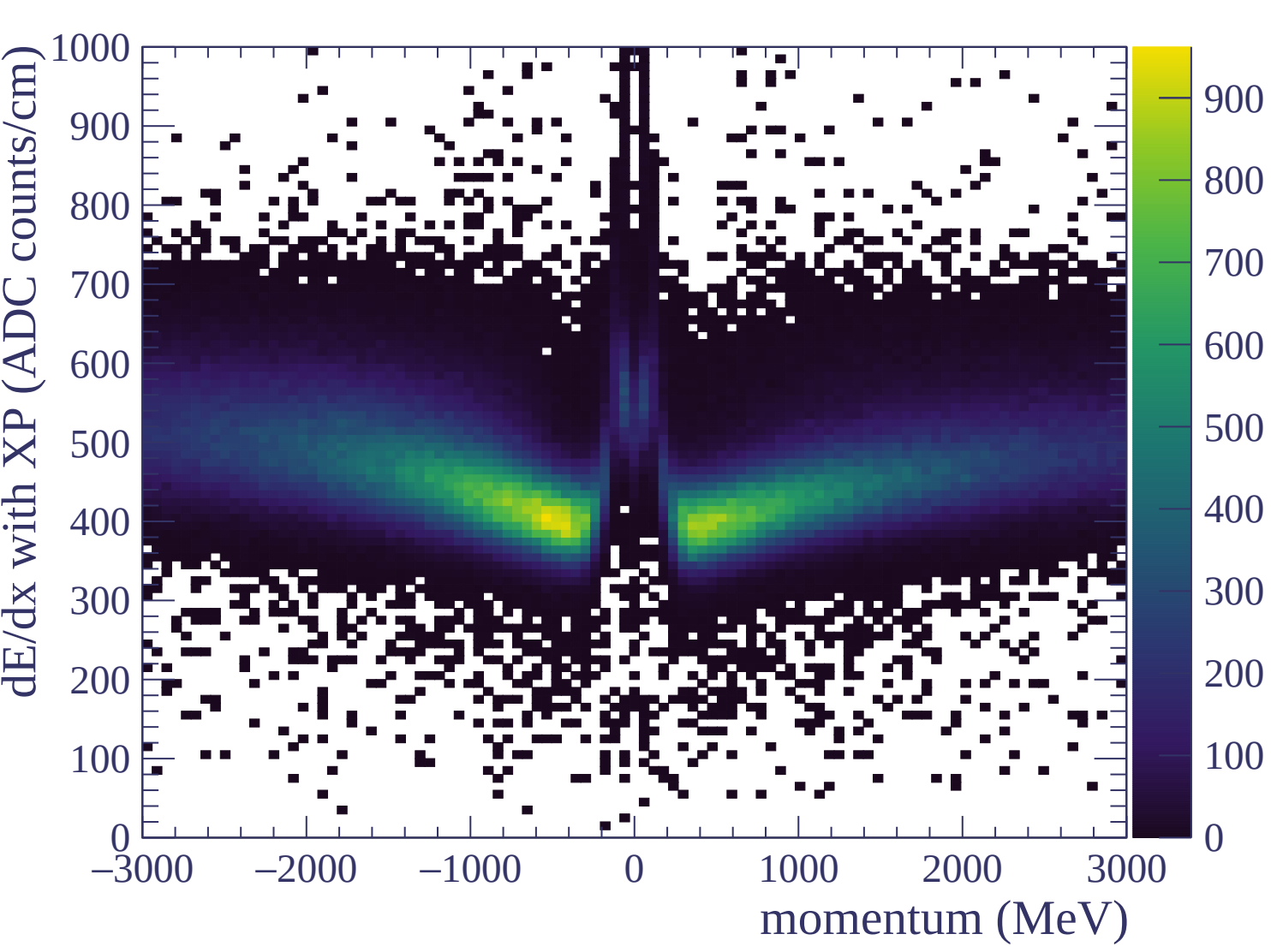
Resolution (%)

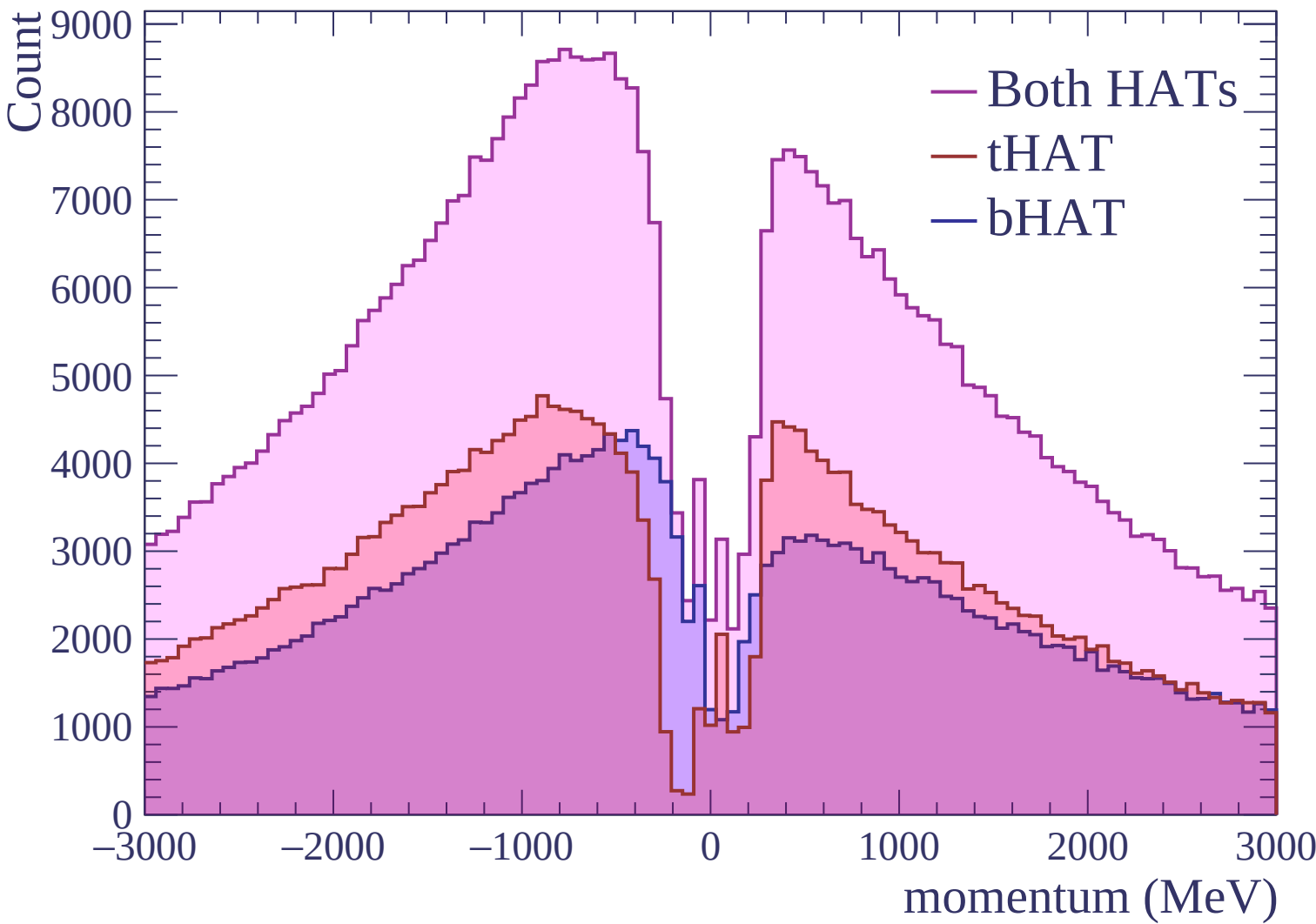








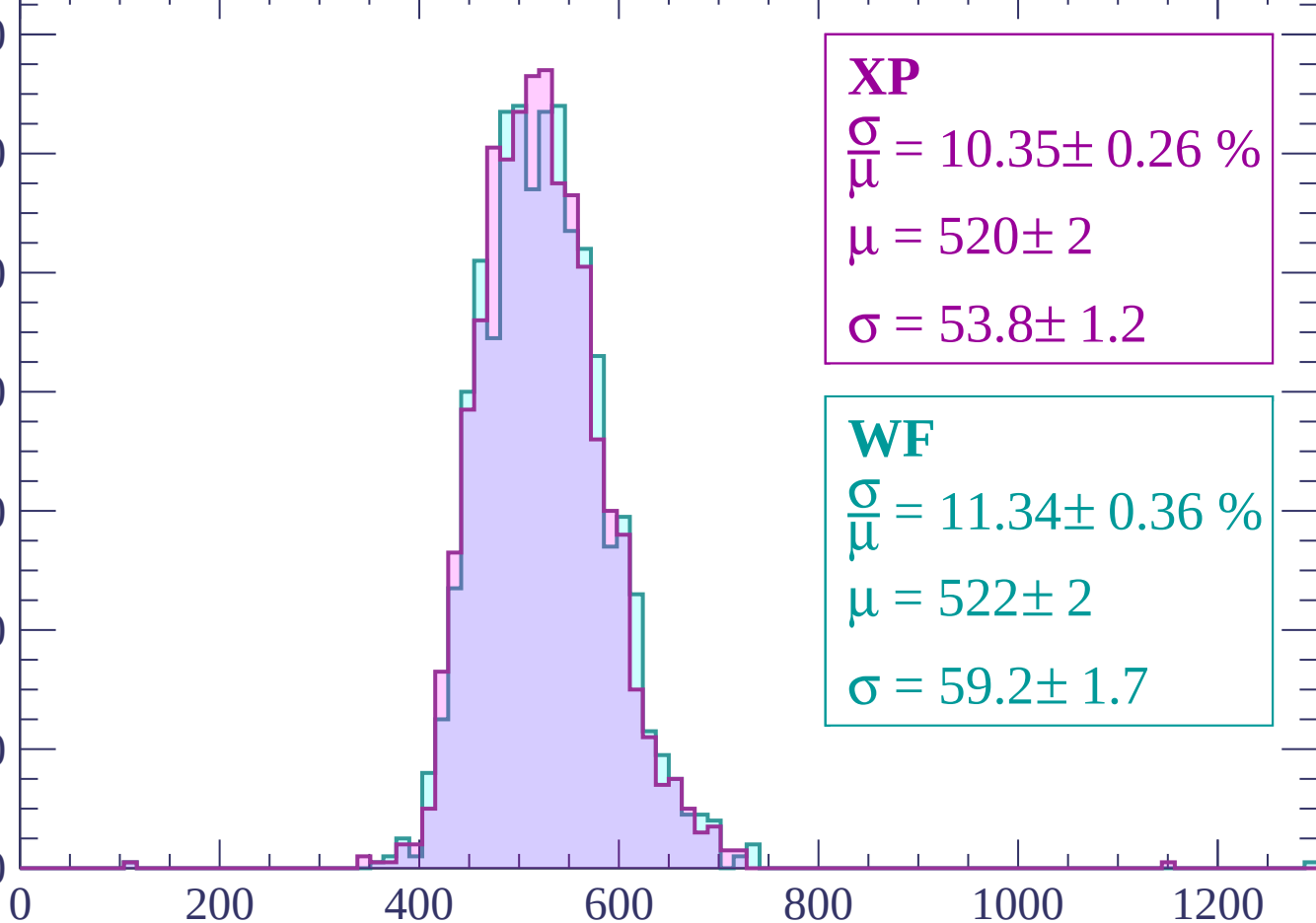




Energy loss | -3000 < p < -2940

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 10.35 \pm 0.26 \%$$

$$\mu = 520 \pm 2$$

$$\sigma = 53.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.34 \pm 0.36 \%$$

$$\mu = 522 \pm 2$$

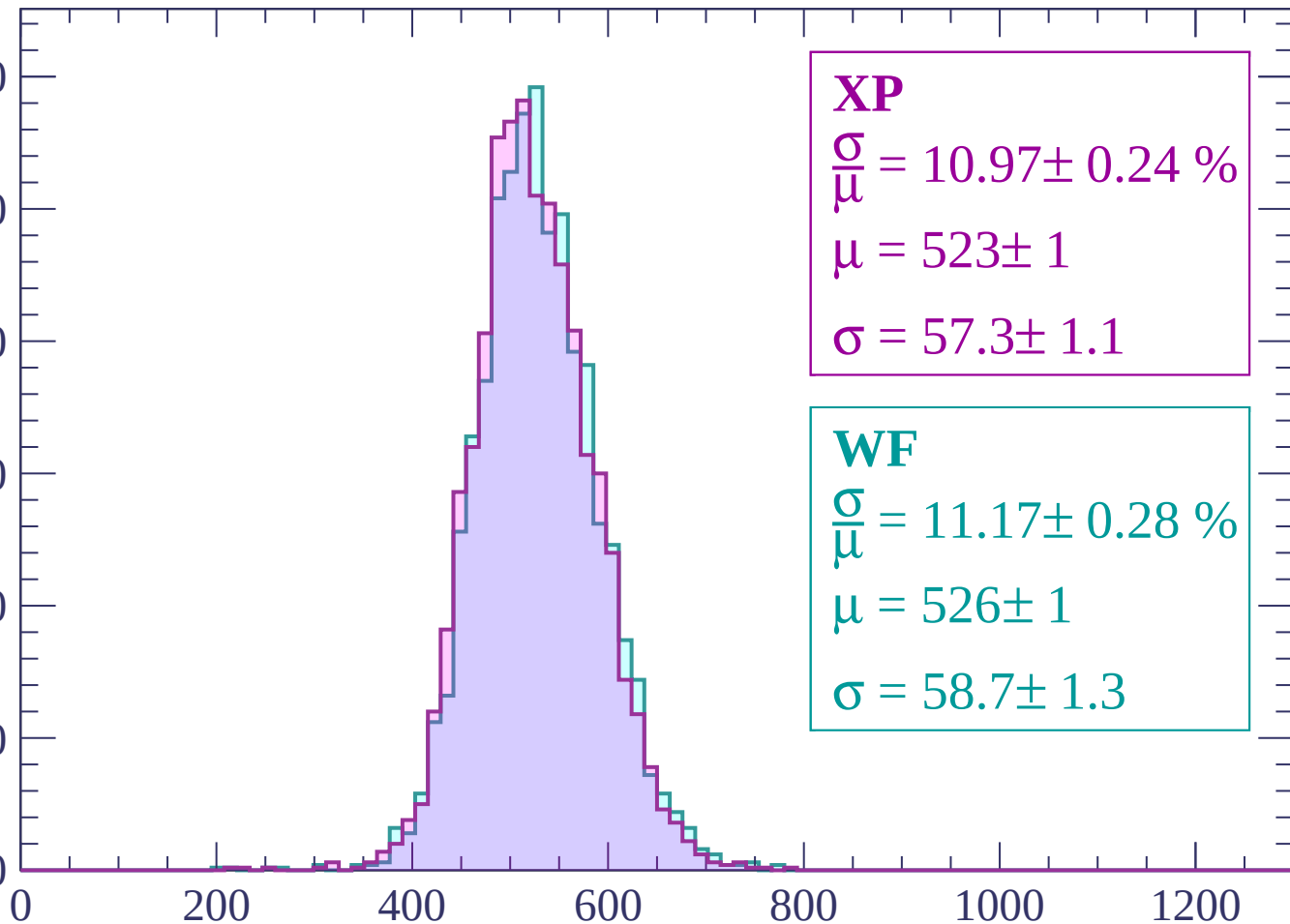
$$\sigma = 59.2 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-2940 < p < -2880$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.97 \pm 0.24 \%$$

$$\mu = 523 \pm 1$$

$$\sigma = 57.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.17 \pm 0.28 \%$$

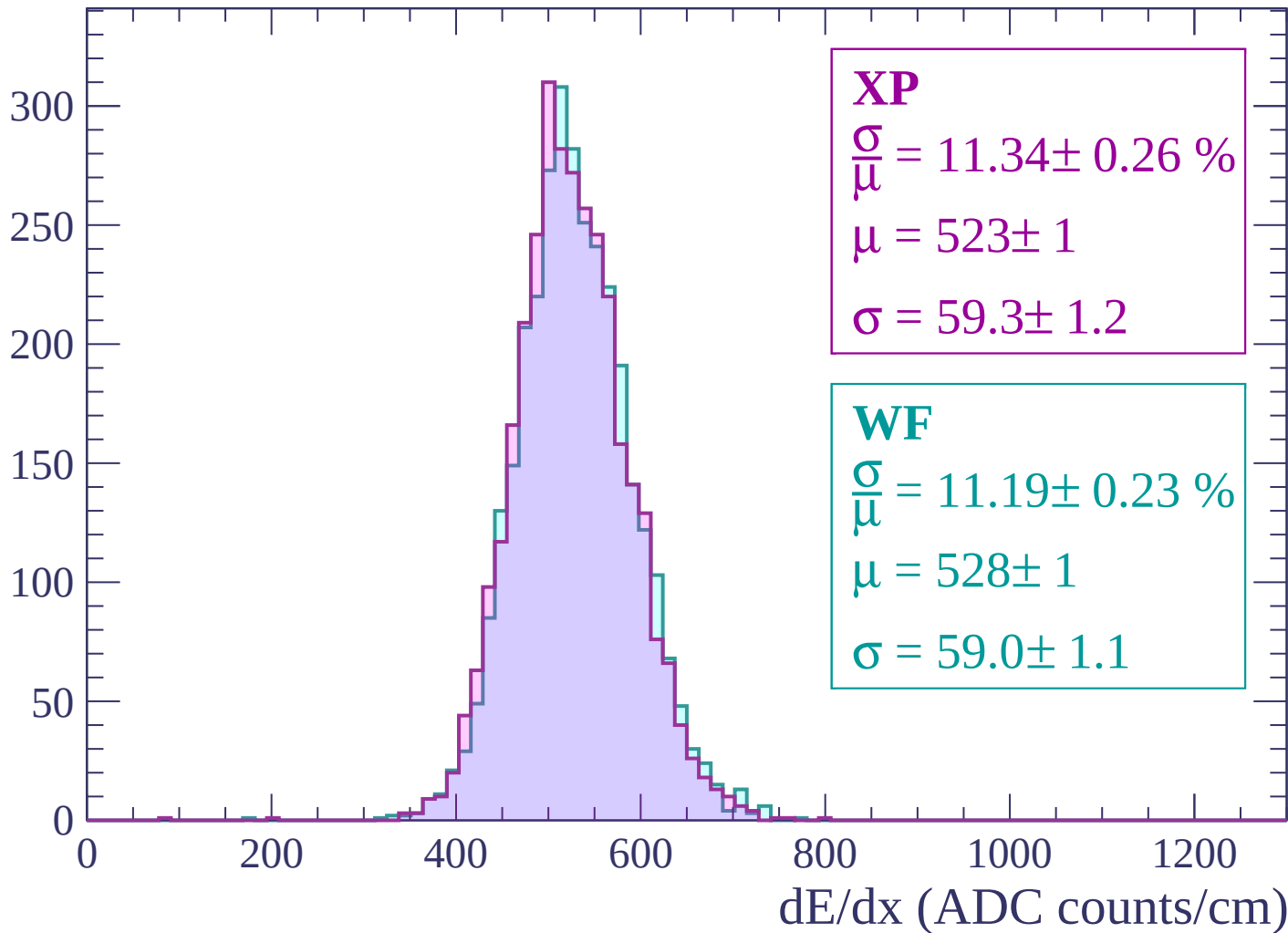
$$\mu = 526 \pm 1$$

$$\sigma = 58.7 \pm 1.3$$

dE/dx (ADC counts/cm)

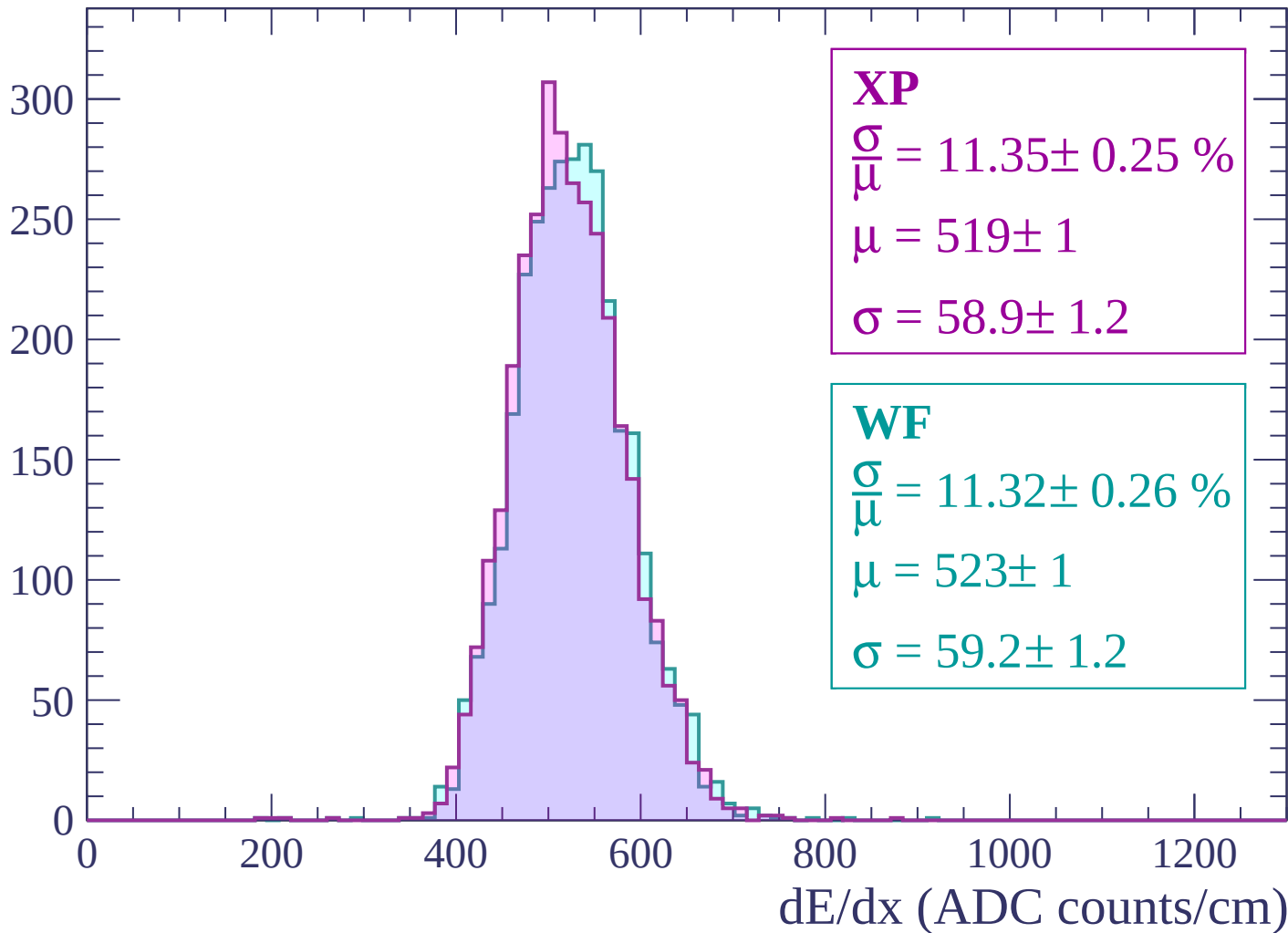
Energy loss | $-2880 < p < -2820$

Count



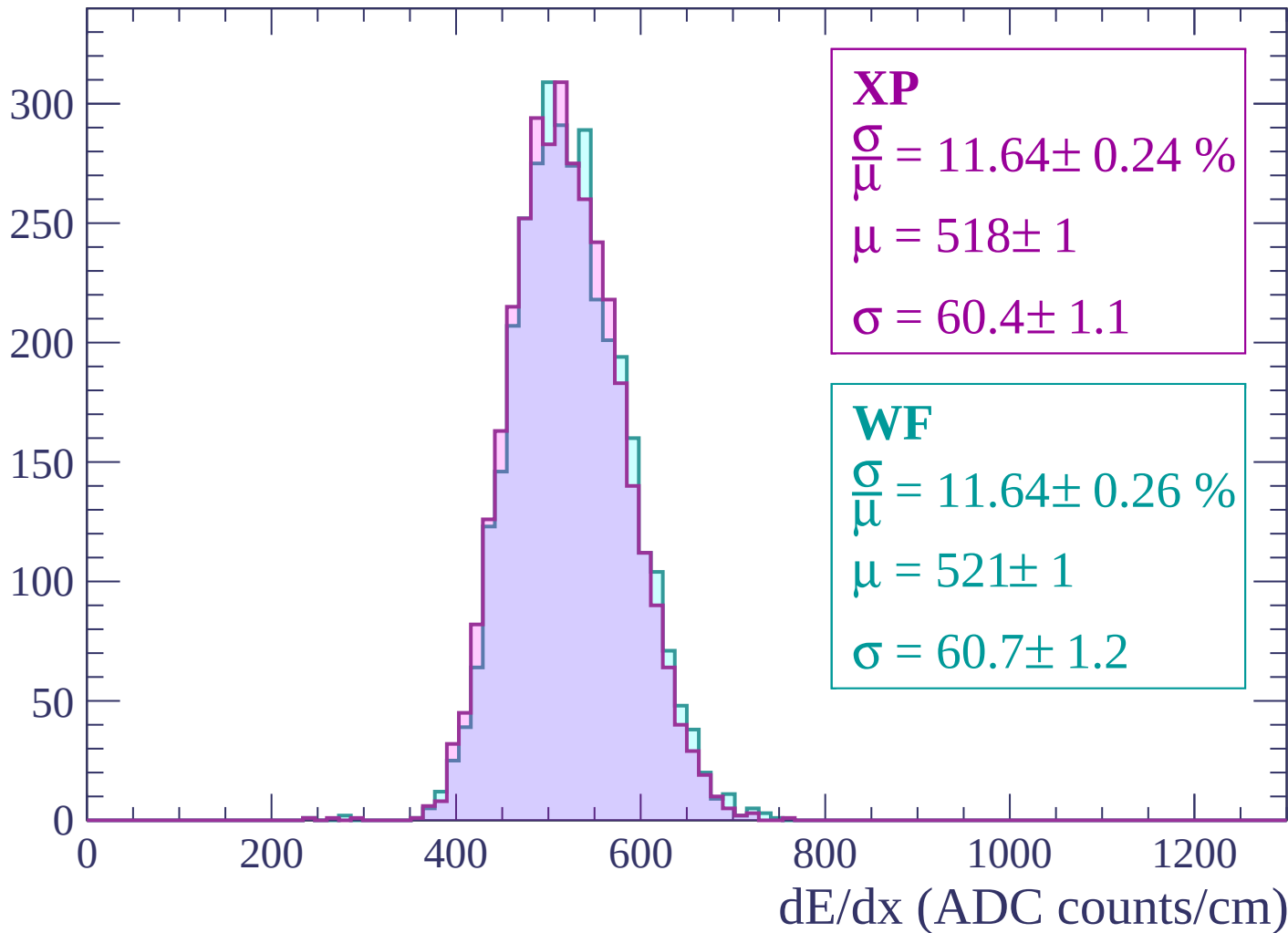
Energy loss | $-2820 < p < -2760$

Count



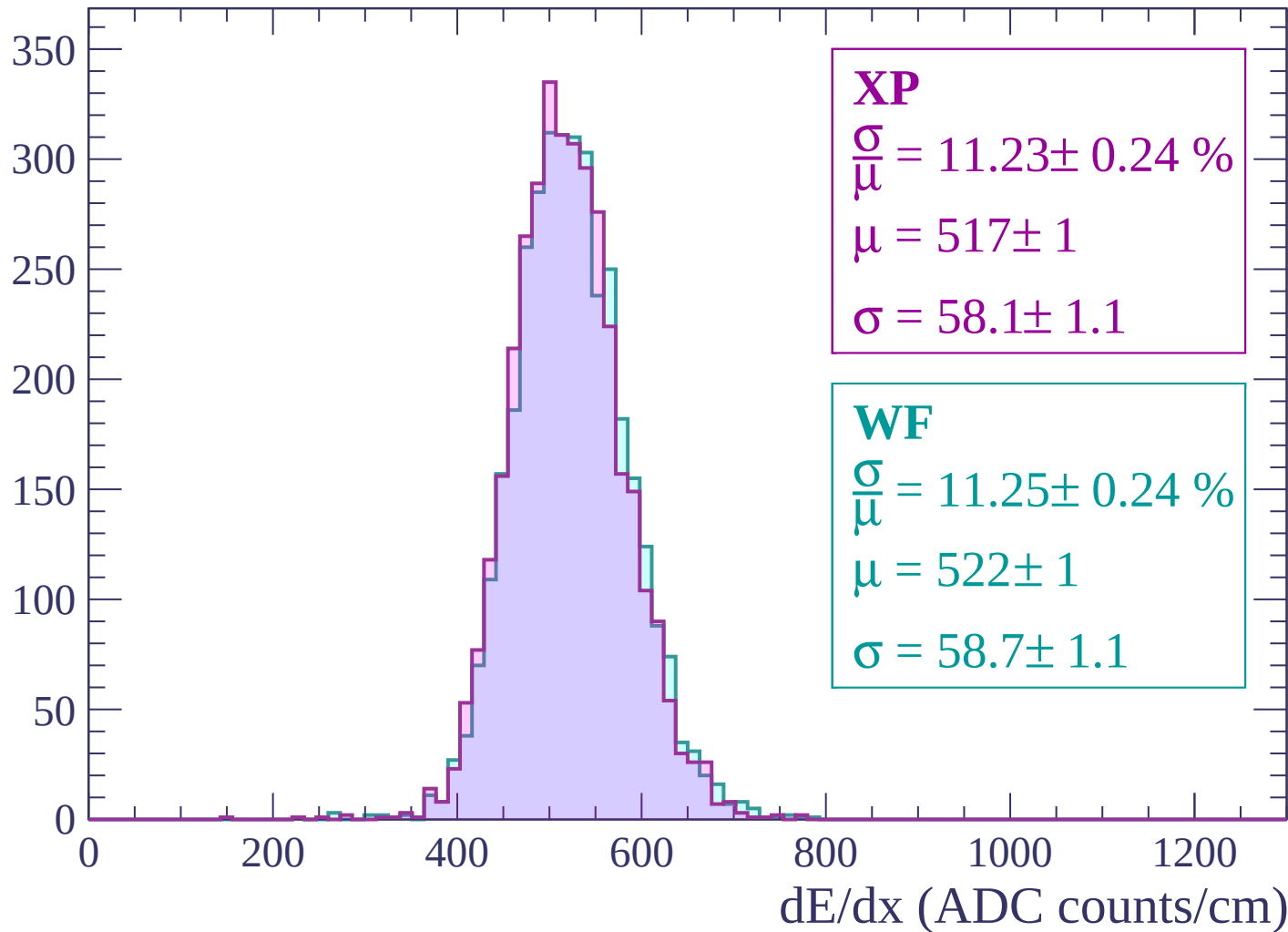
Energy loss | -2760 < p < -2700

Count



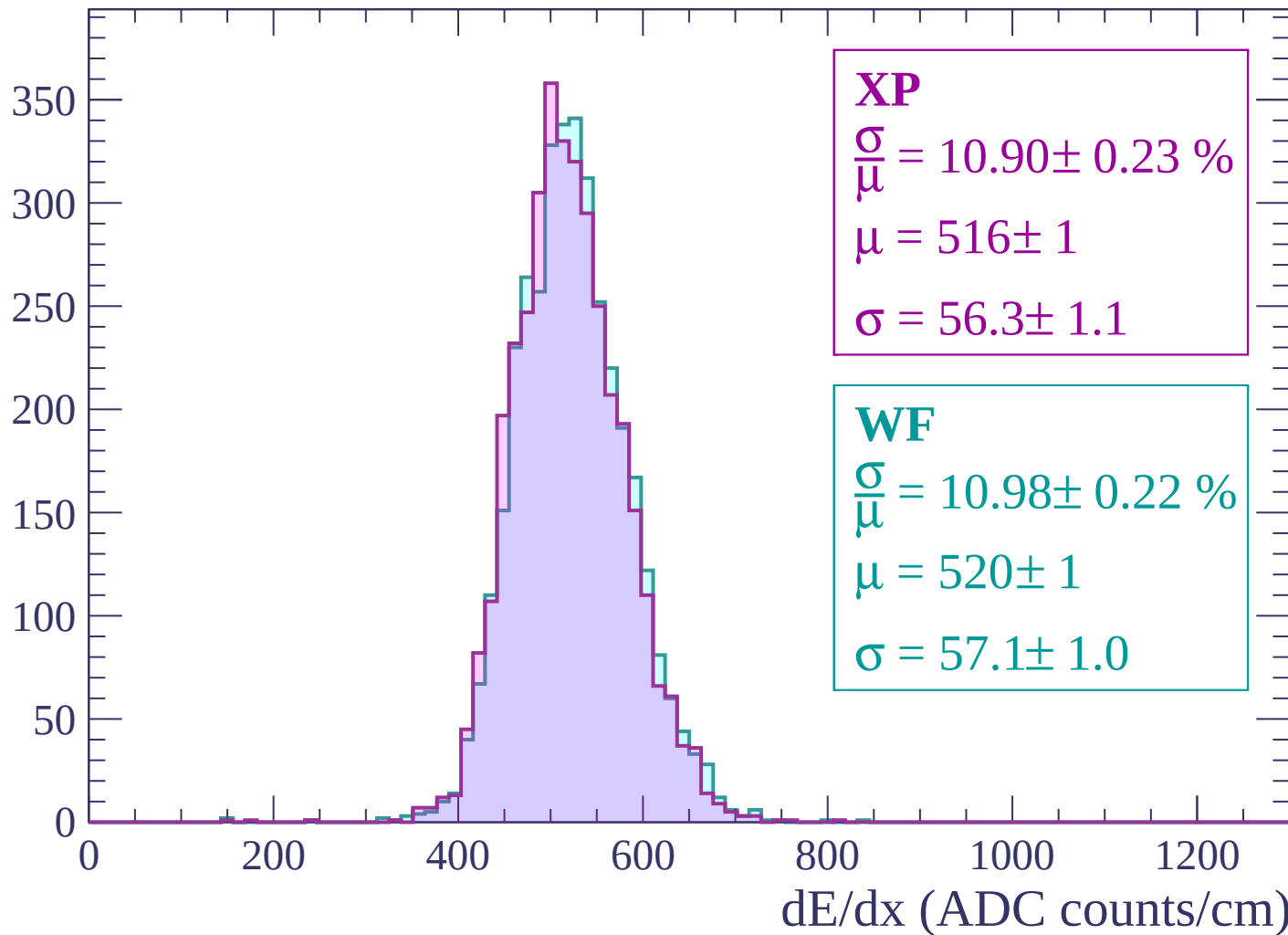
Energy loss | -2700 < p < -2640

Count



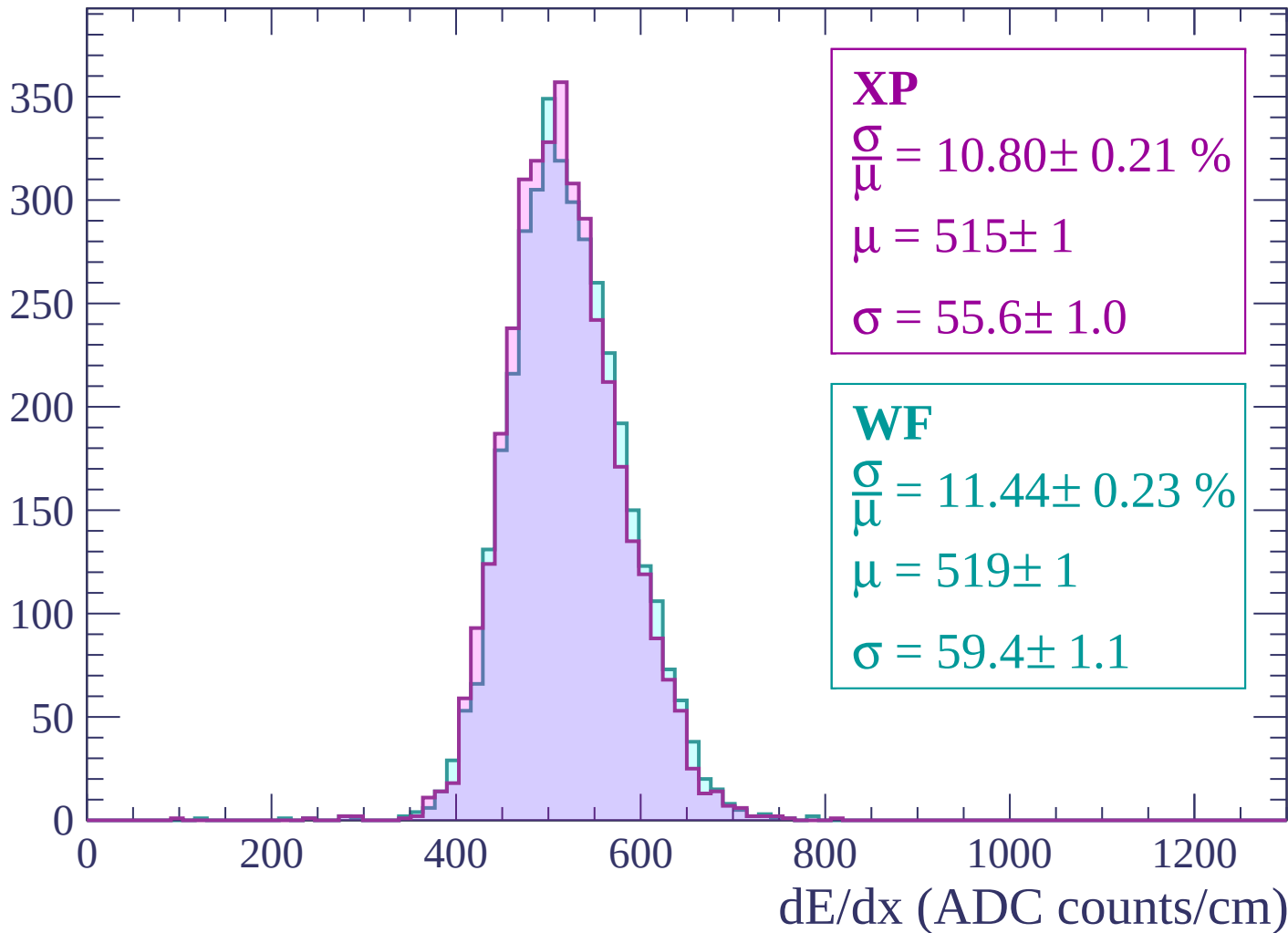
Energy loss | $-2640 < p < -2580$

Count



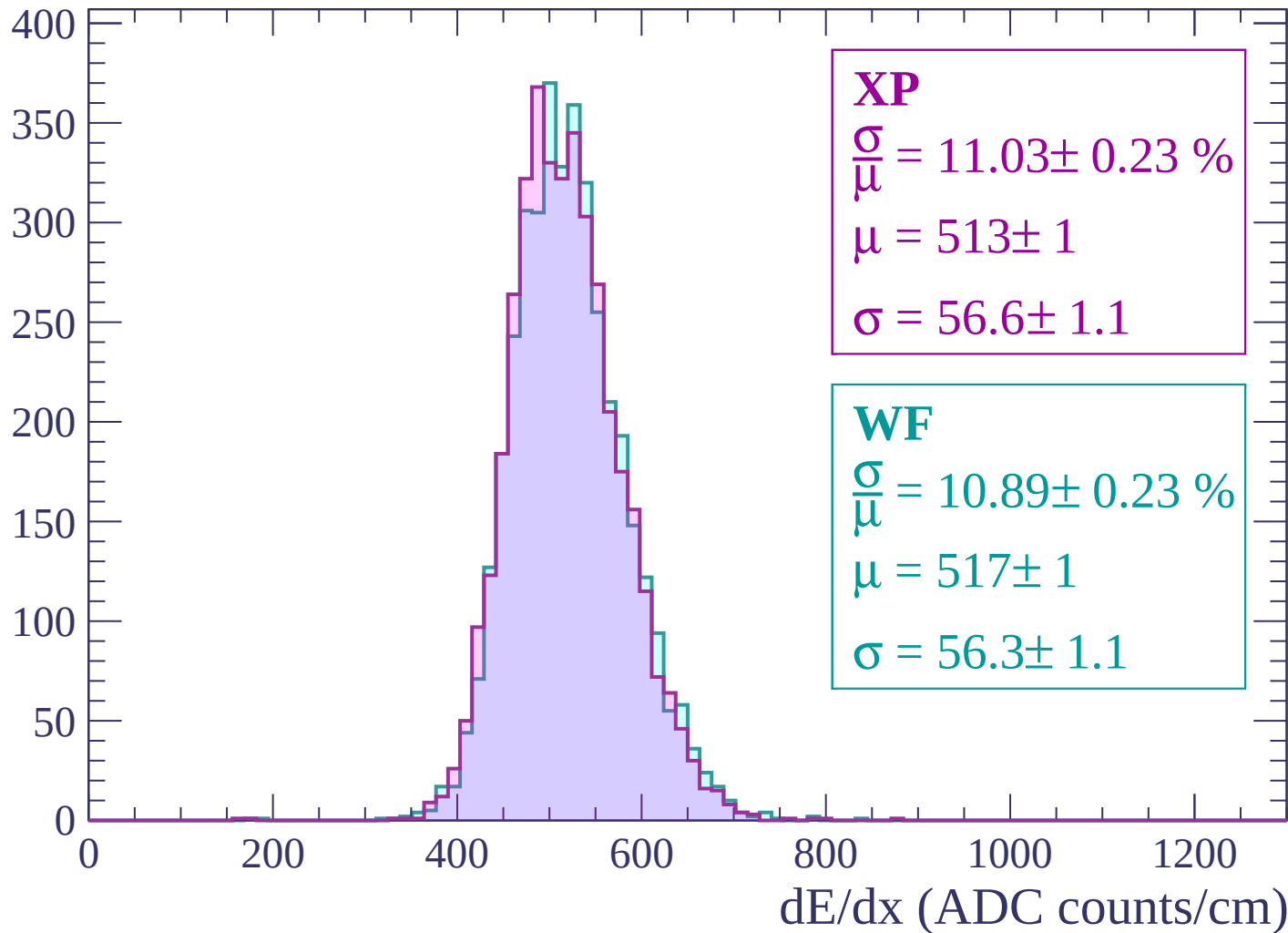
Energy loss | -2580 < p < -2520

Count



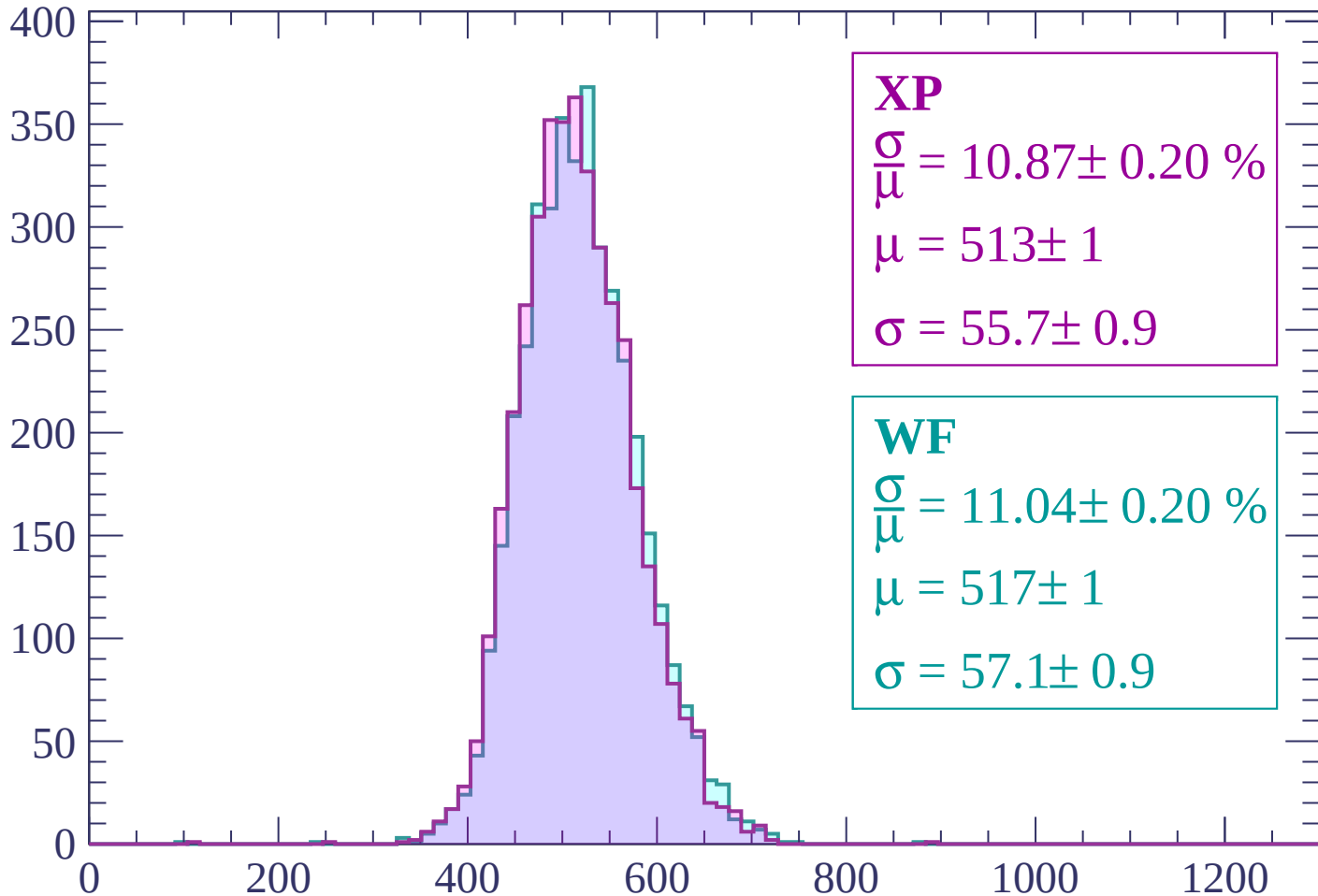
Energy loss | -2520 < p < -2460

Count



Energy loss | -2460 < p < -2400

Count



XP

$$\frac{\sigma}{\mu} = 10.87 \pm 0.20 \%$$

$$\mu = 513 \pm 1$$

$$\sigma = 55.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.04 \pm 0.20 \%$$

$$\mu = 517 \pm 1$$

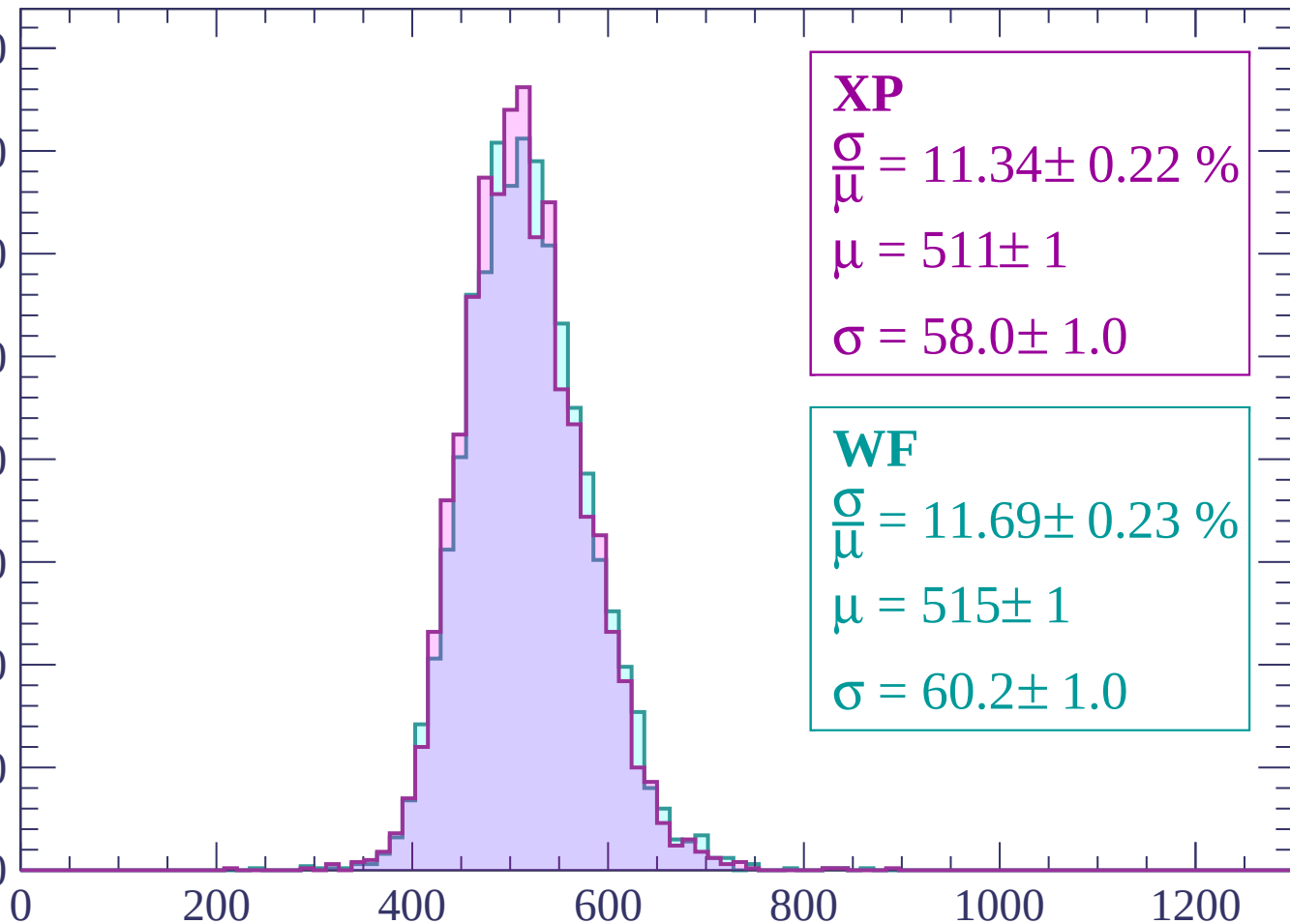
$$\sigma = 57.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -2400 < p < -2340

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.34 \pm 0.22 \%$$

$$\mu = 511 \pm 1$$

$$\sigma = 58.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.69 \pm 0.23 \%$$

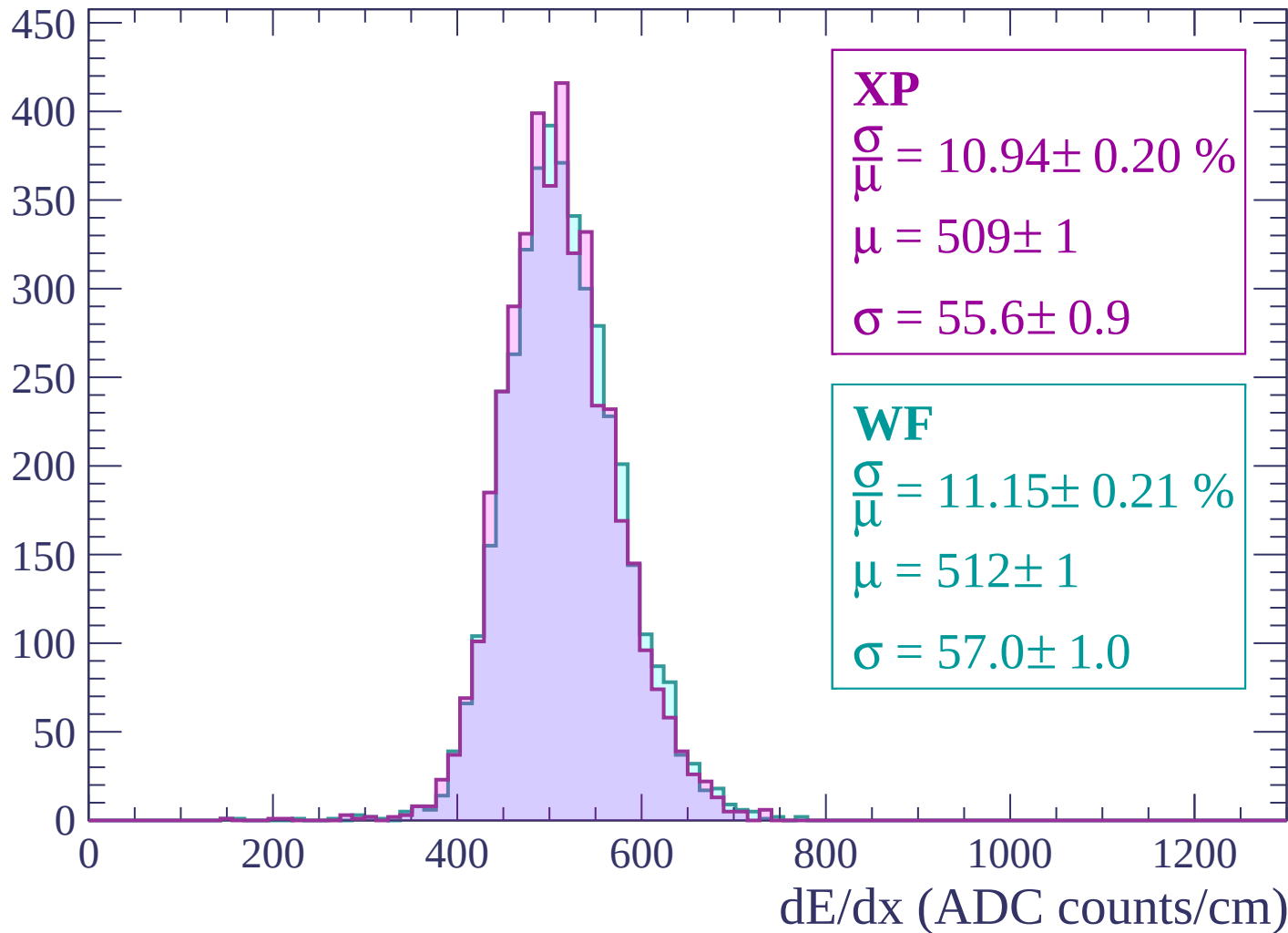
$$\mu = 515 \pm 1$$

$$\sigma = 60.2 \pm 1.0$$

dE/dx (ADC counts/cm)

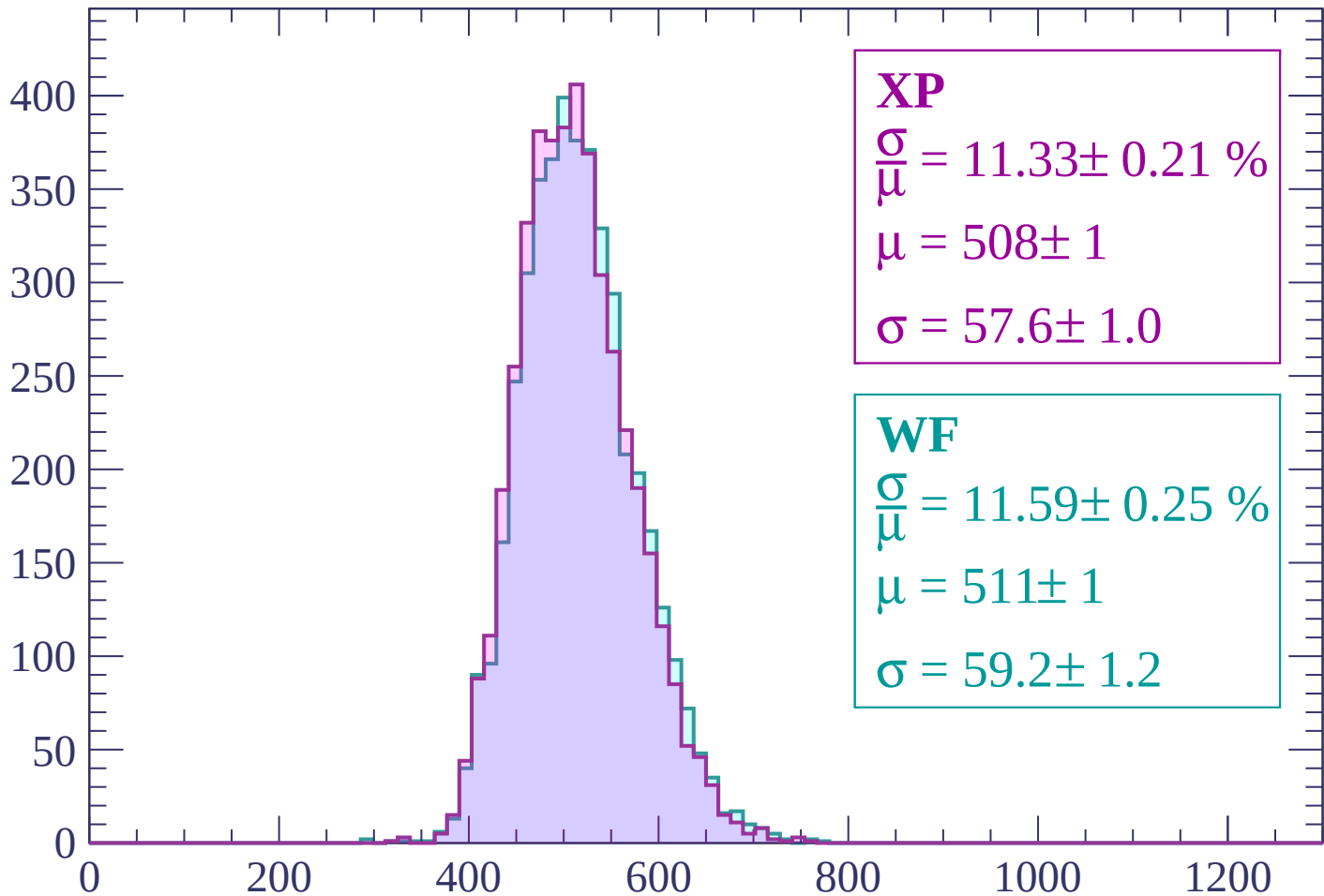
Energy loss | -2340 < p < -2280

Count



Energy loss | $-2280 < p < -2220$

Count



XP

$$\frac{\sigma}{\mu} = 11.33 \pm 0.21 \%$$

$$\mu = 508 \pm 1$$

$$\sigma = 57.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.59 \pm 0.25 \%$$

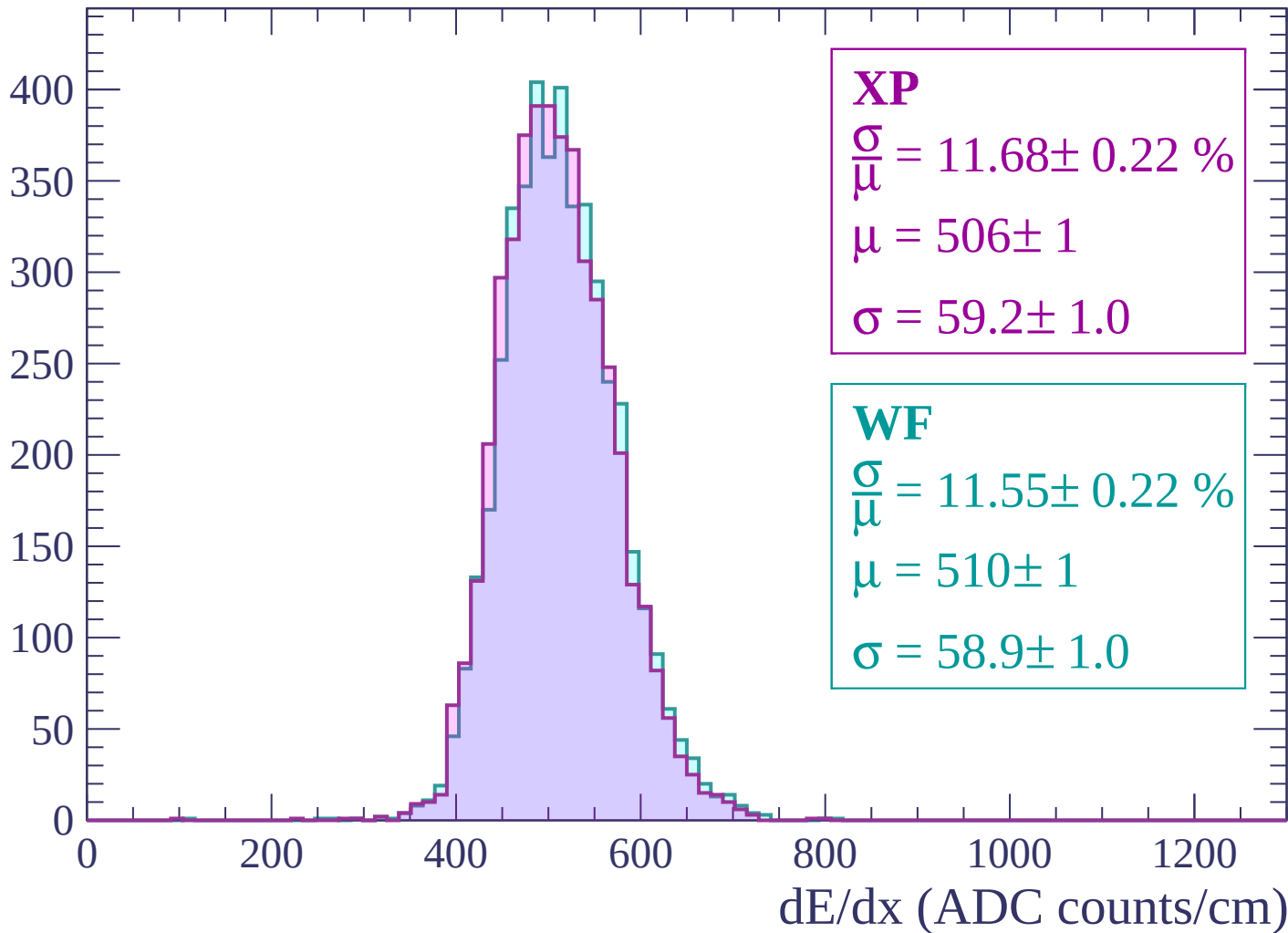
$$\mu = 511 \pm 1$$

$$\sigma = 59.2 \pm 1.2$$

dE/dx (ADC counts/cm)

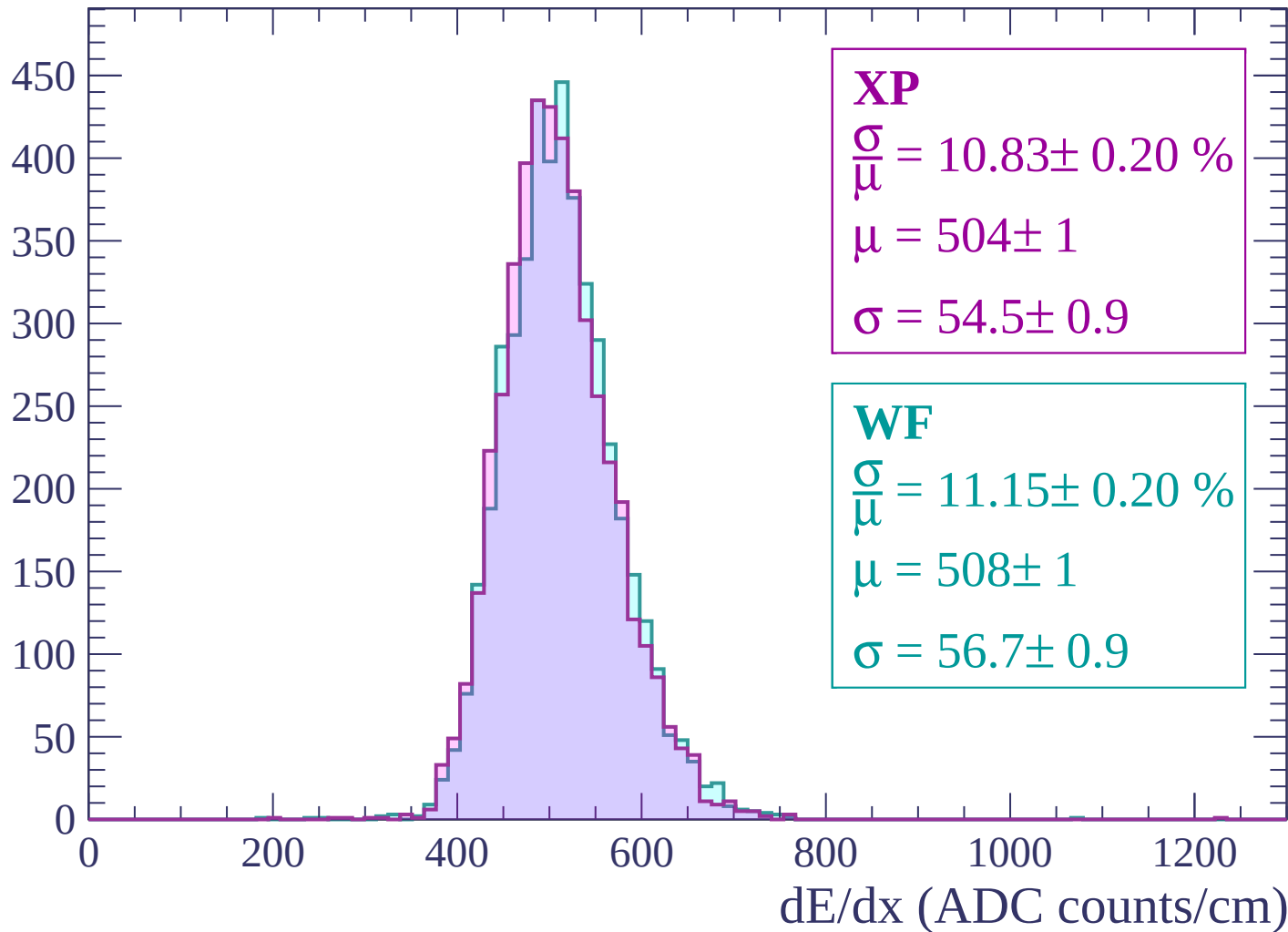
Energy loss | -2220 < p < -2160

Count



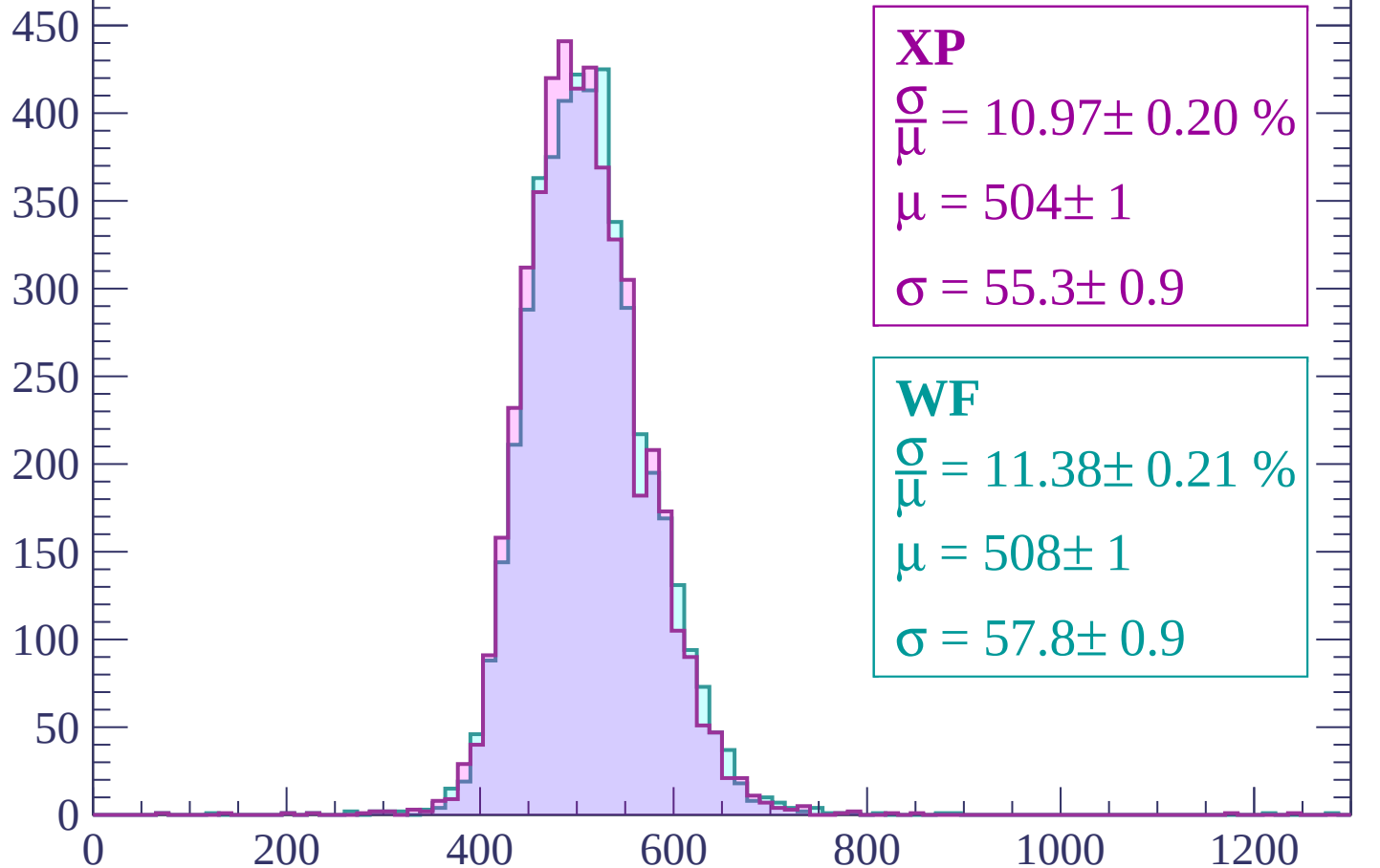
Energy loss | -2160 < p < -2100

Count



Energy loss | -2100 < p < -2040

Count



XP

$$\frac{\sigma}{\mu} = 10.97 \pm 0.20 \%$$

$$\mu = 504 \pm 1$$

$$\sigma = 55.3 \pm 0.9$$

WF

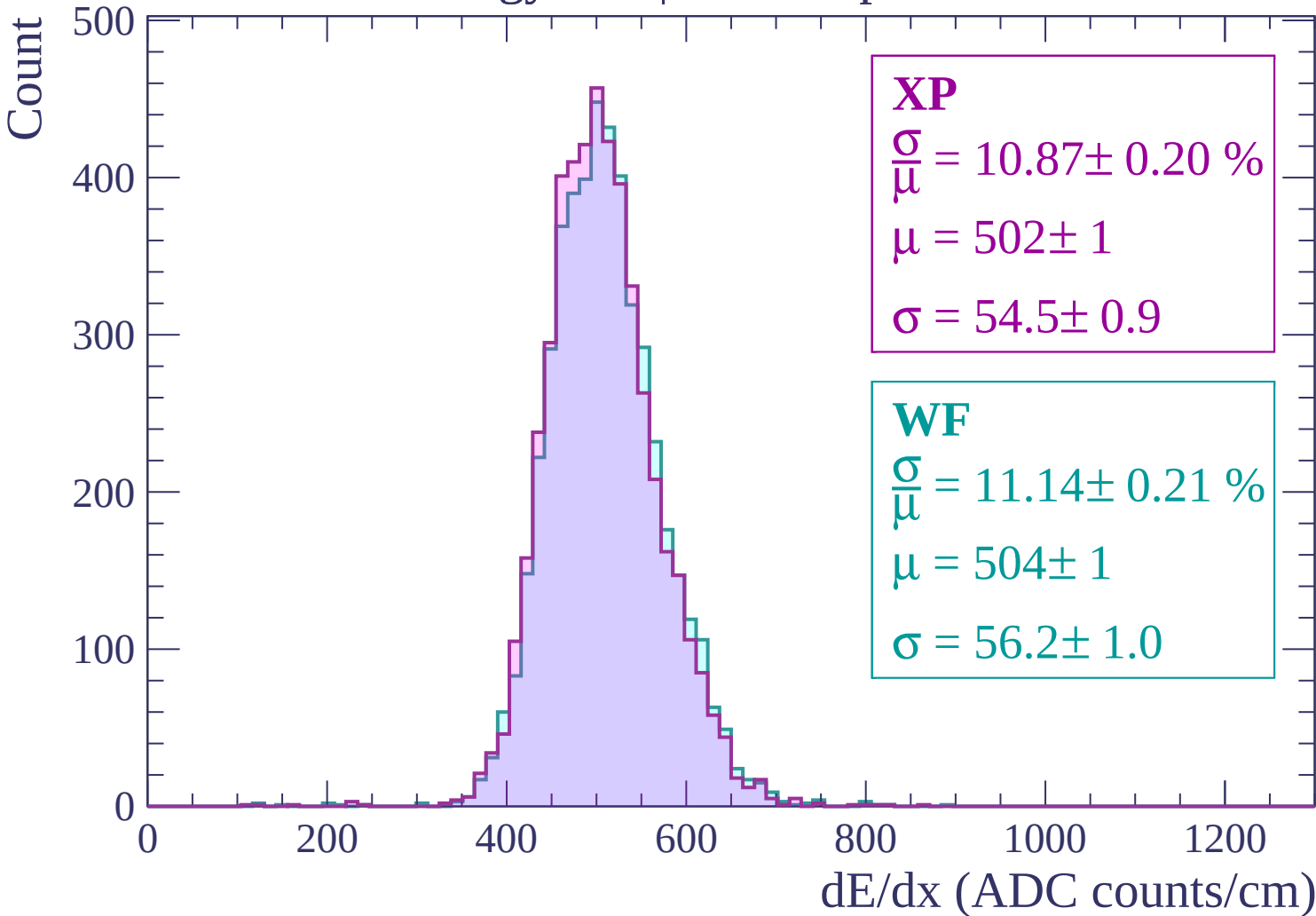
$$\frac{\sigma}{\mu} = 11.38 \pm 0.21 \%$$

$$\mu = 508 \pm 1$$

$$\sigma = 57.8 \pm 0.9$$

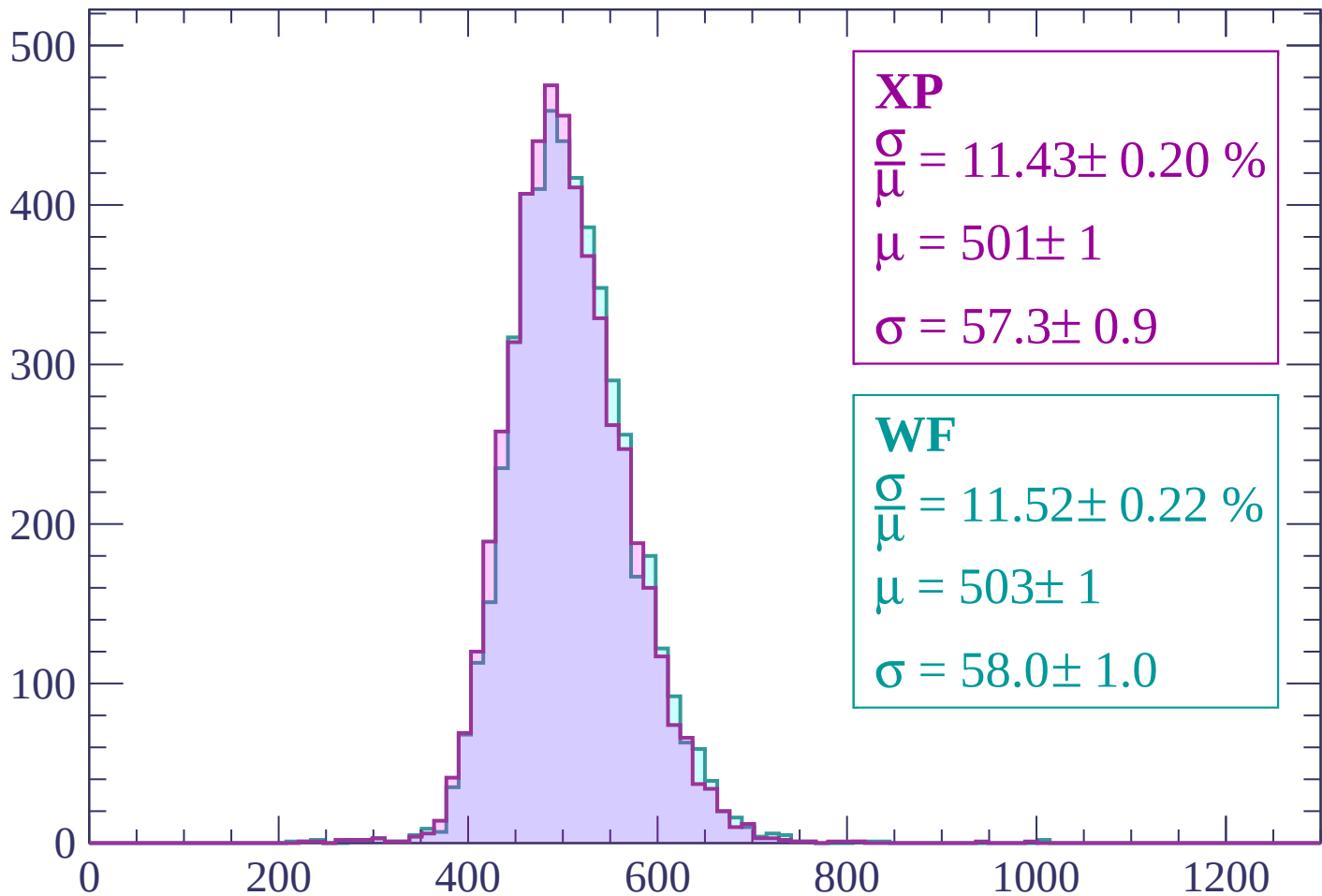
dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980



Energy loss | -1980 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 11.43 \pm 0.20 \%$$

$$\mu = 501 \pm 1$$

$$\sigma = 57.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$$

$$\mu = 503 \pm 1$$

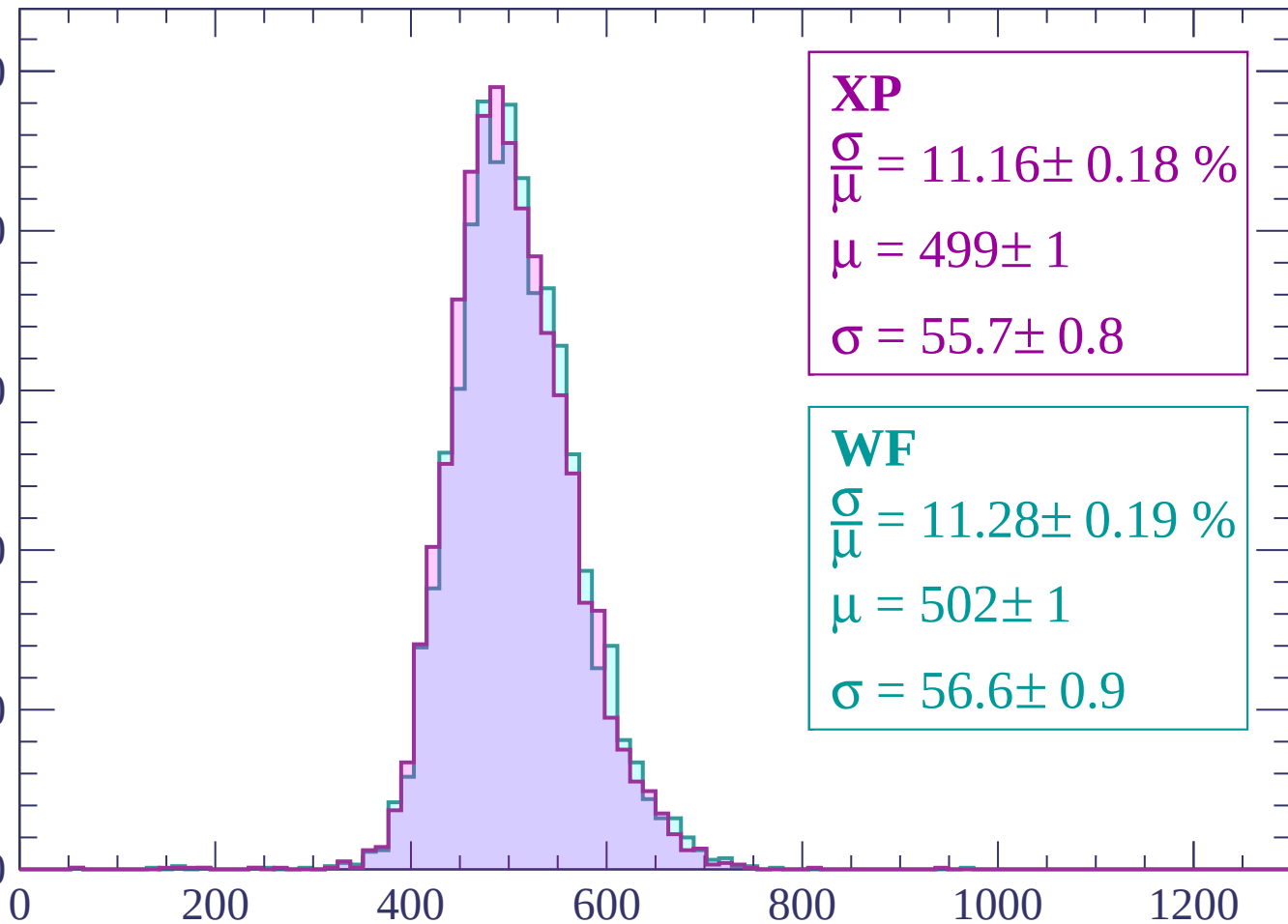
$$\sigma = 58.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 11.16 \pm 0.18 \%$$

$$\mu = 499 \pm 1$$

$$\sigma = 55.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.28 \pm 0.19 \%$$

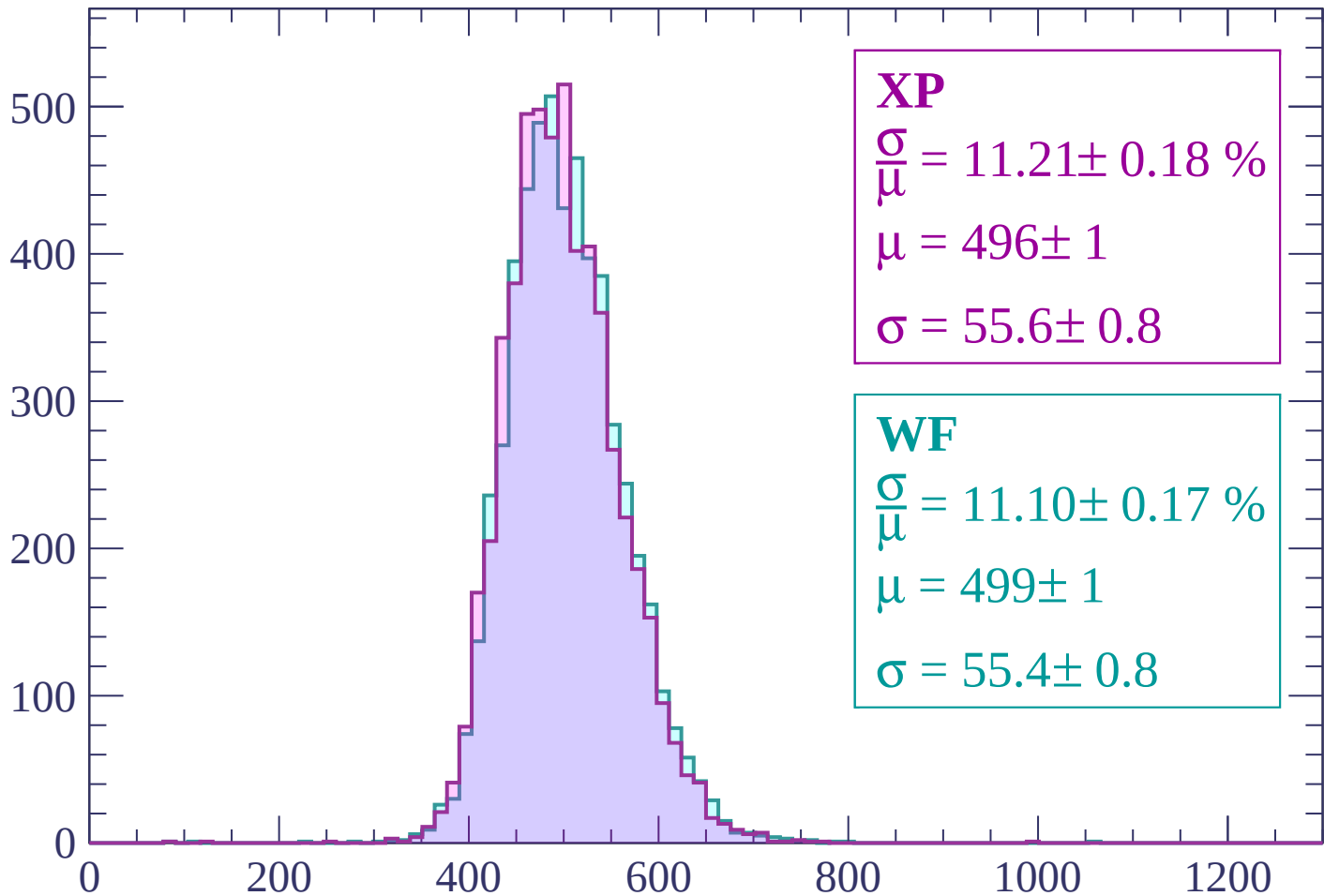
$$\mu = 502 \pm 1$$

$$\sigma = 56.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1860 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 11.21 \pm 0.18 \%$$

$$\mu = 496 \pm 1$$

$$\sigma = 55.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.10 \pm 0.17 \%$$

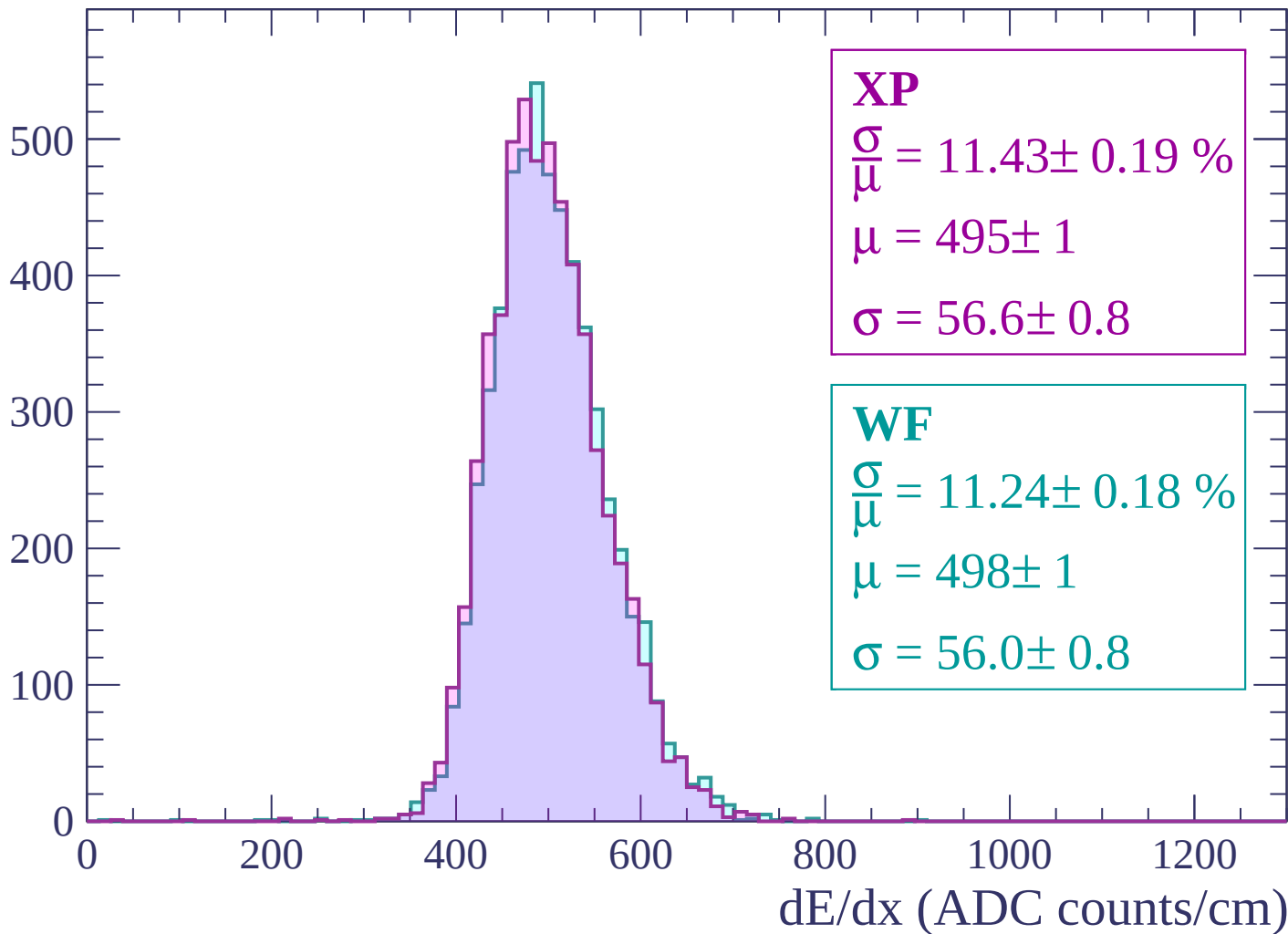
$$\mu = 499 \pm 1$$

$$\sigma = 55.4 \pm 0.8$$

dE/dx (ADC counts/cm)

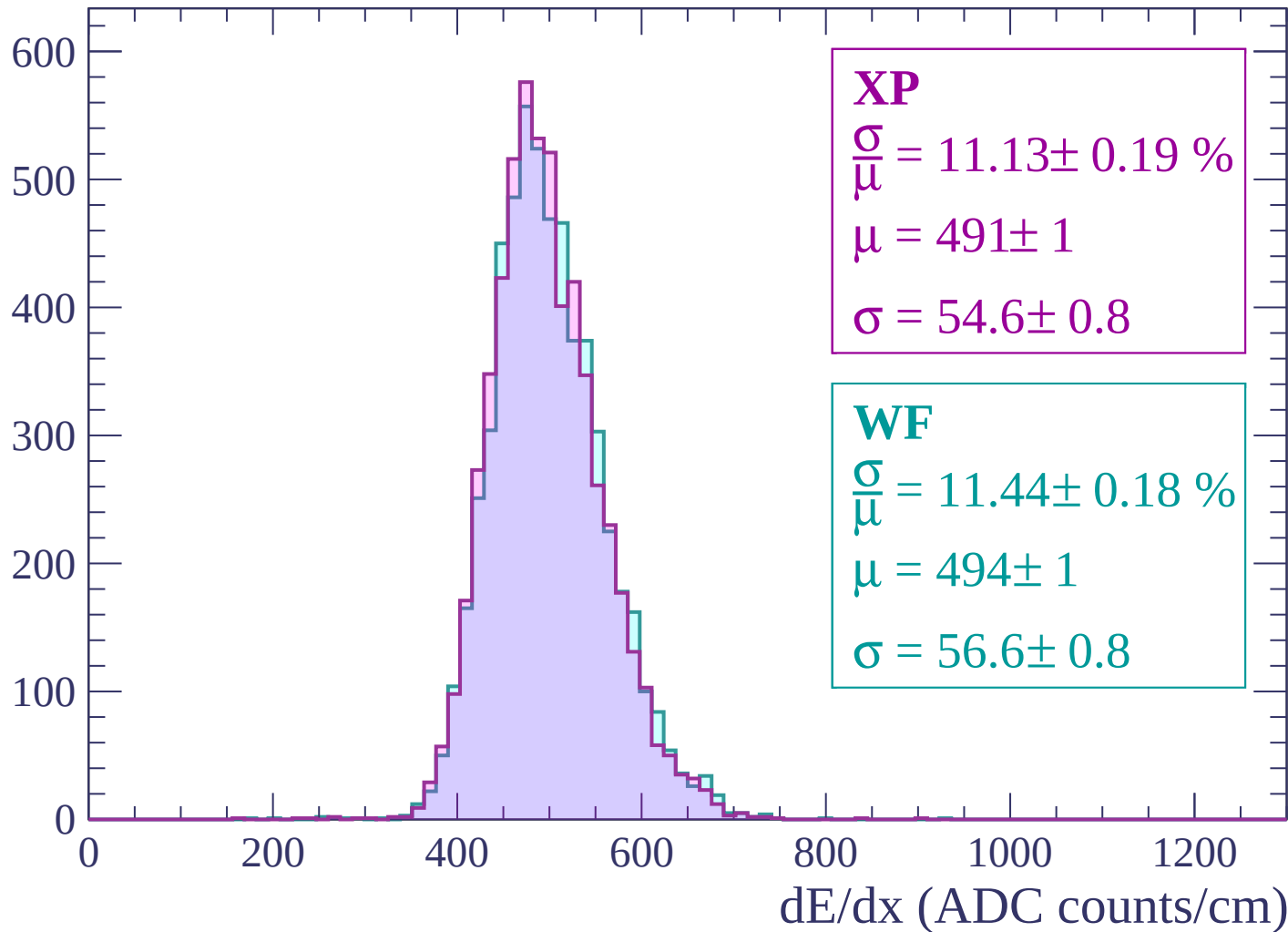
Energy loss | $-1800 < p < -1740$

Count



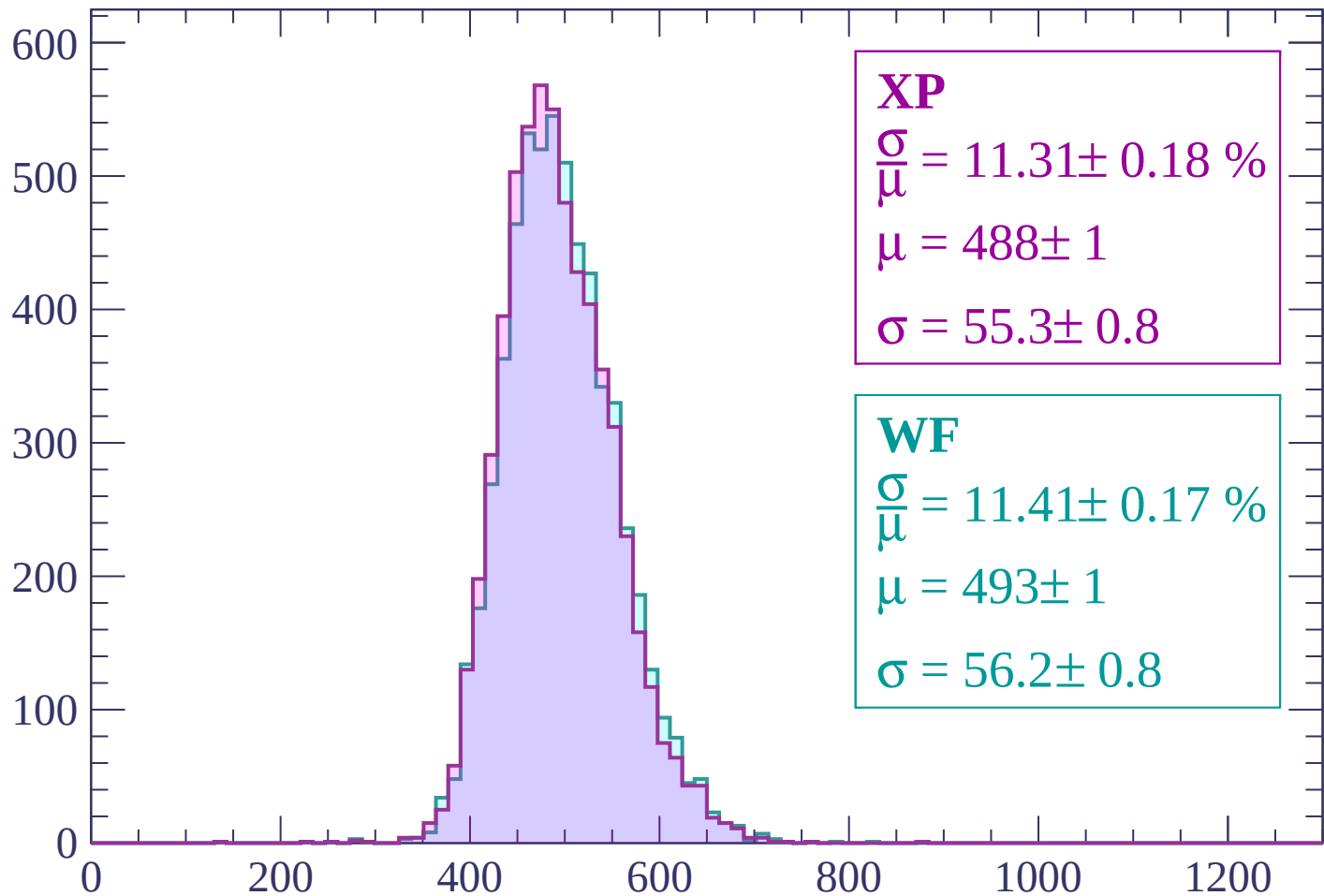
Energy loss | -1740 < p < -1680

Count



Energy loss | $-1680 < p < -1620$

Count



XP

$$\frac{\sigma}{\mu} = 11.31 \pm 0.18 \%$$

$$\mu = 488 \pm 1$$

$$\sigma = 55.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.41 \pm 0.17 \%$$

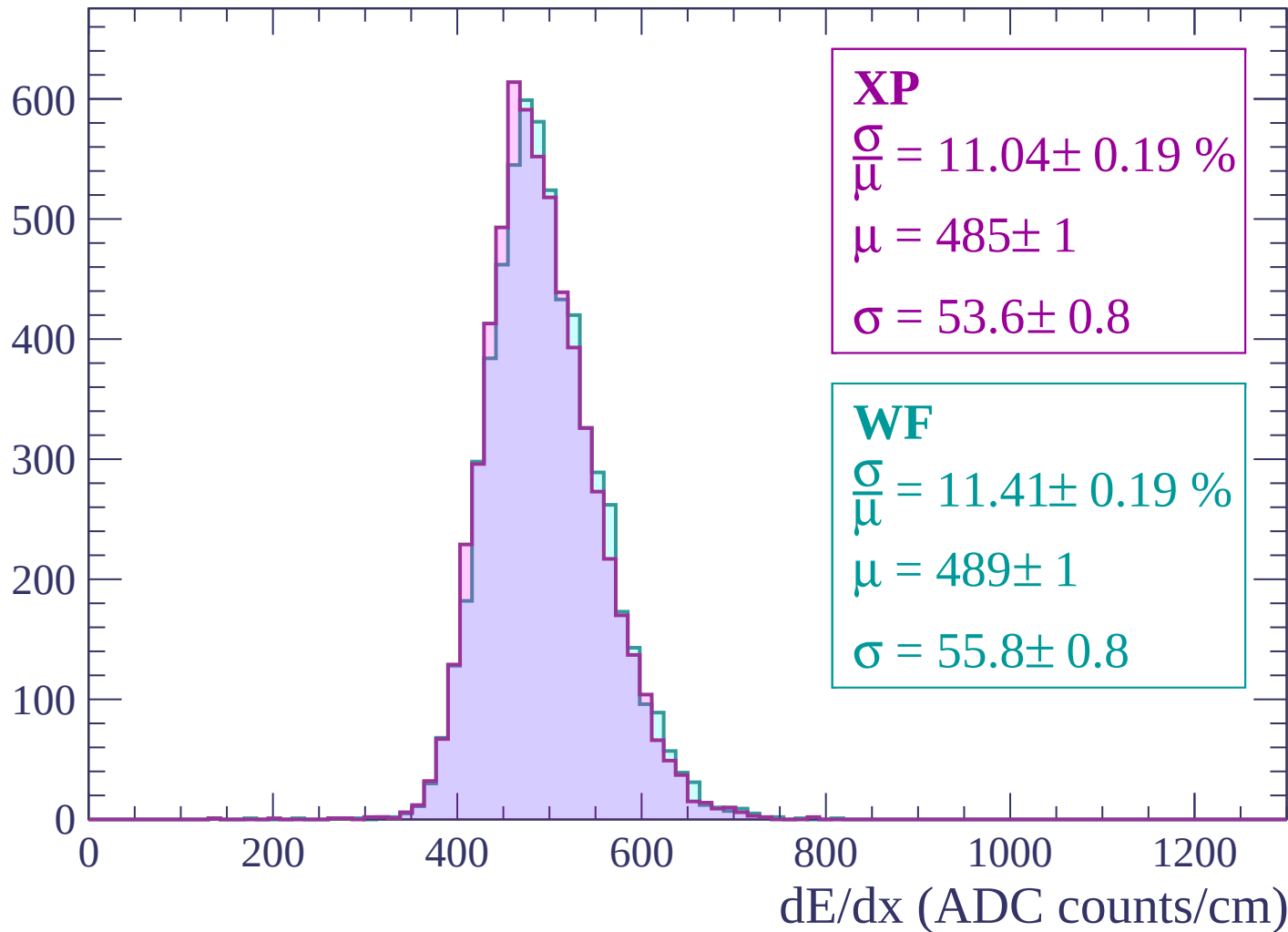
$$\mu = 493 \pm 1$$

$$\sigma = 56.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

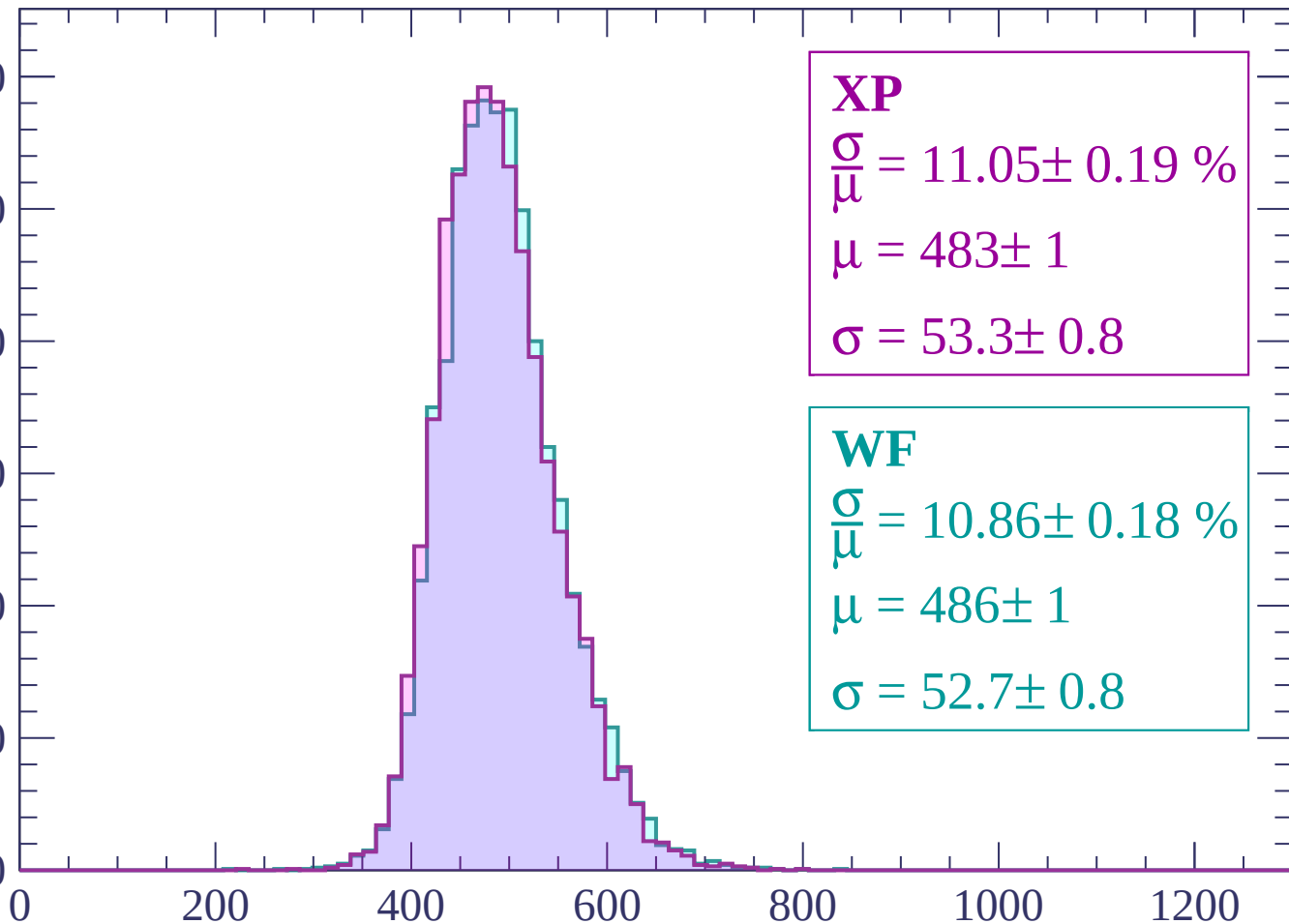
Count



Energy loss | -1560 < p < -1500

Count

600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 11.05 \pm 0.19 \%$$

$$\mu = 483 \pm 1$$

$$\sigma = 53.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.18 \%$$

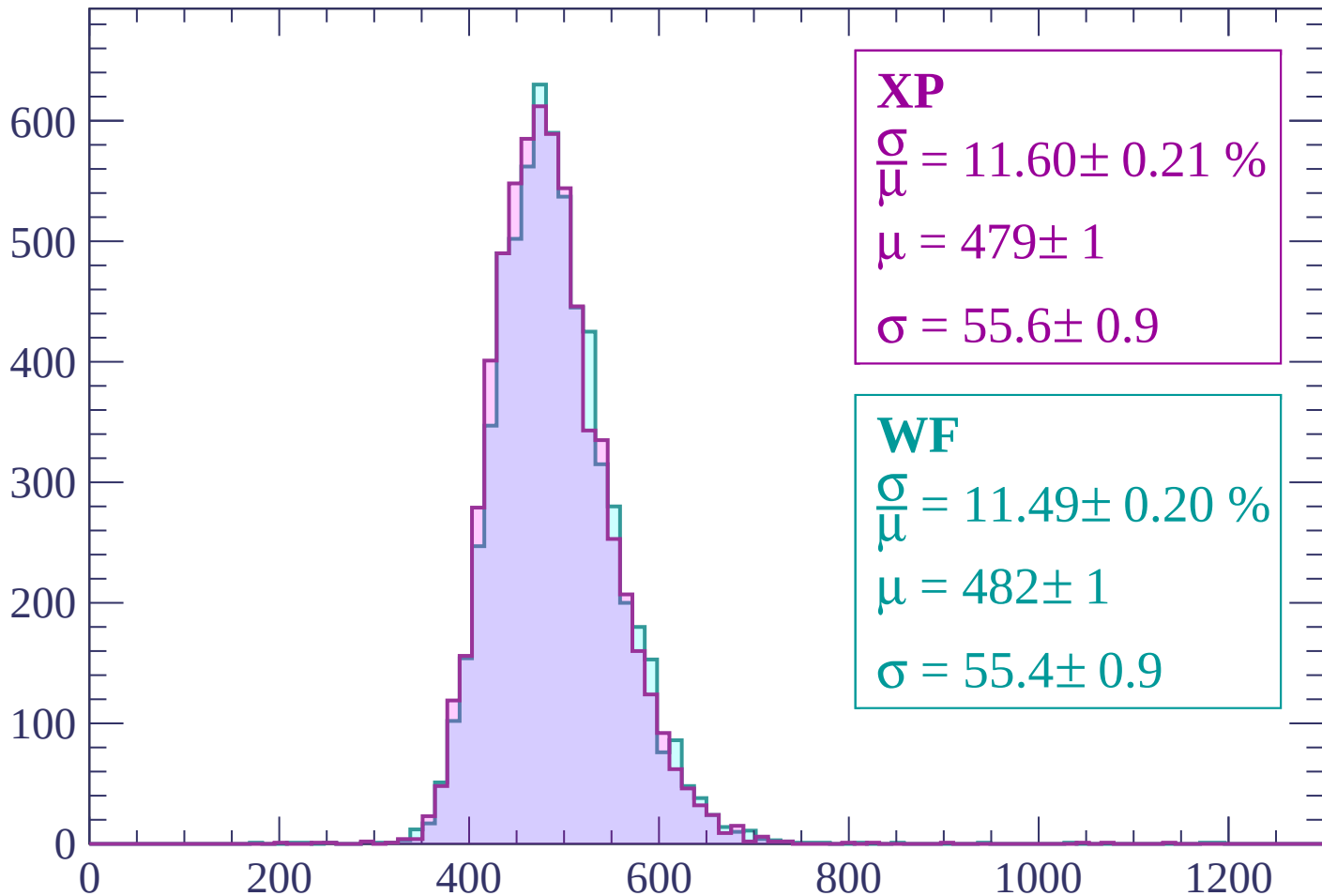
$$\mu = 486 \pm 1$$

$$\sigma = 52.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP

$$\frac{\sigma}{\mu} = 11.60 \pm 0.21 \%$$

$$\mu = 479 \pm 1$$

$$\sigma = 55.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.49 \pm 0.20 \%$$

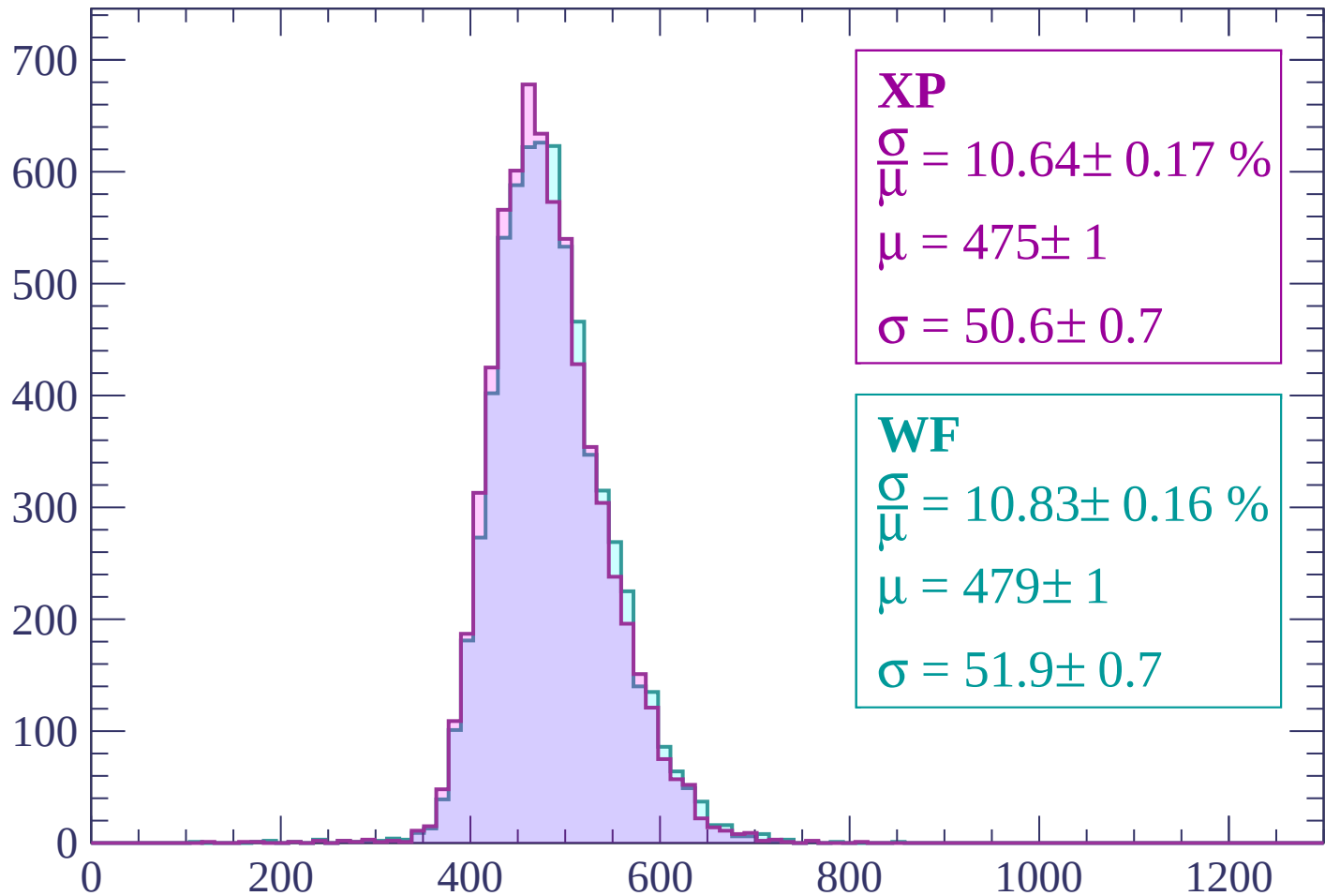
$$\mu = 482 \pm 1$$

$$\sigma = 55.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1440 < p < -1380

Count



XP

$$\frac{\sigma}{\mu} = 10.64 \pm 0.17 \%$$

$$\mu = 475 \pm 1$$

$$\sigma = 50.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.83 \pm 0.16 \%$$

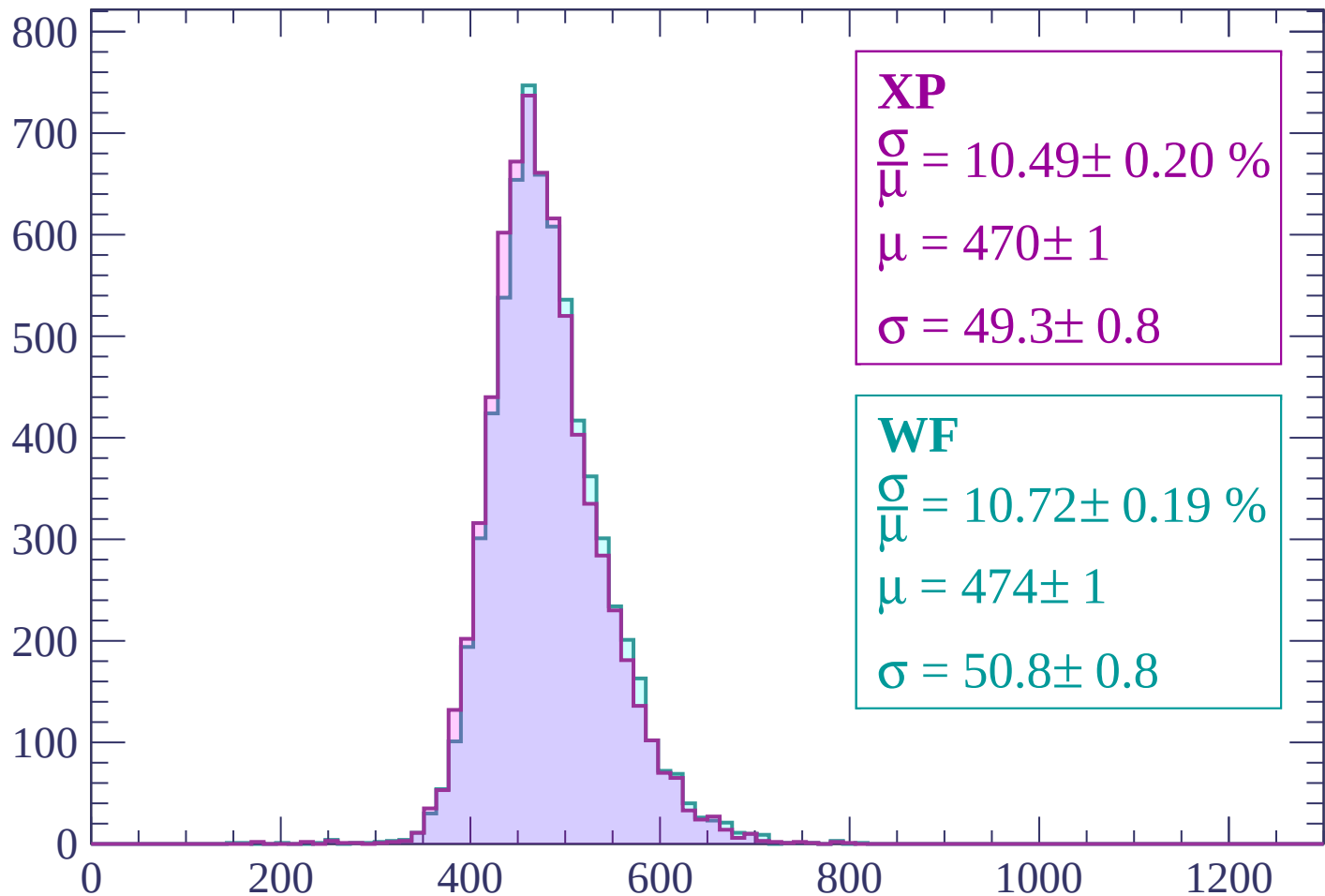
$$\mu = 479 \pm 1$$

$$\sigma = 51.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP

$$\frac{\sigma}{\mu} = 10.49 \pm 0.20 \%$$

$$\mu = 470 \pm 1$$

$$\sigma = 49.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.72 \pm 0.19 \%$$

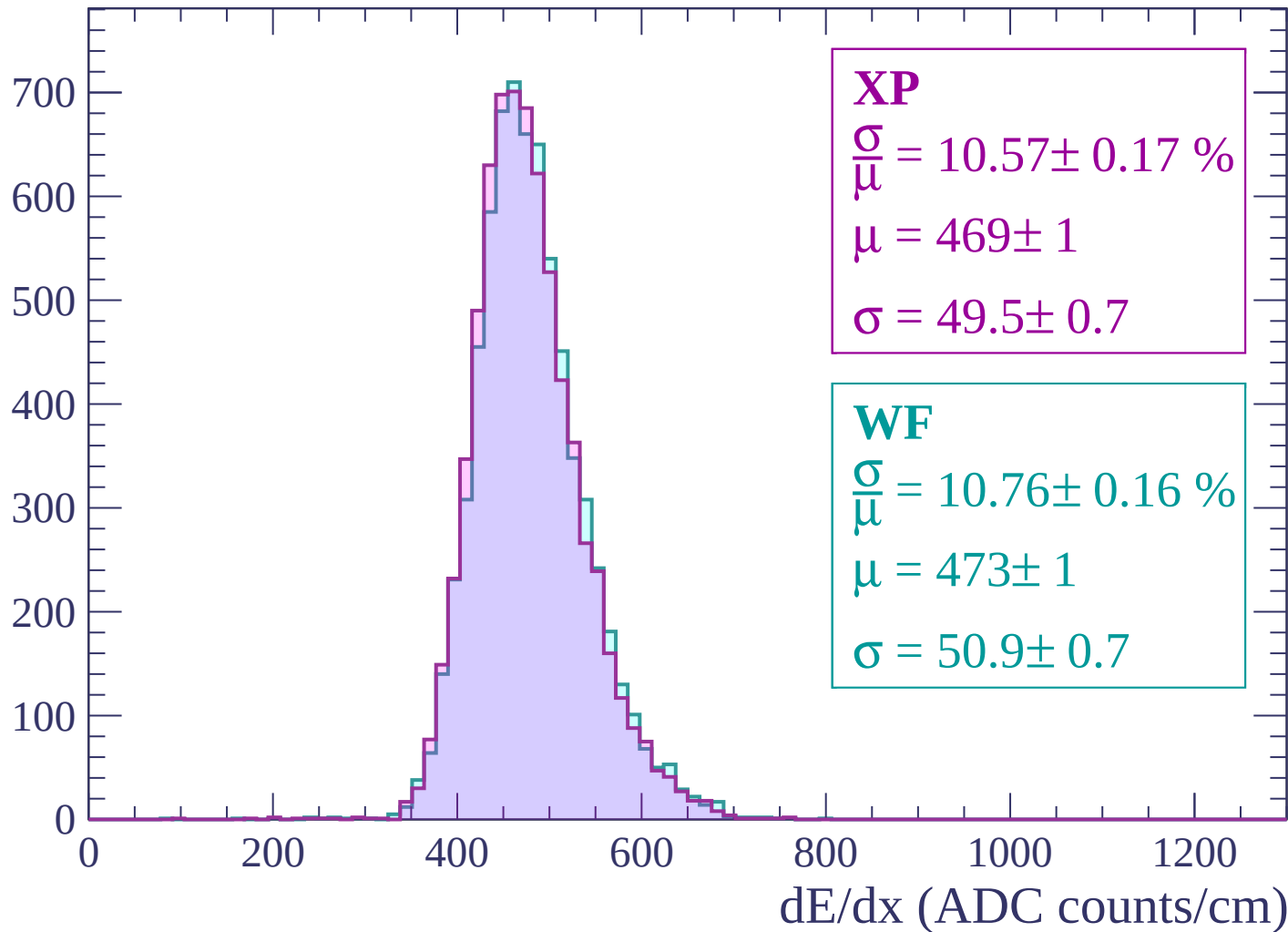
$$\mu = 474 \pm 1$$

$$\sigma = 50.8 \pm 0.8$$

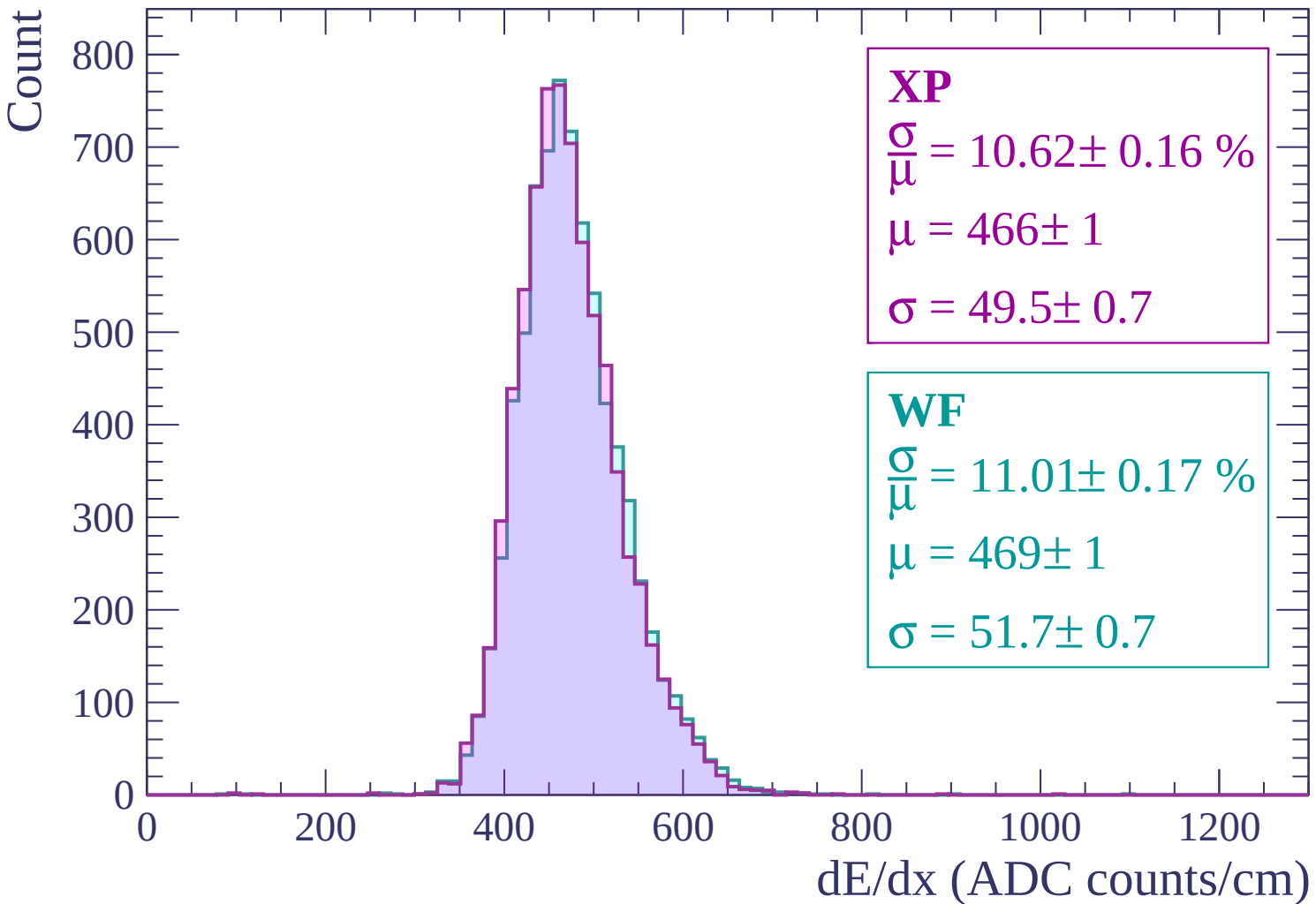
dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count



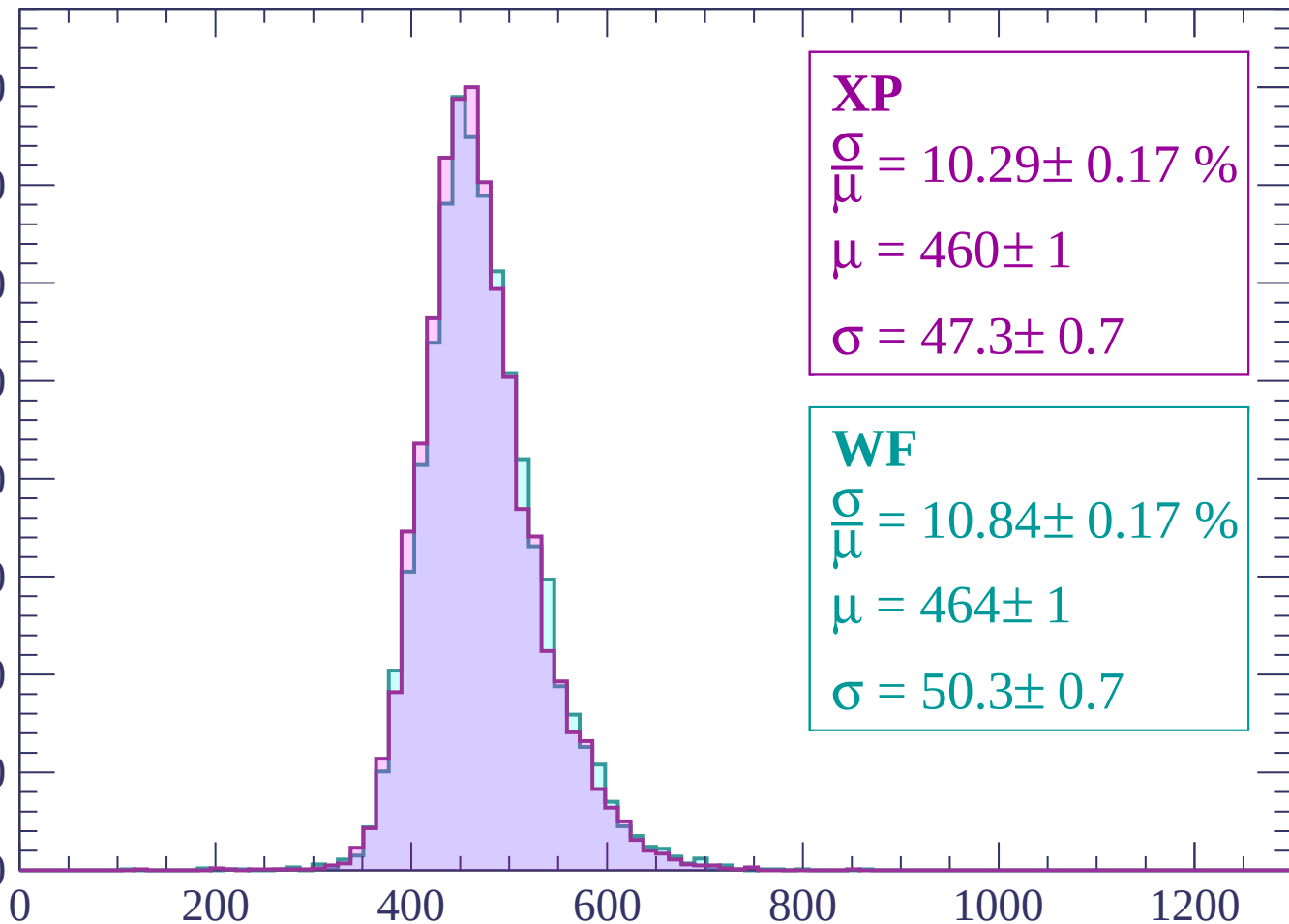
Energy loss | -1260 < p < -1200



Energy loss | -1200 < p < -1140

Count

800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 10.29 \pm 0.17 \%$$

$$\mu = 460 \pm 1$$

$$\sigma = 47.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.84 \pm 0.17 \%$$

$$\mu = 464 \pm 1$$

$$\sigma = 50.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1140 < p < -1080$

Count

800
700
600
500
400
300
200
100
0

XP

$$\frac{\sigma}{\mu} = 10.23 \pm 0.16 \%$$

$$\mu = 458 \pm 1$$

$$\sigma = 46.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.65 \pm 0.16 \%$$

$$\mu = 461 \pm 1$$

$$\sigma = 49.1 \pm 0.7$$

0

200

400

600

800

1000

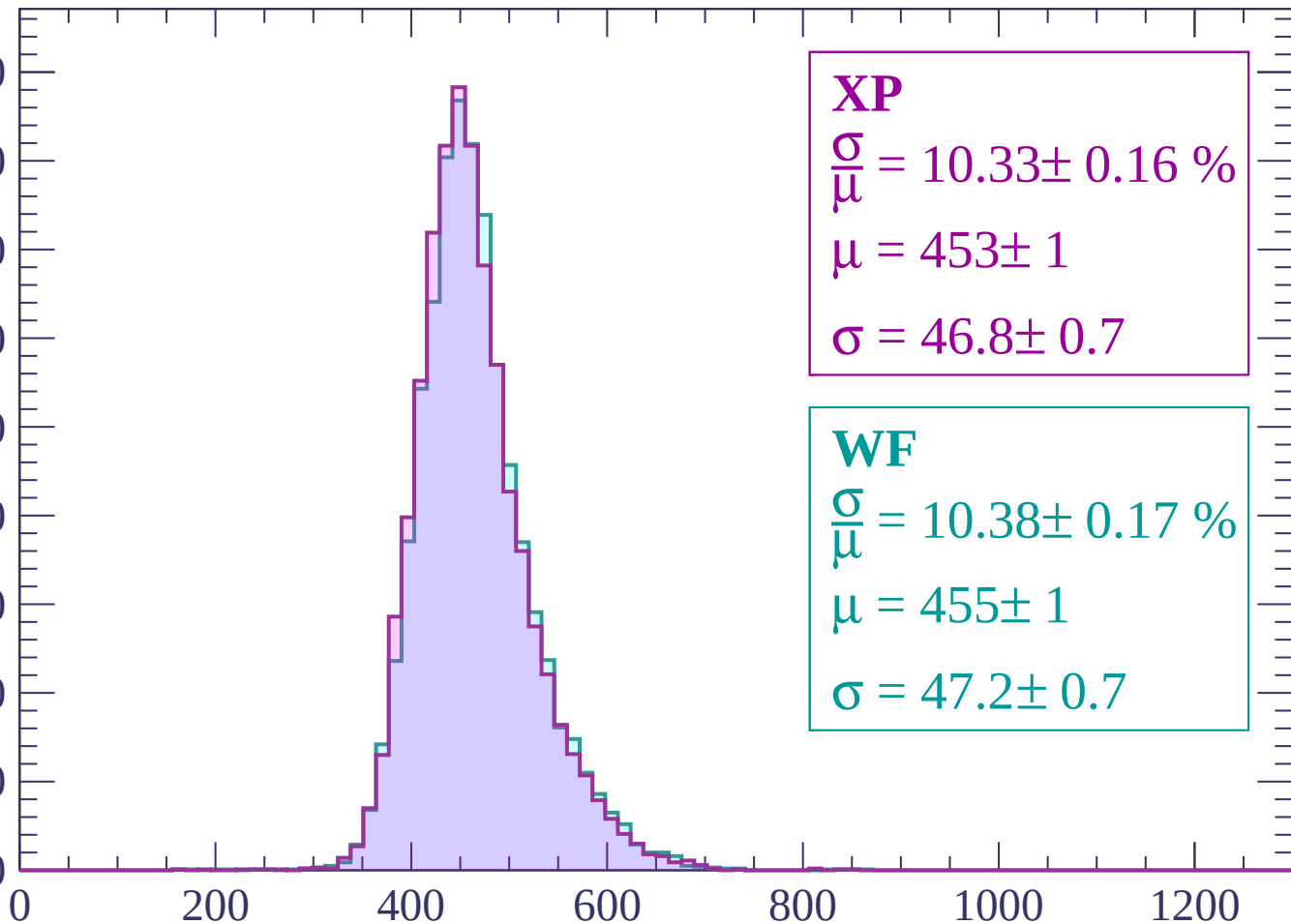
1200

dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1020

Count

900
800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.16 \%$$

$$\mu = 453 \pm 1$$

$$\sigma = 46.8 \pm 0.7$$

WF

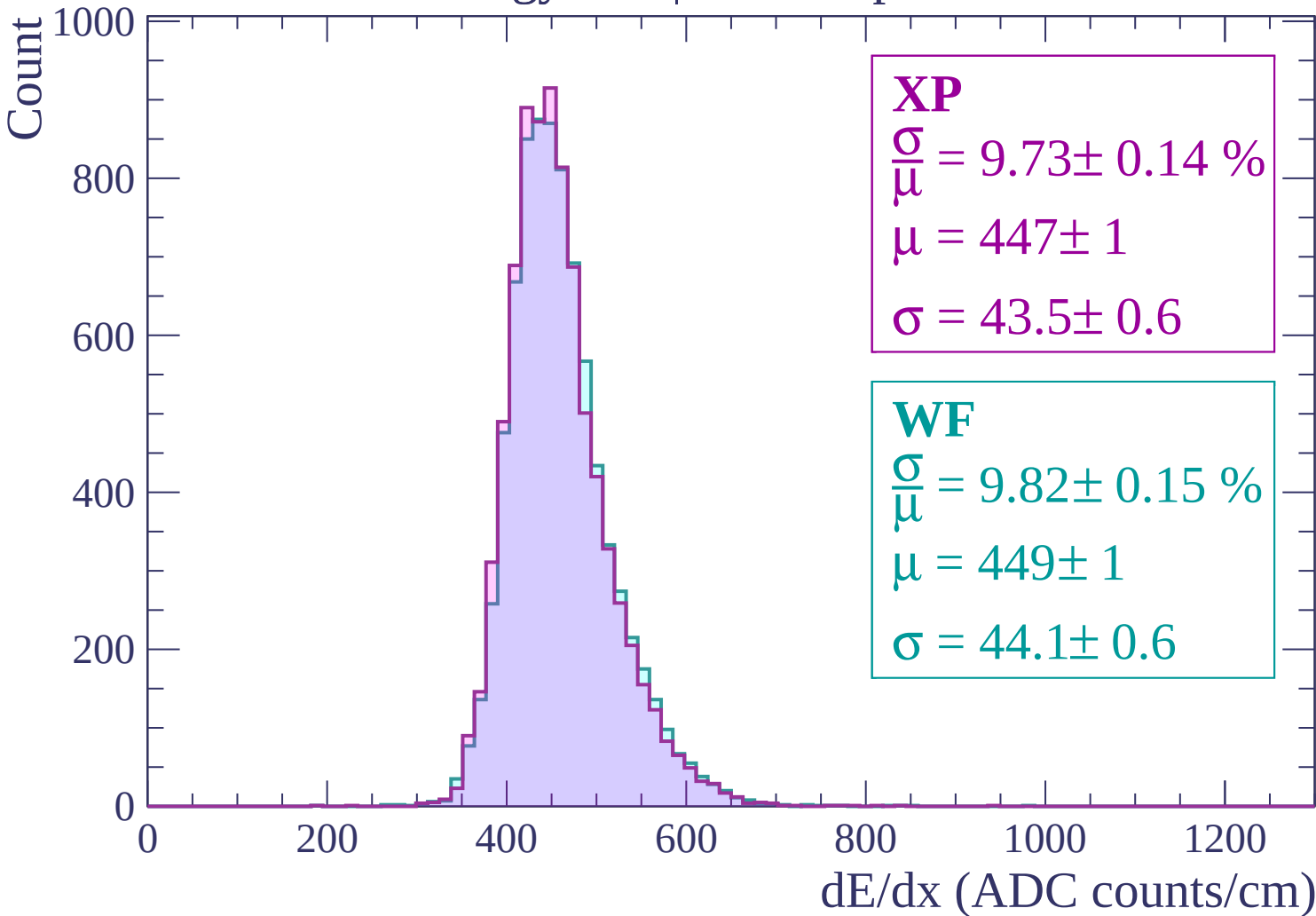
$$\frac{\sigma}{\mu} = 10.38 \pm 0.17 \%$$

$$\mu = 455 \pm 1$$

$$\sigma = 47.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$



Energy loss | $-960 < p < -900$

Count

1000

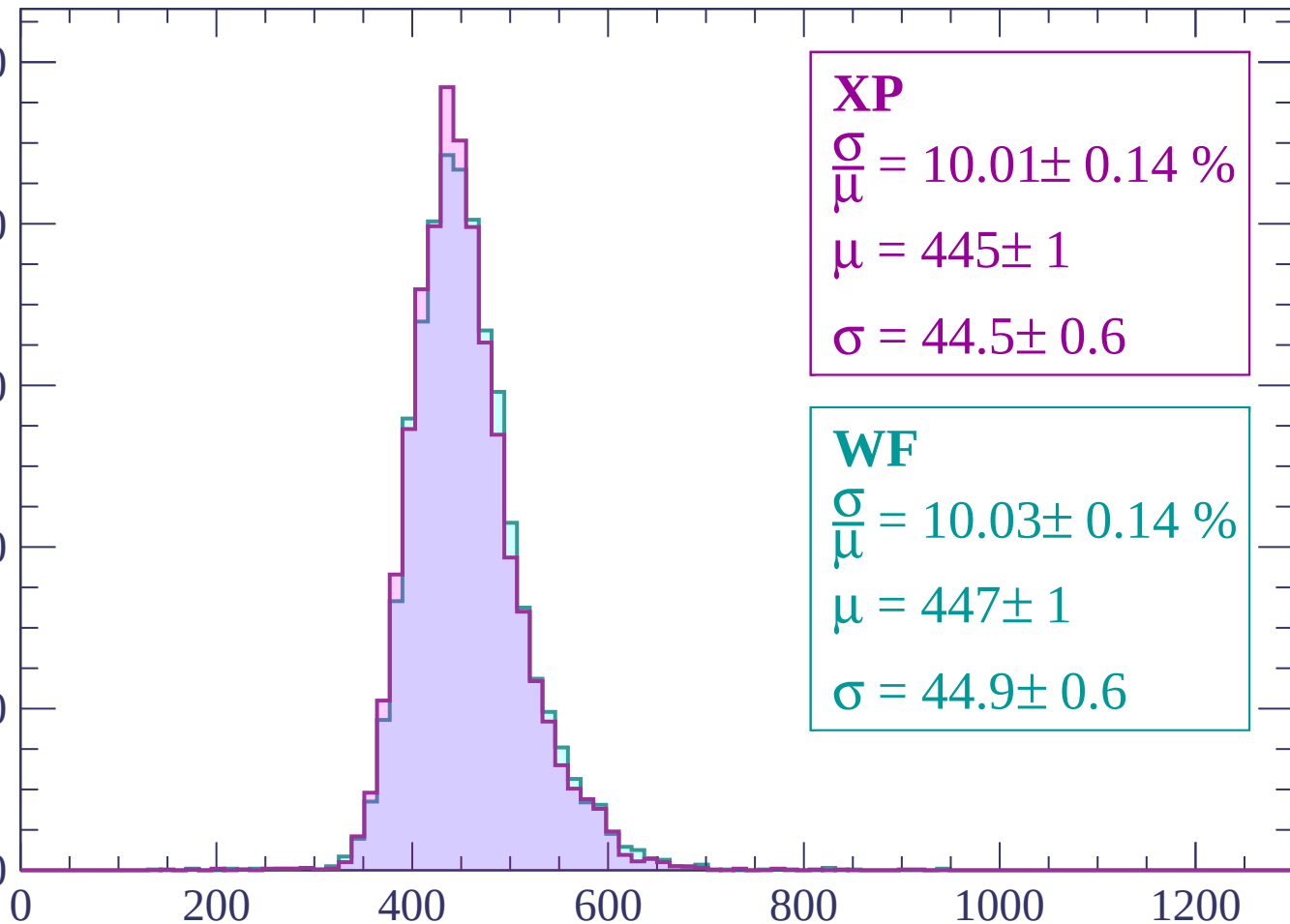
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 10.01 \pm 0.14 \%$$

$$\mu = 445 \pm 1$$

$$\sigma = 44.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.03 \pm 0.14 \%$$

$$\mu = 447 \pm 1$$

$$\sigma = 44.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count

1000

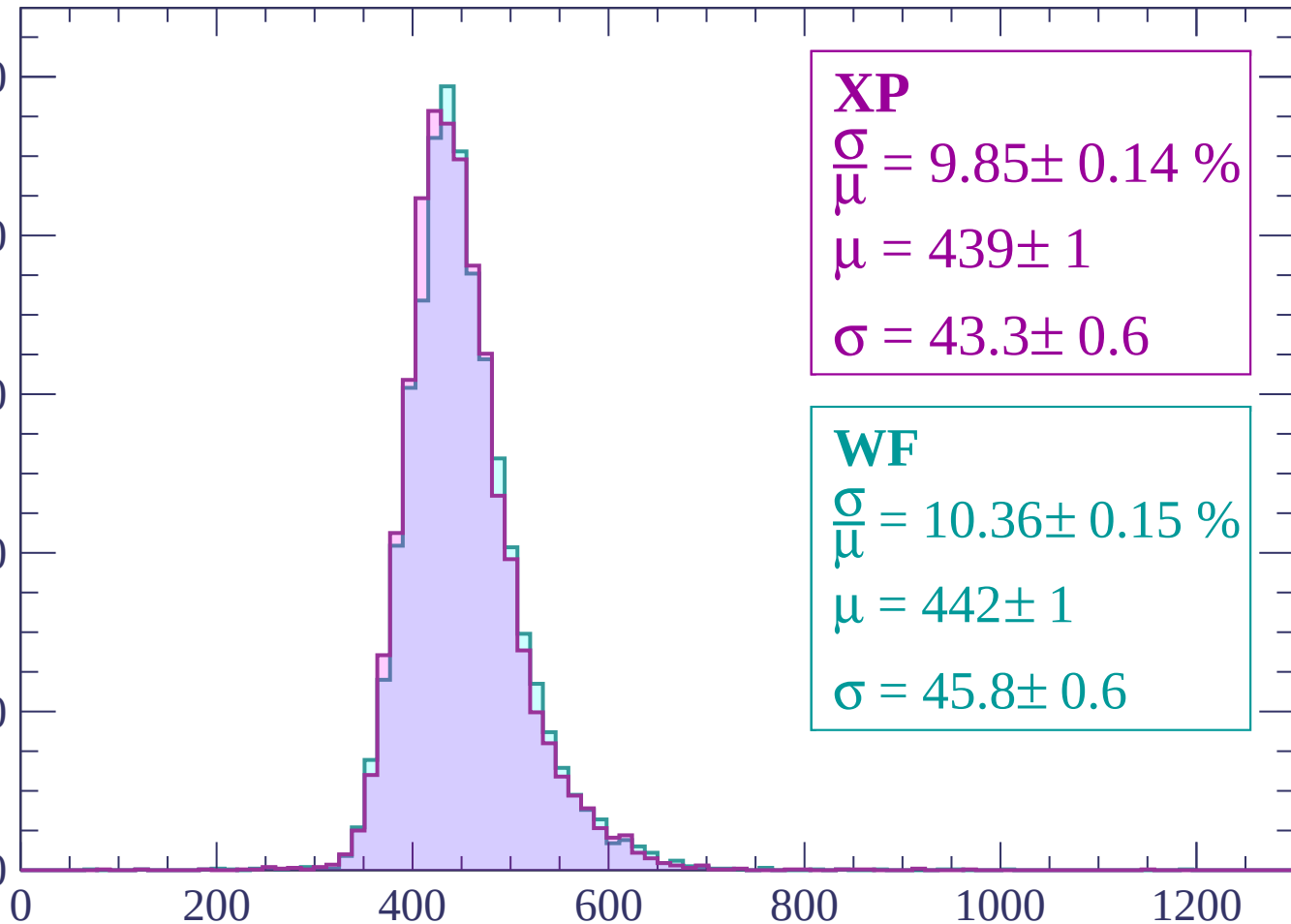
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.85 \pm 0.14 \%$$

$$\mu = 439 \pm 1$$

$$\sigma = 43.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.36 \pm 0.15 \%$$

$$\mu = 442 \pm 1$$

$$\sigma = 45.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1000

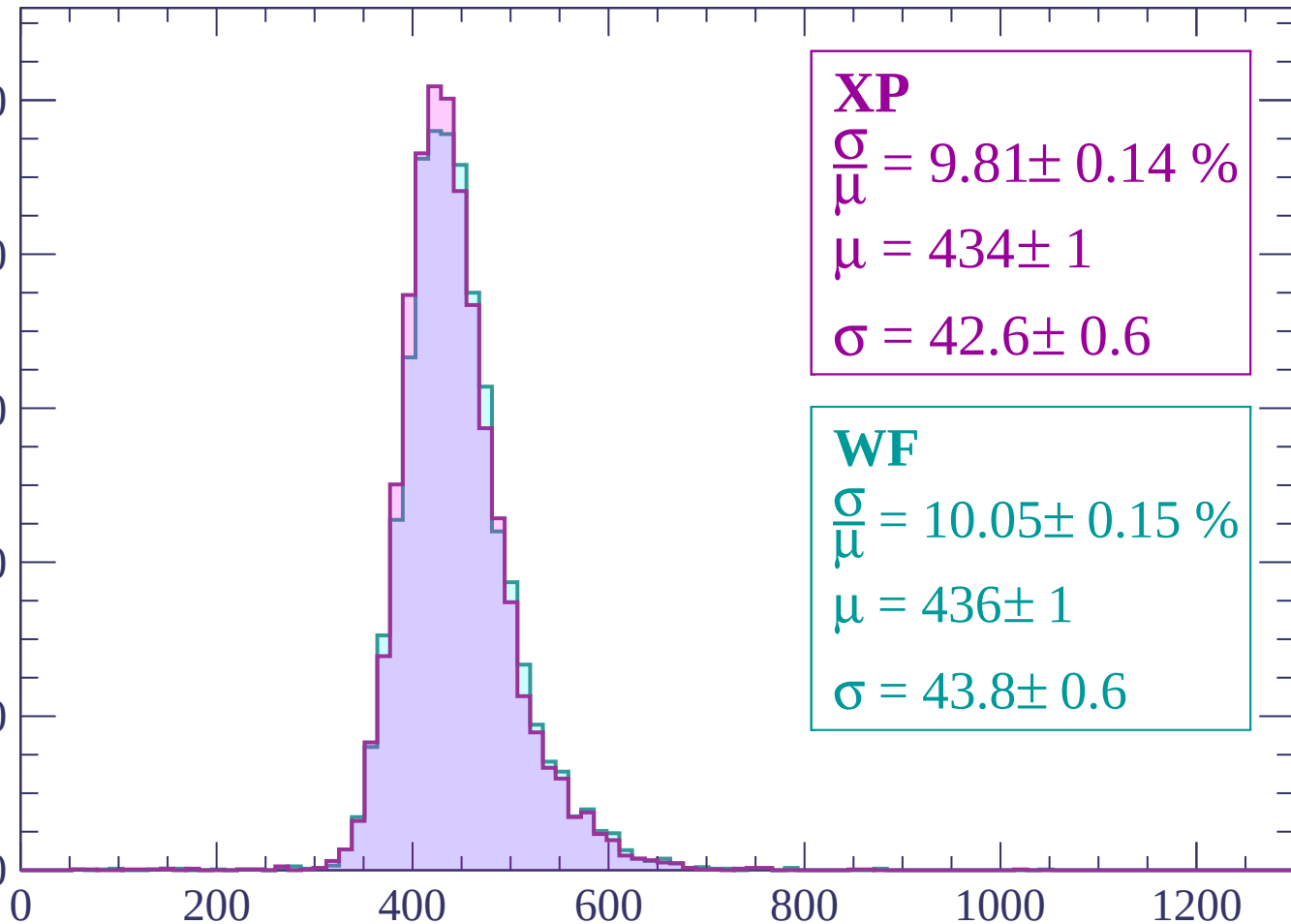
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.81 \pm 0.14 \%$$

$$\mu = 434 \pm 1$$

$$\sigma = 42.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.05 \pm 0.15 \%$$

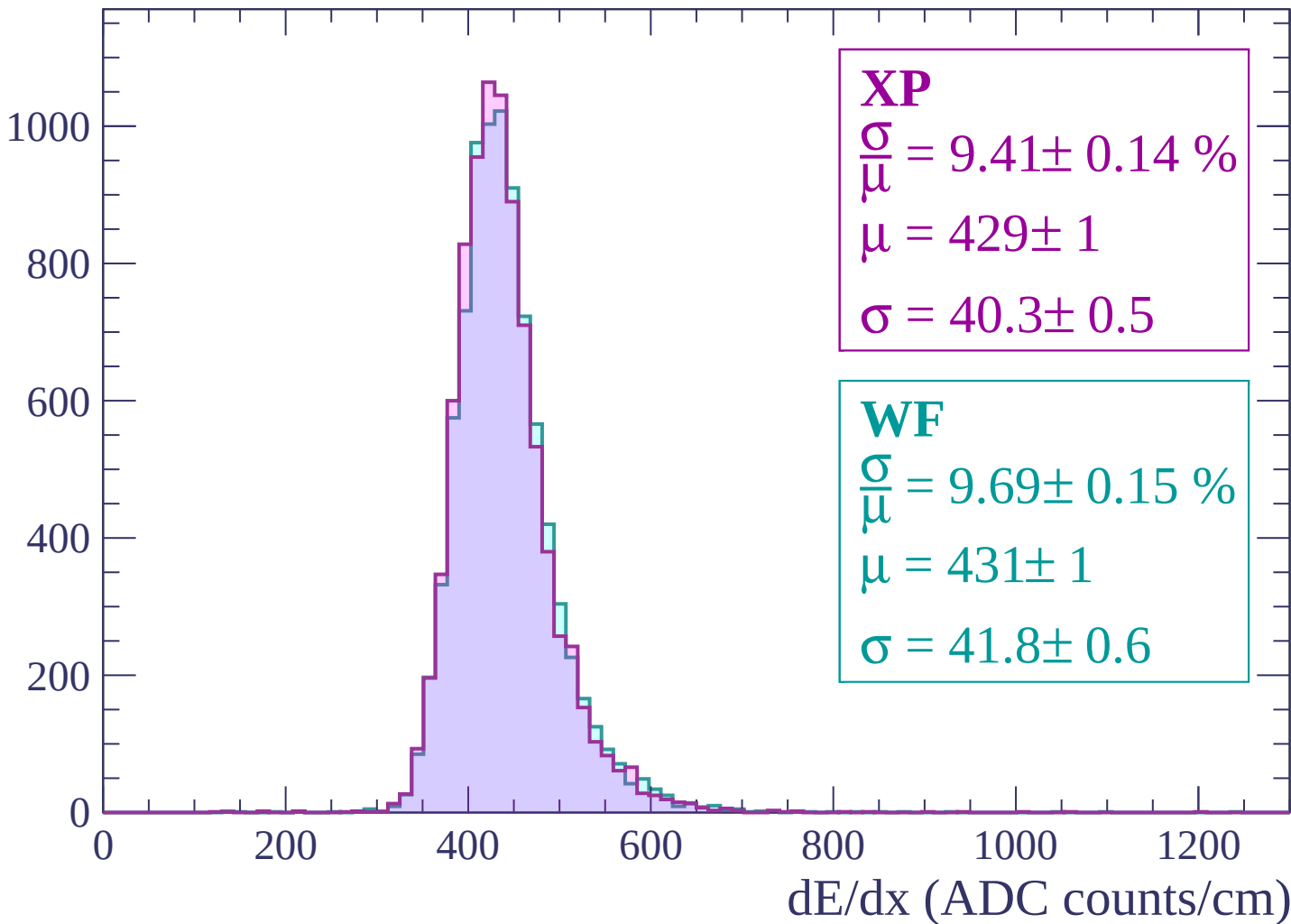
$$\mu = 436 \pm 1$$

$$\sigma = 43.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



Energy loss | $-720 < p < -660$

Count

1000

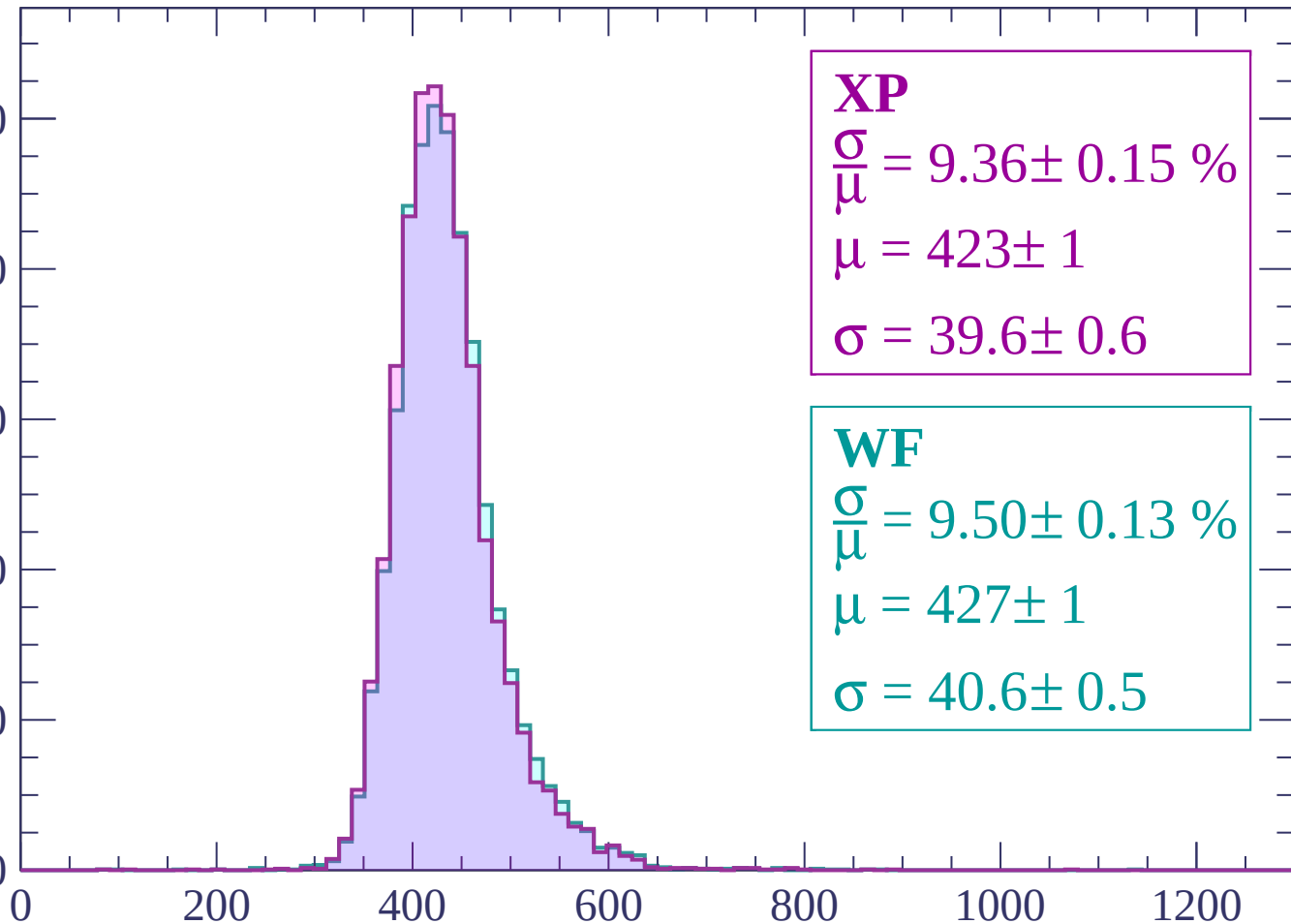
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.15 \%$$

$$\mu = 423 \pm 1$$

$$\sigma = 39.6 \pm 0.6$$

WF

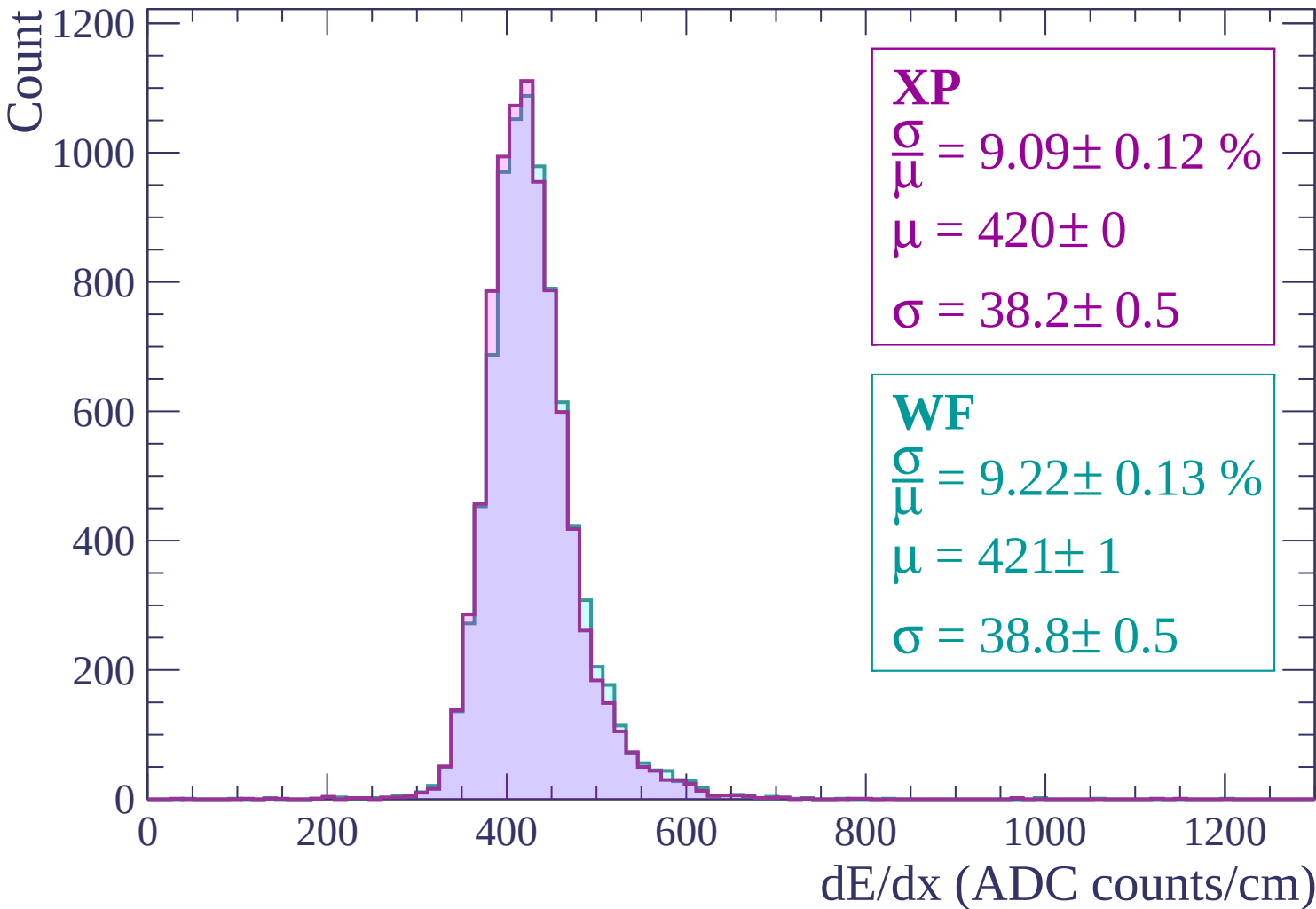
$$\frac{\sigma}{\mu} = 9.50 \pm 0.13 \%$$

$$\mu = 427 \pm 1$$

$$\sigma = 40.6 \pm 0.5$$

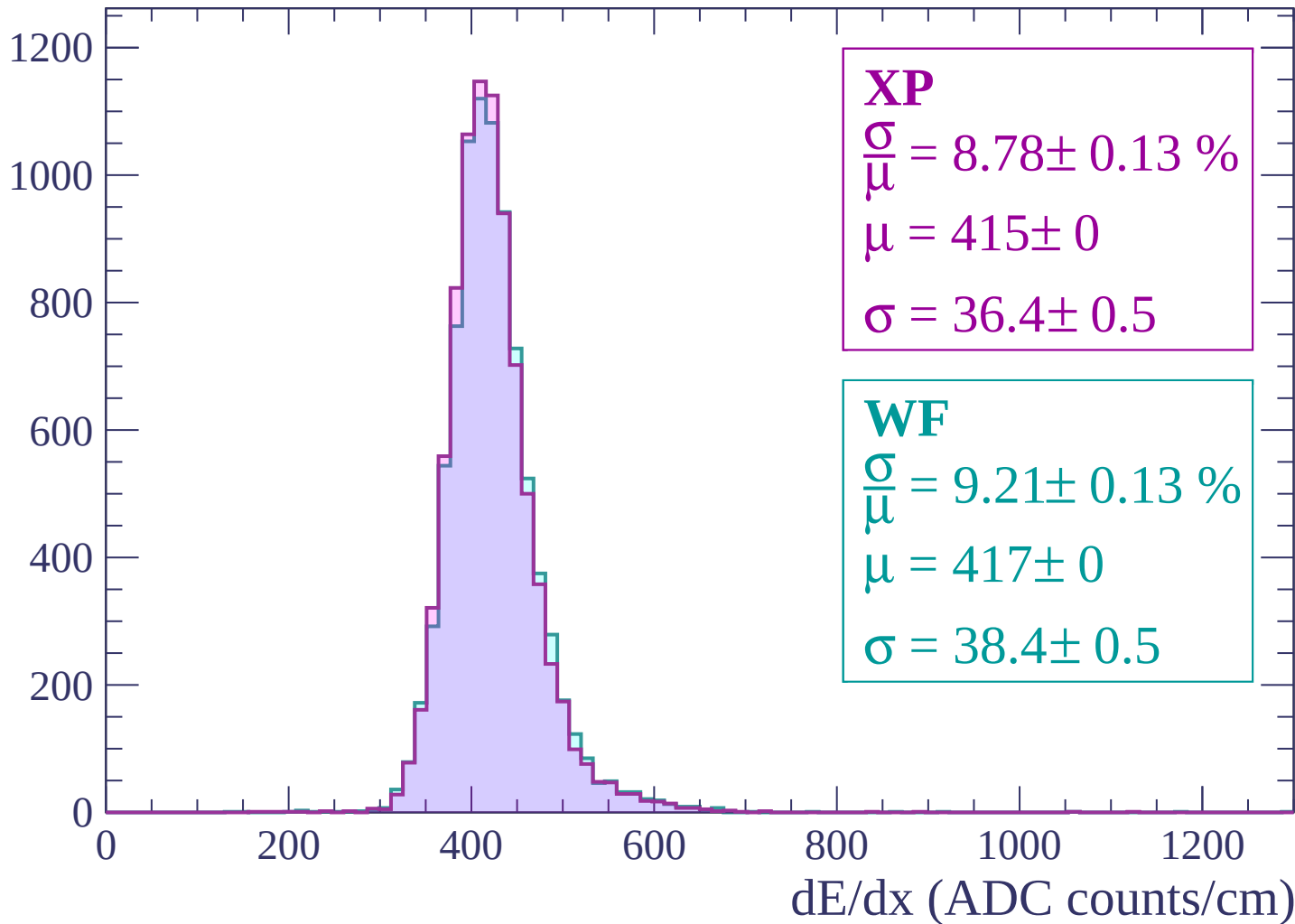
dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count



Energy loss | $-540 < p < -480$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.12 \%$$

$$\mu = 410 \pm 0$$

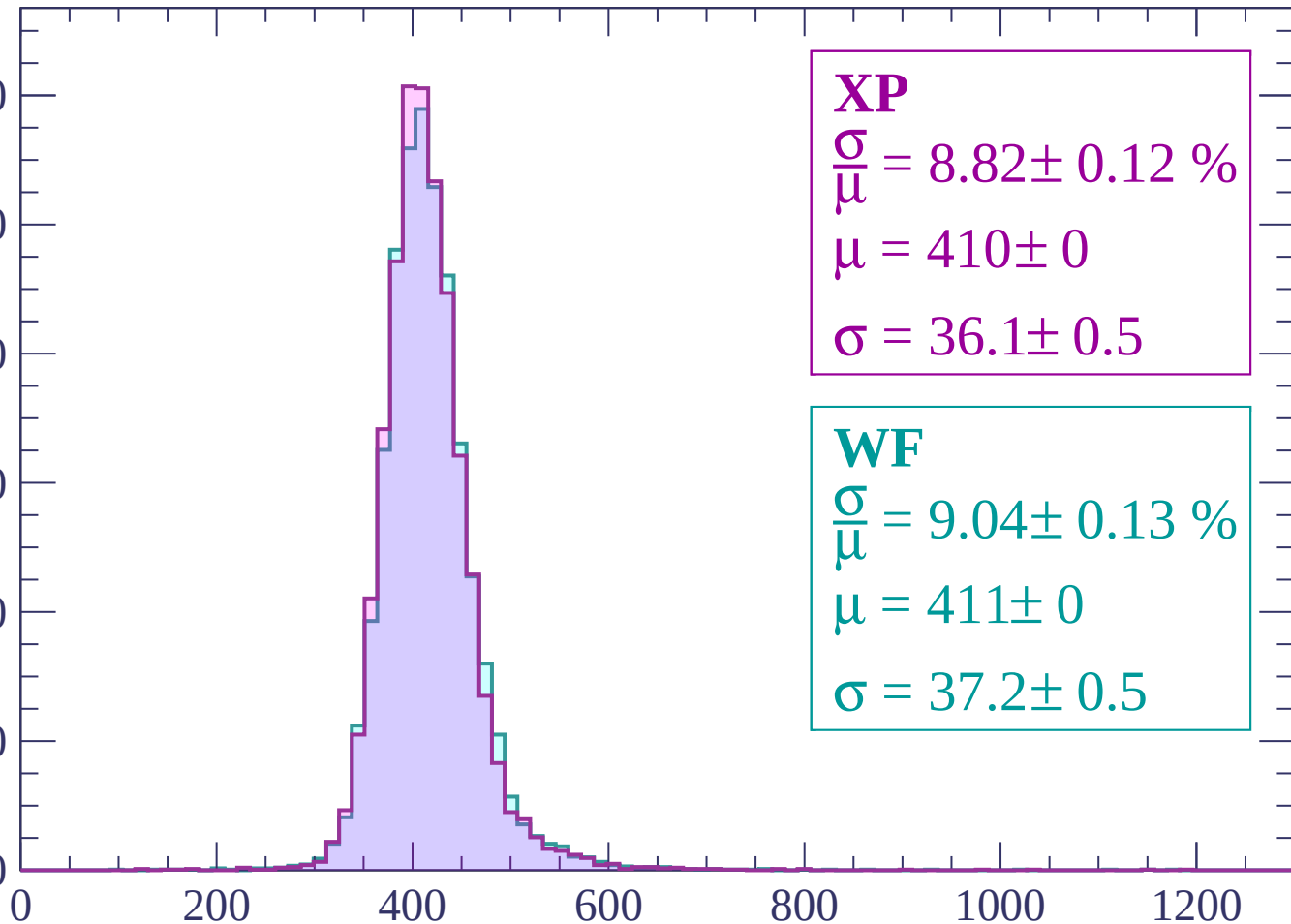
$$\sigma = 36.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.04 \pm 0.13 \%$$

$$\mu = 411 \pm 0$$

$$\sigma = 37.2 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.13 \%$$

$$\mu = 406 \pm 0$$

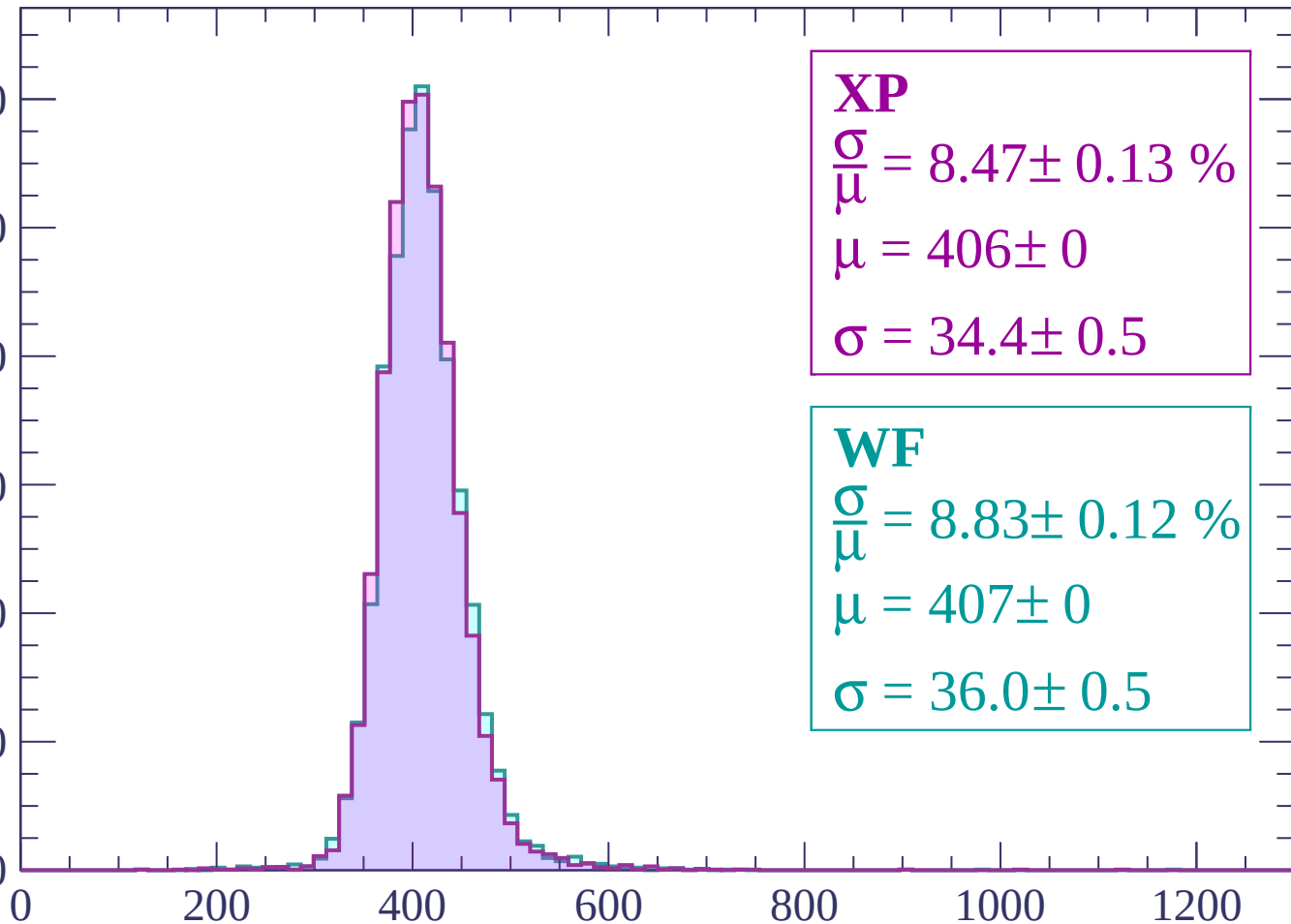
$$\sigma = 34.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.83 \pm 0.12 \%$$

$$\mu = 407 \pm 0$$

$$\sigma = 36.0 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 8.63 \pm 0.13 \%$$

$$\mu = 402 \pm 0$$

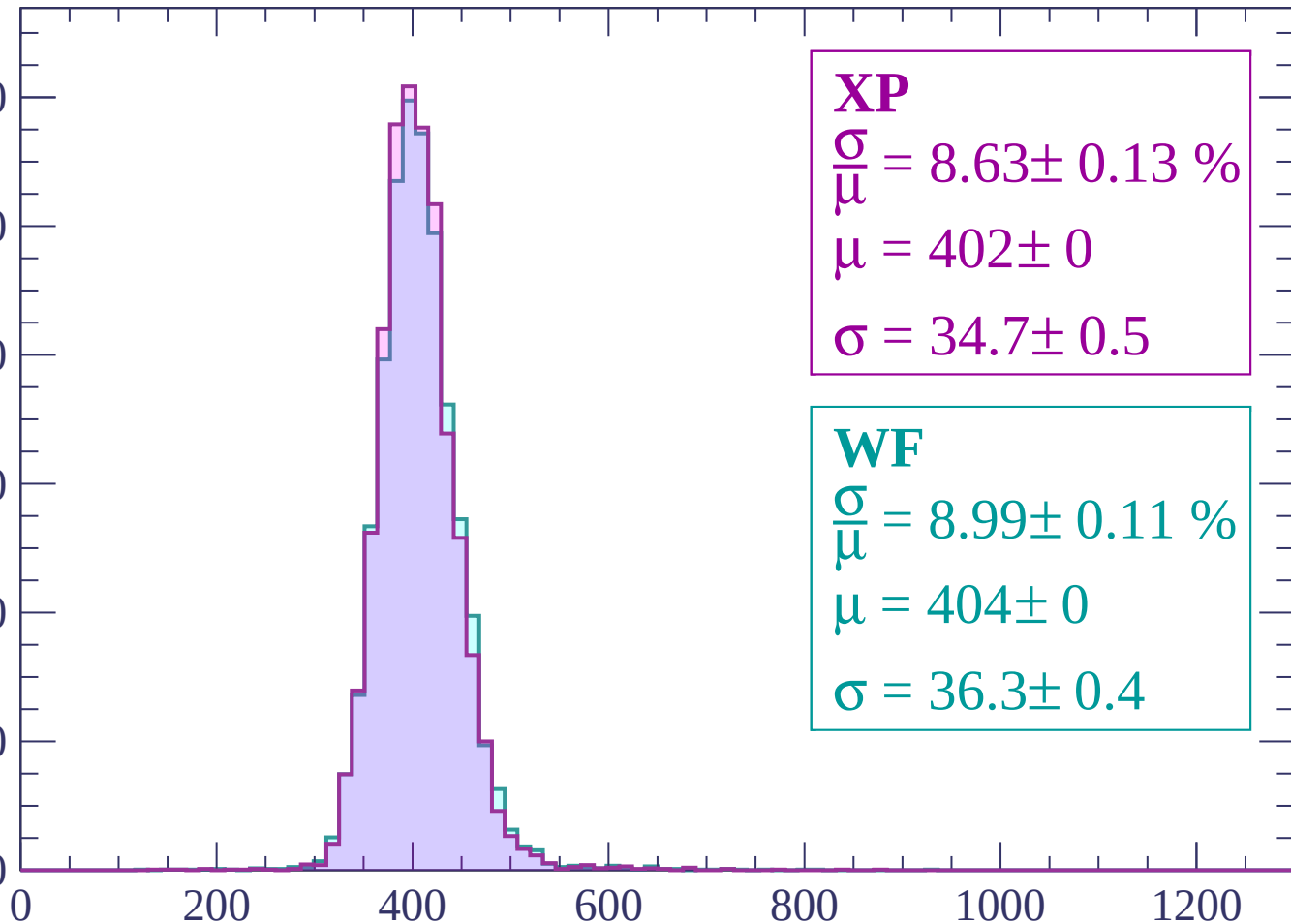
$$\sigma = 34.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.11 \%$$

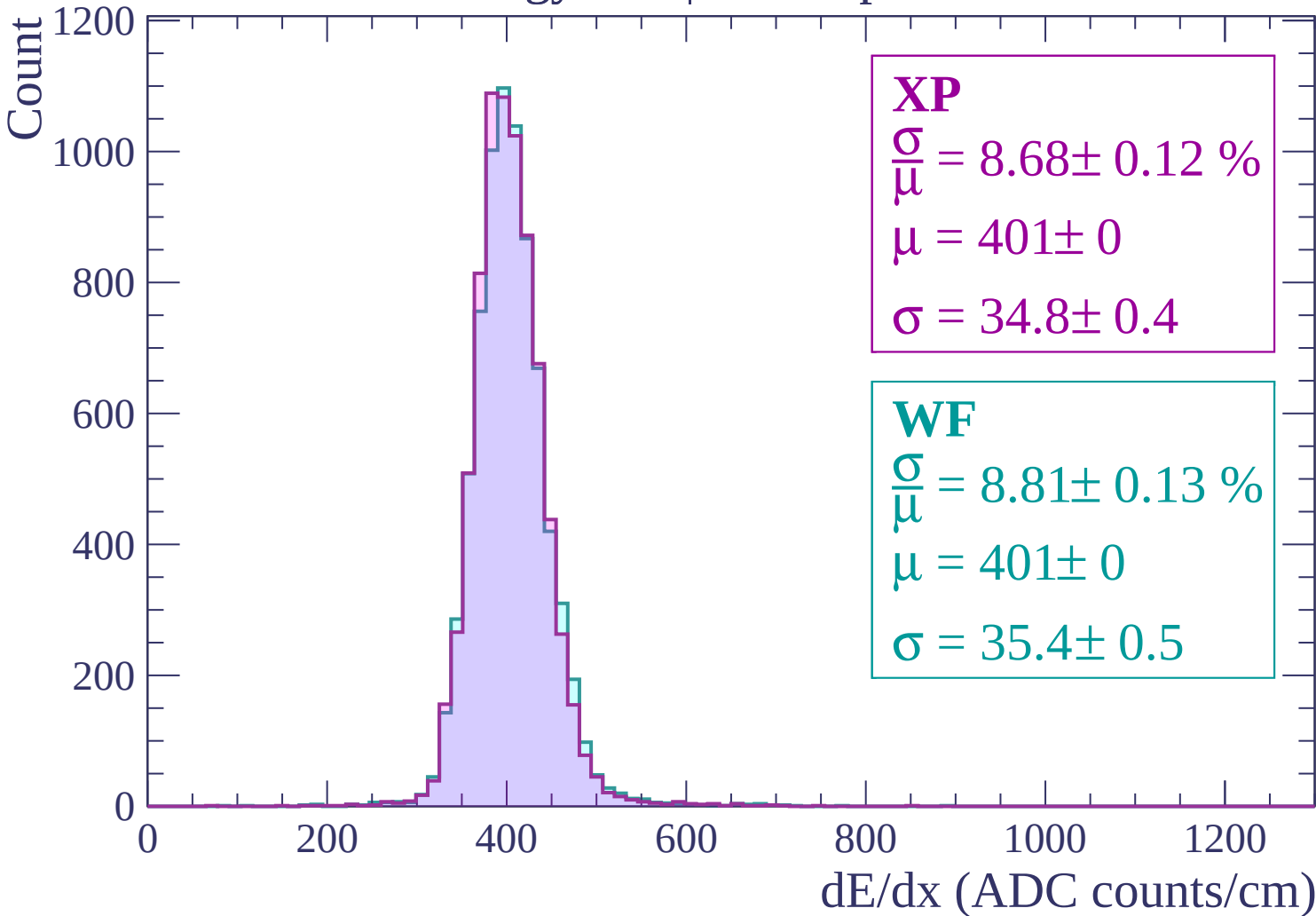
$$\mu = 404 \pm 0$$

$$\sigma = 36.3 \pm 0.4$$



dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$



Energy loss | $-300 < p < -240$

Count

1000

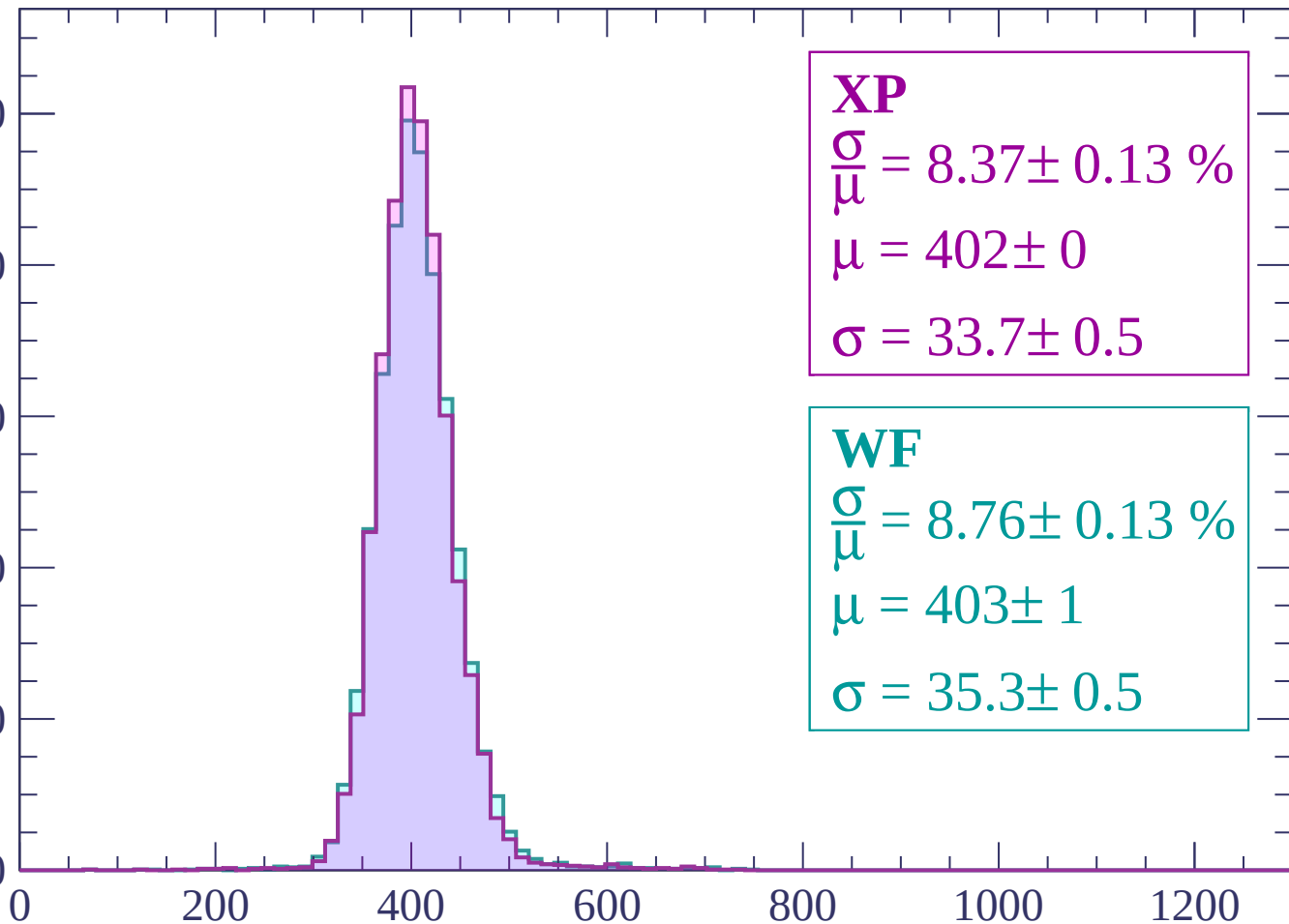
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.13 \%$$

$$\mu = 402 \pm 0$$

$$\sigma = 33.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.76 \pm 0.13 \%$$

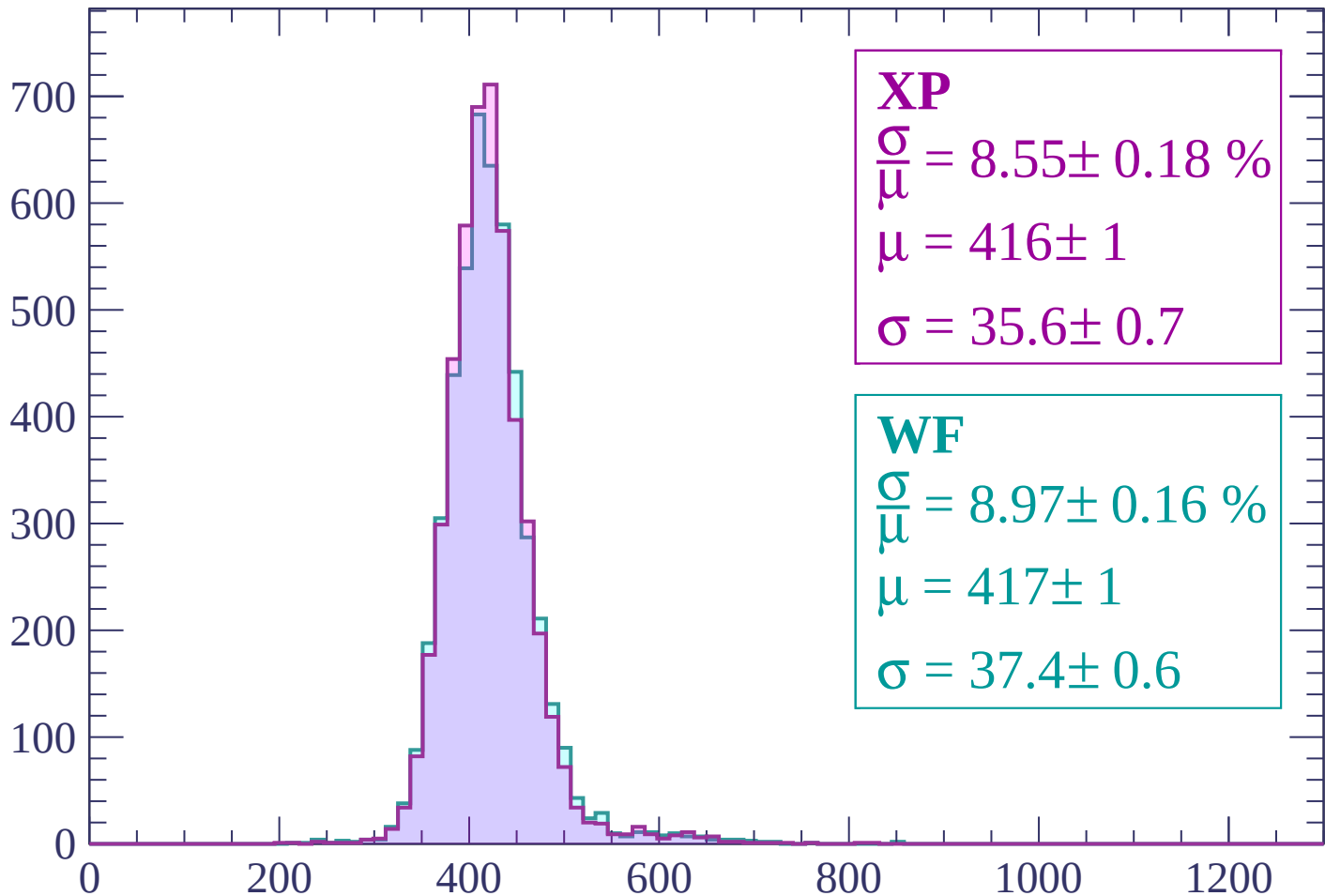
$$\mu = 403 \pm 1$$

$$\sigma = 35.3 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.18 \%$$

$$\mu = 416 \pm 1$$

$$\sigma = 35.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 0.16 \%$$

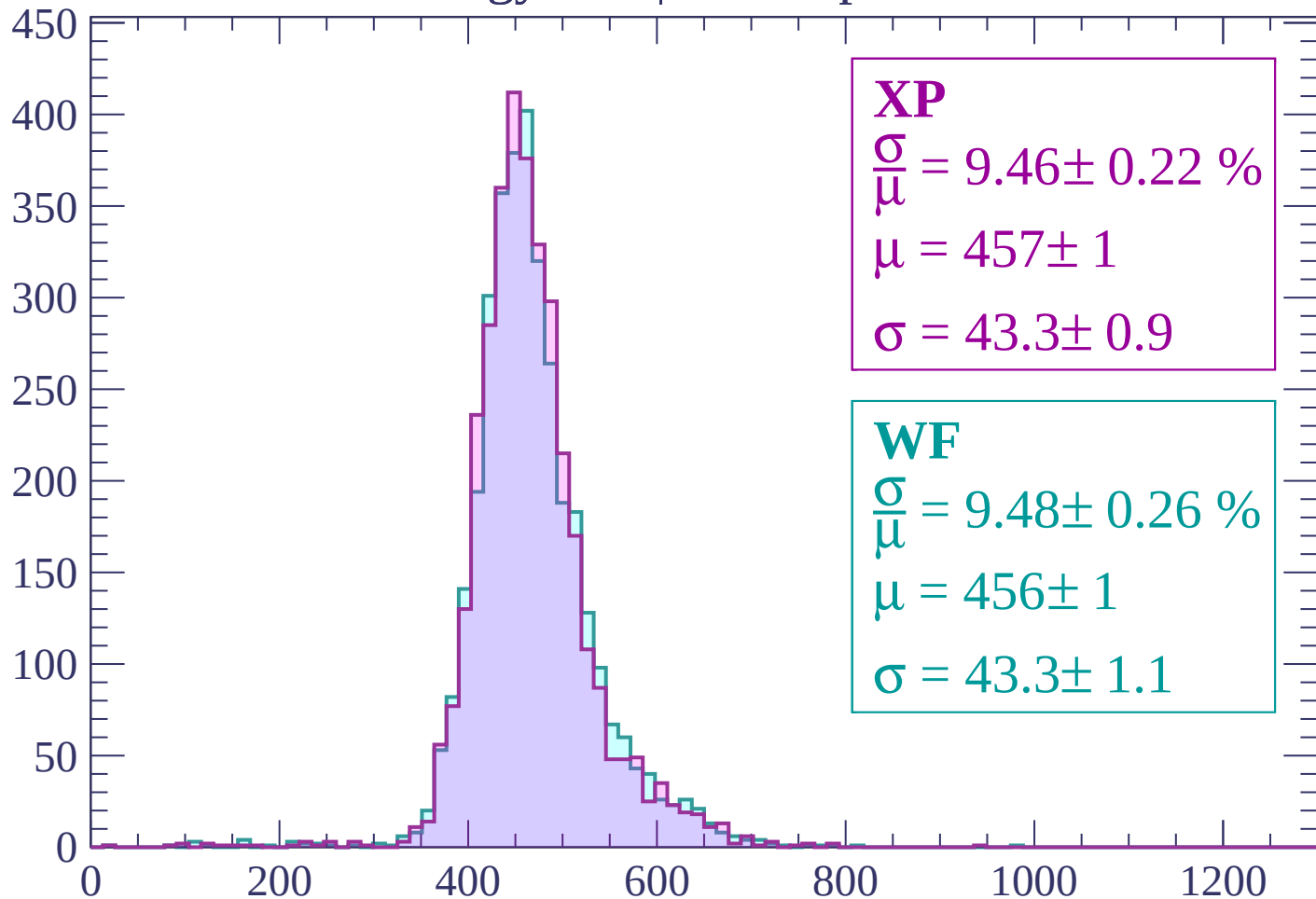
$$\mu = 417 \pm 1$$

$$\sigma = 37.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -180 < p < -120

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.22 \%$$

$$\mu = 457 \pm 1$$

$$\sigma = 43.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.26 \%$$

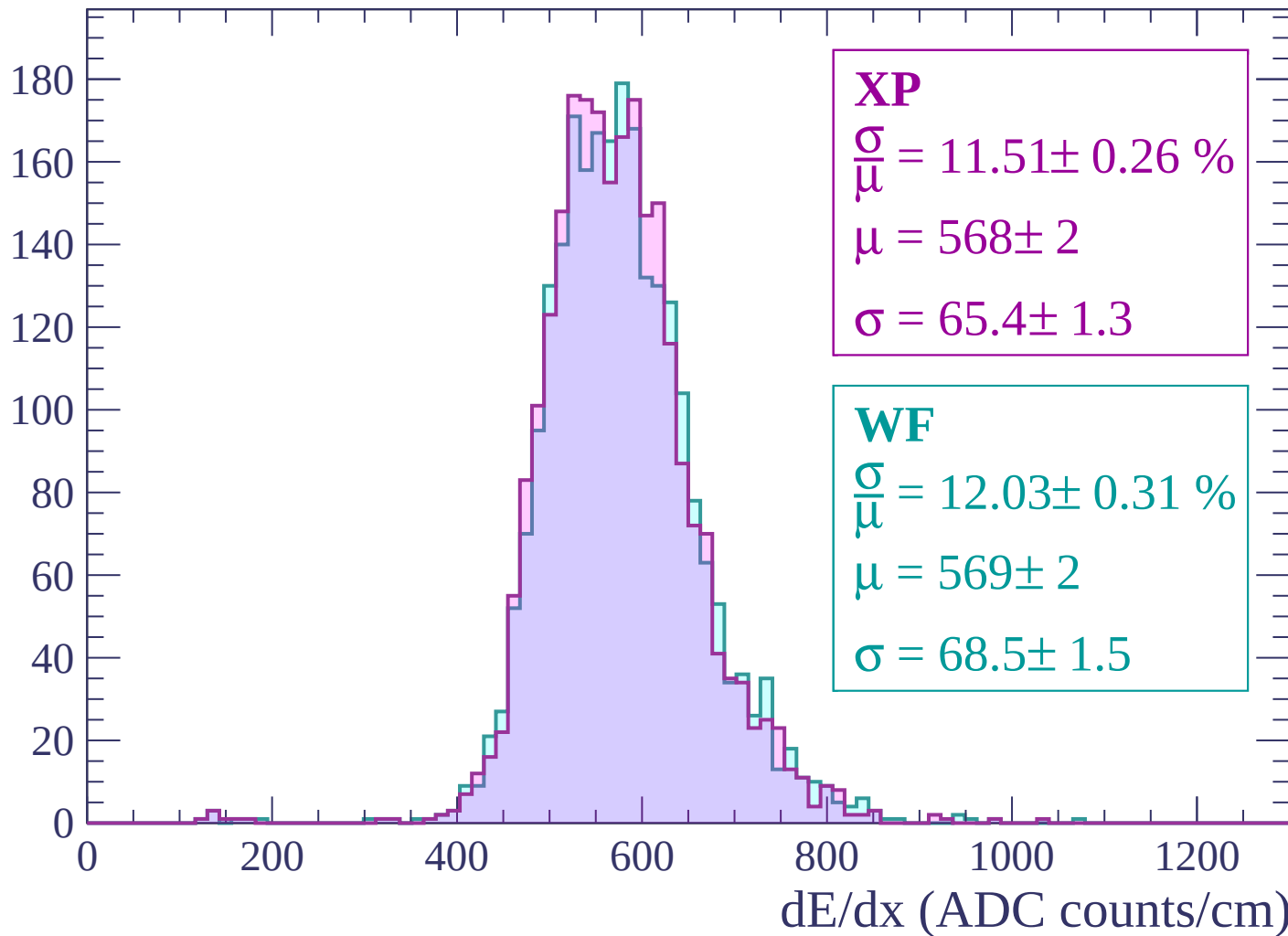
$$\mu = 456 \pm 1$$

$$\sigma = 43.3 \pm 1.1$$

dE/dx (ADC counts/cm)

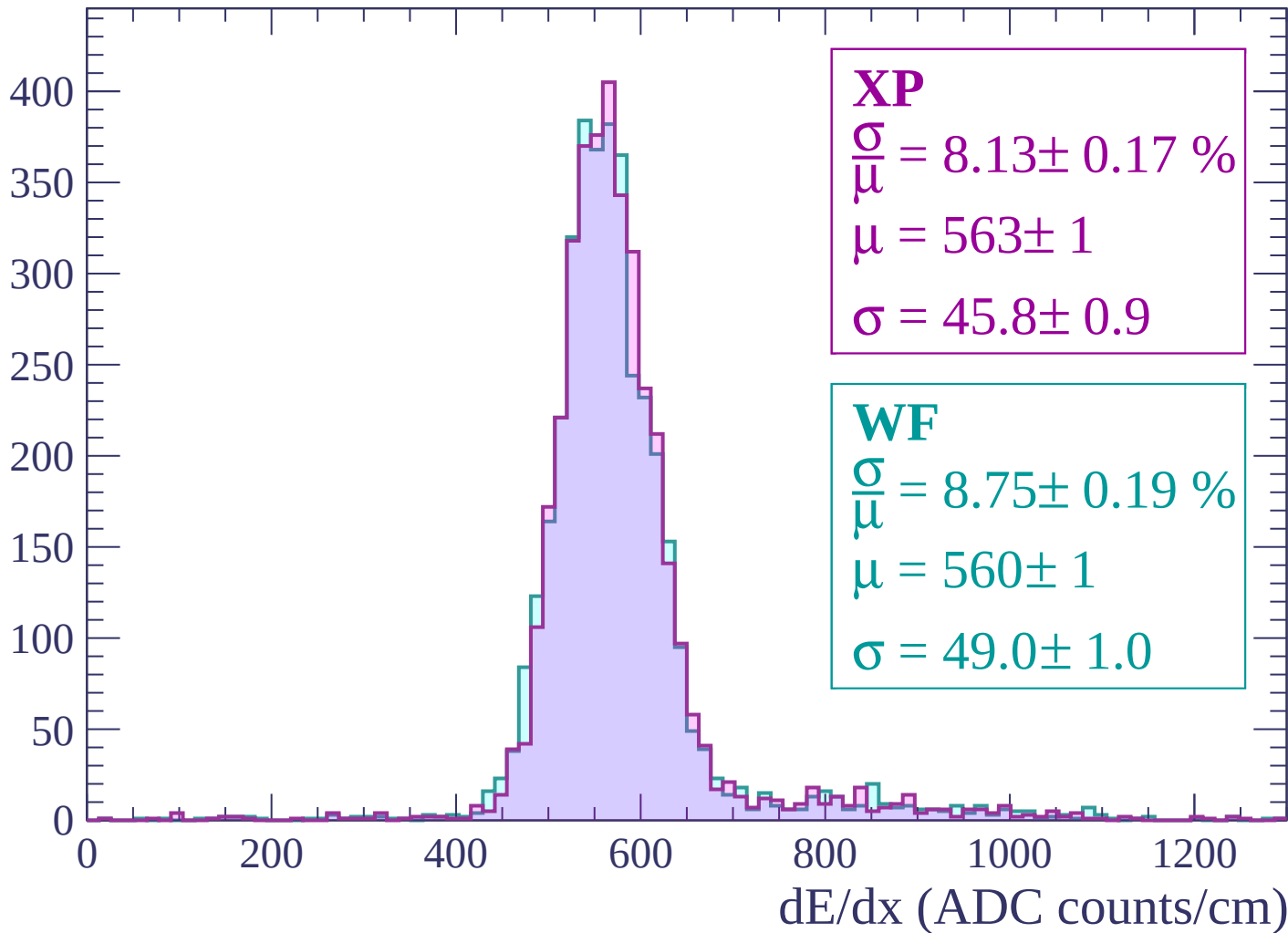
Energy loss | $-120 < p < -60$

Count



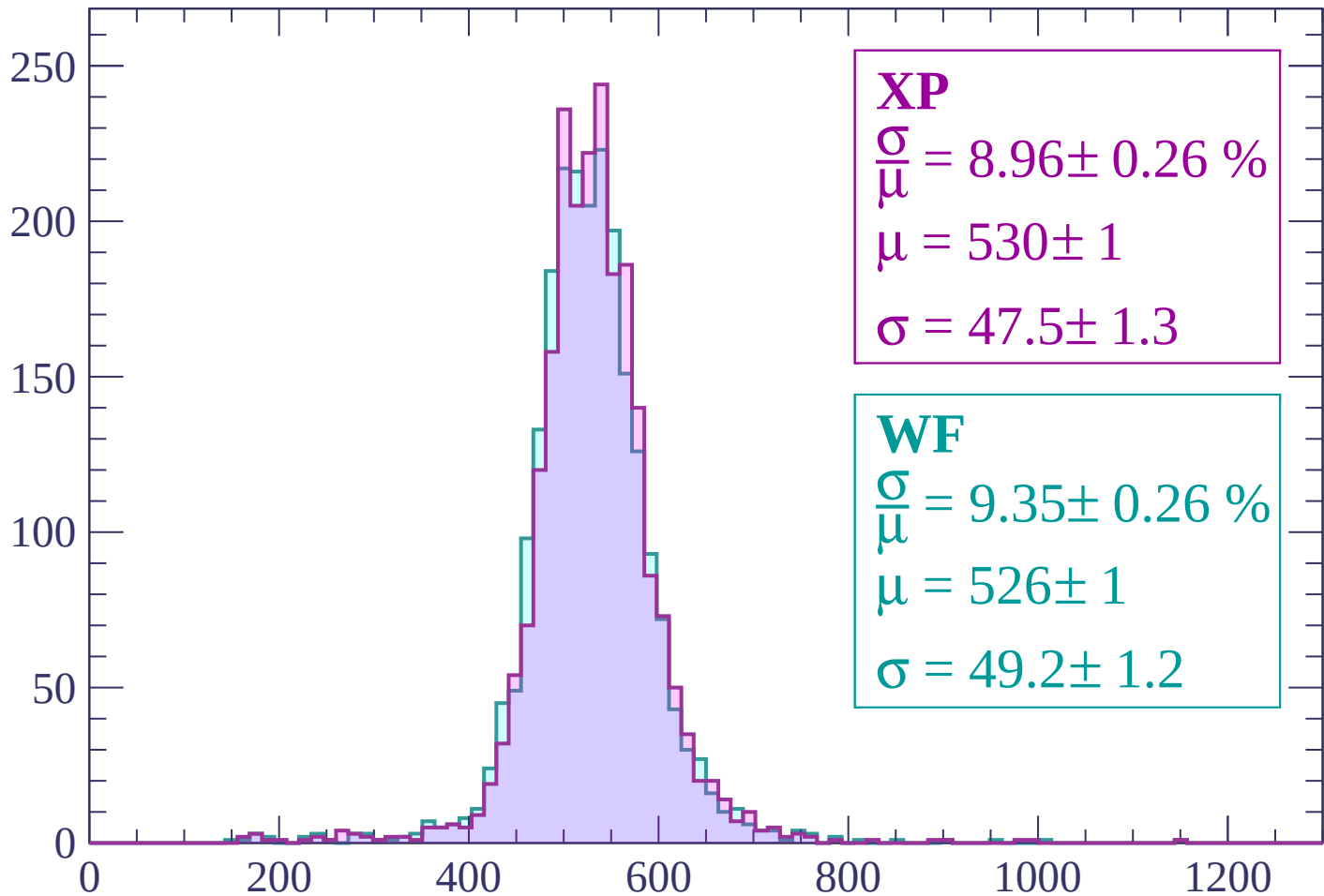
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP

$$\frac{\sigma}{\mu} = 8.96 \pm 0.26 \%$$

$$\mu = 530 \pm 1$$

$$\sigma = 47.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.35 \pm 0.26 \%$$

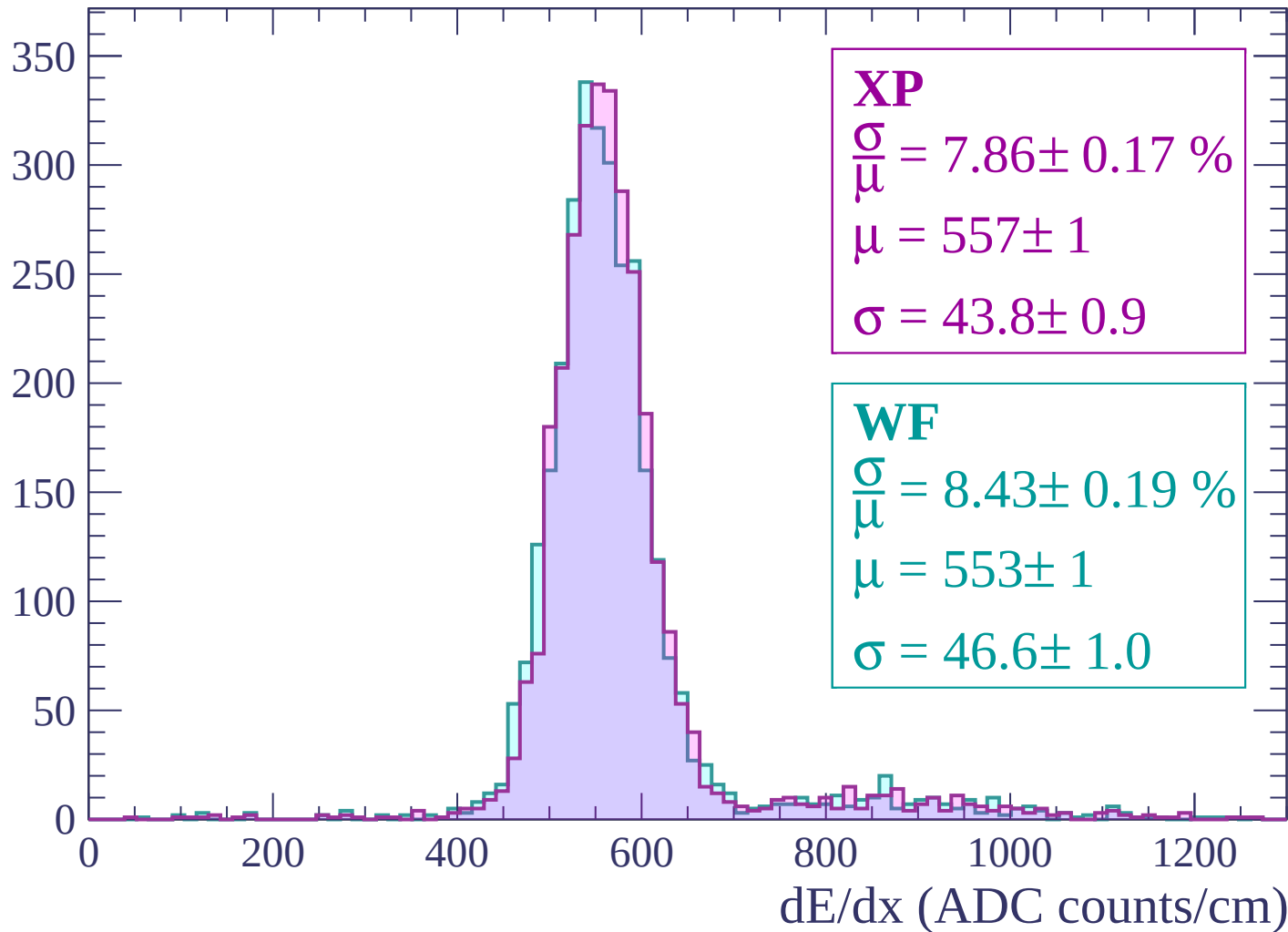
$$\mu = 526 \pm 1$$

$$\sigma = 49.2 \pm 1.2$$

dE/dx (ADC counts/cm)

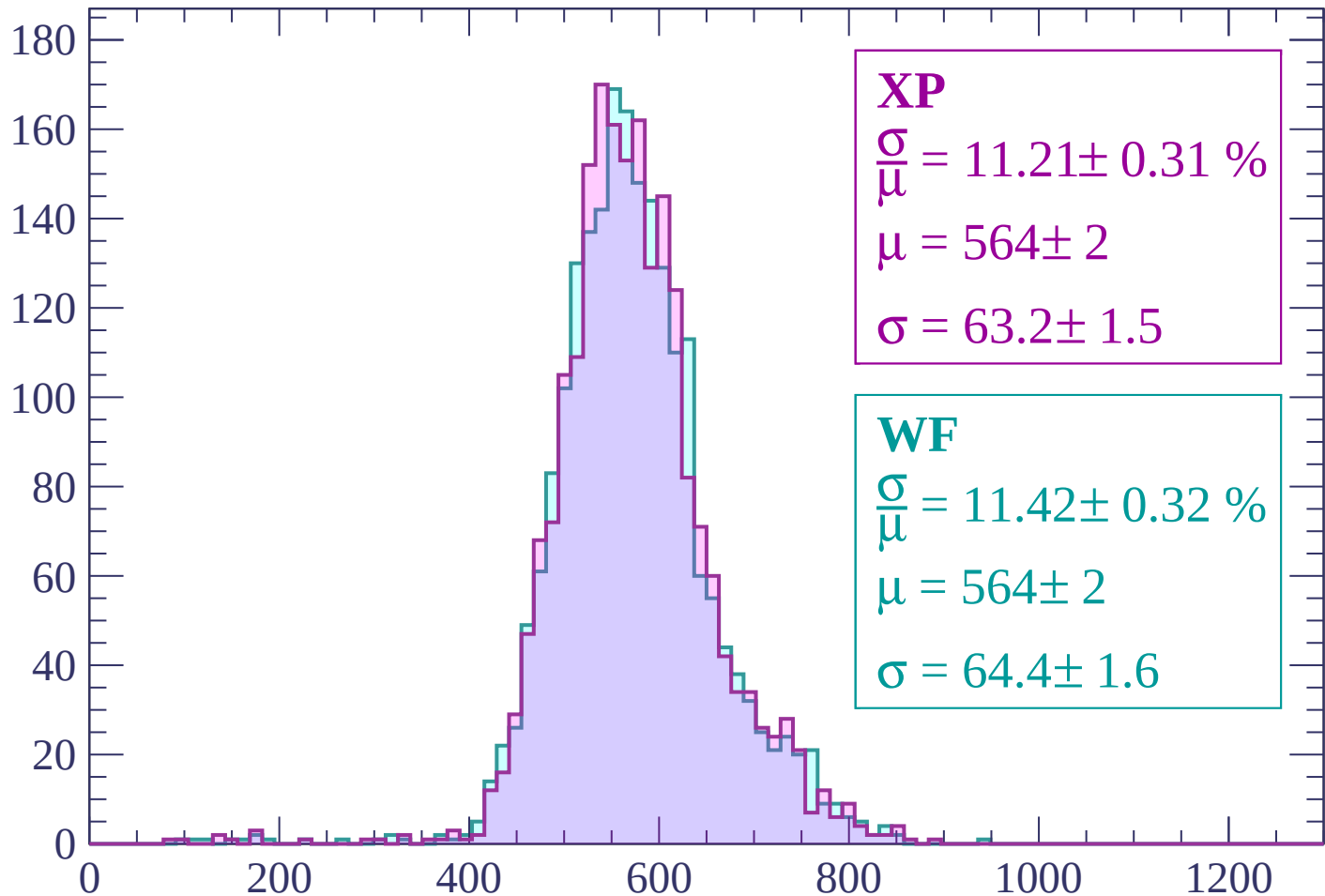
Energy loss | $60 < p < 120$

Count



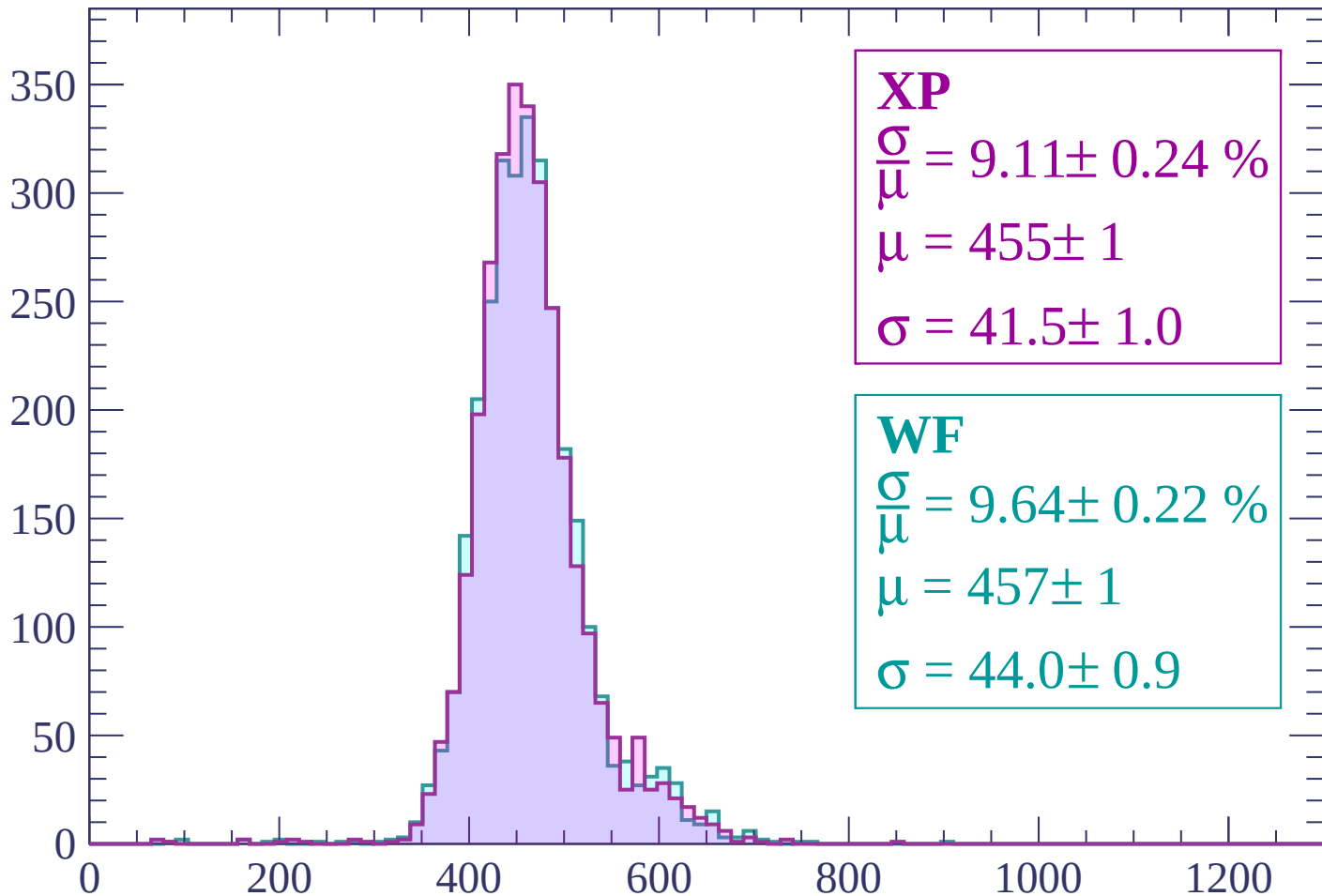
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.24 \%$$

$$\mu = 455 \pm 1$$

$$\sigma = 41.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.22 \%$$

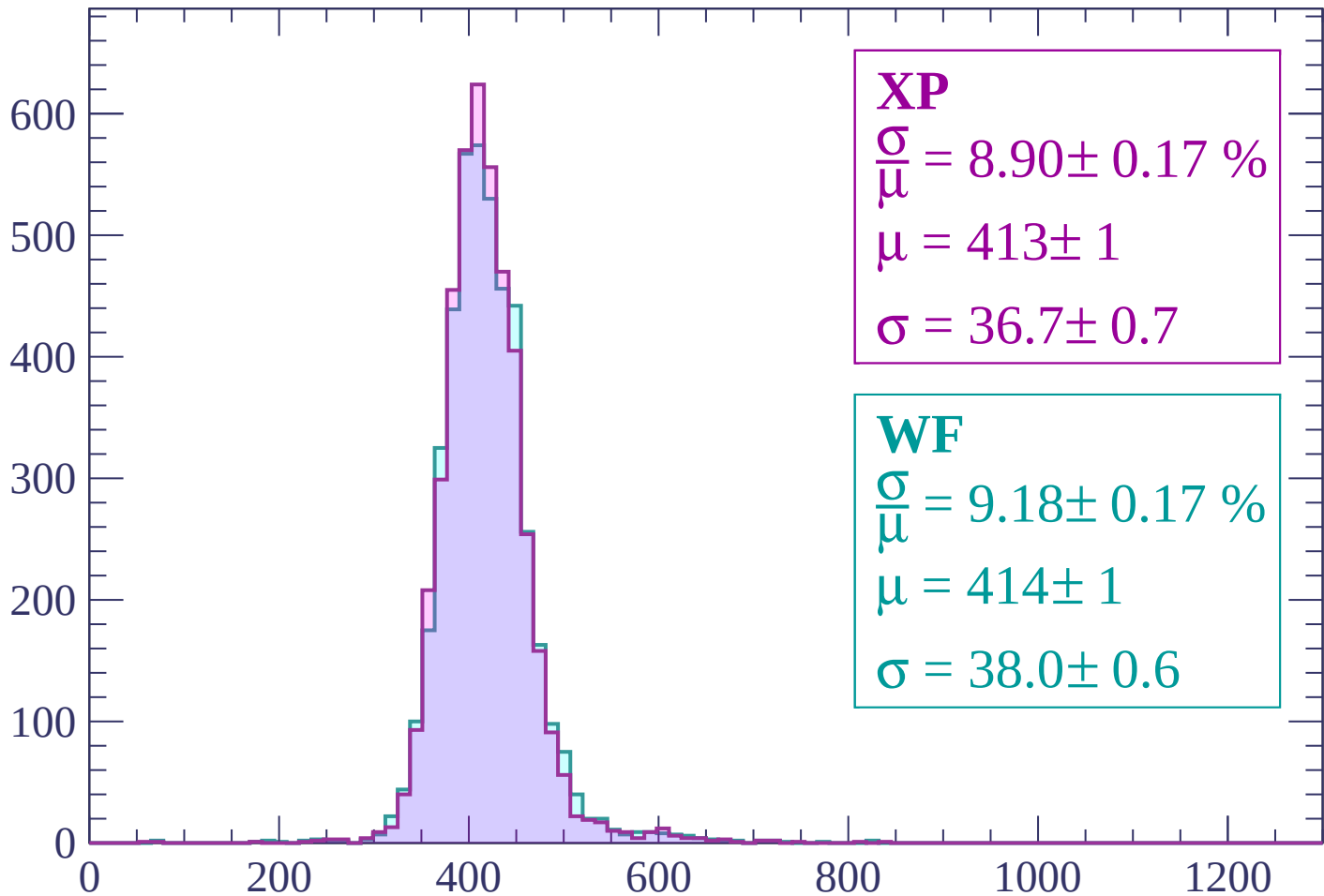
$$\mu = 457 \pm 1$$

$$\sigma = 44.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count

1000

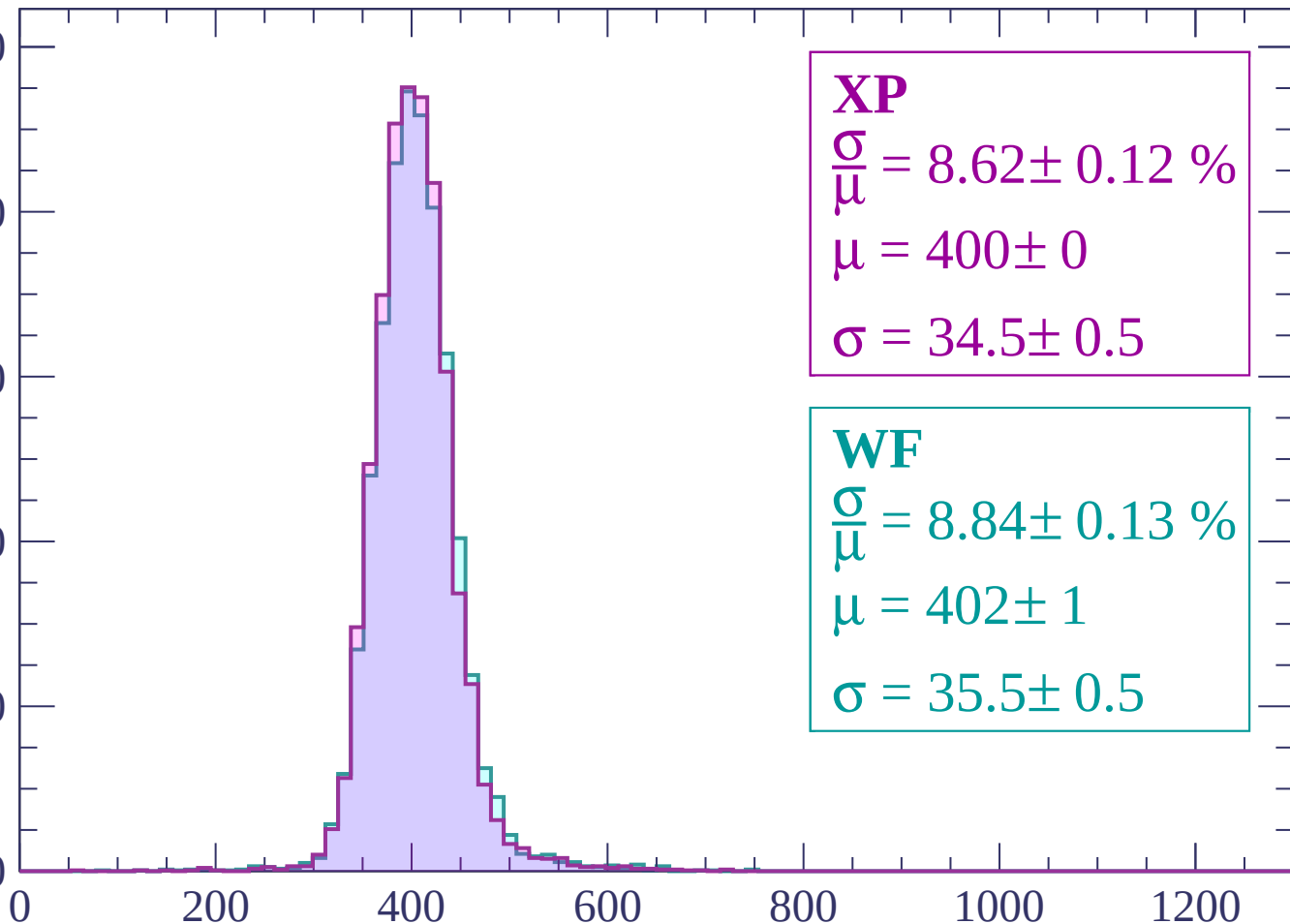
800

600

400

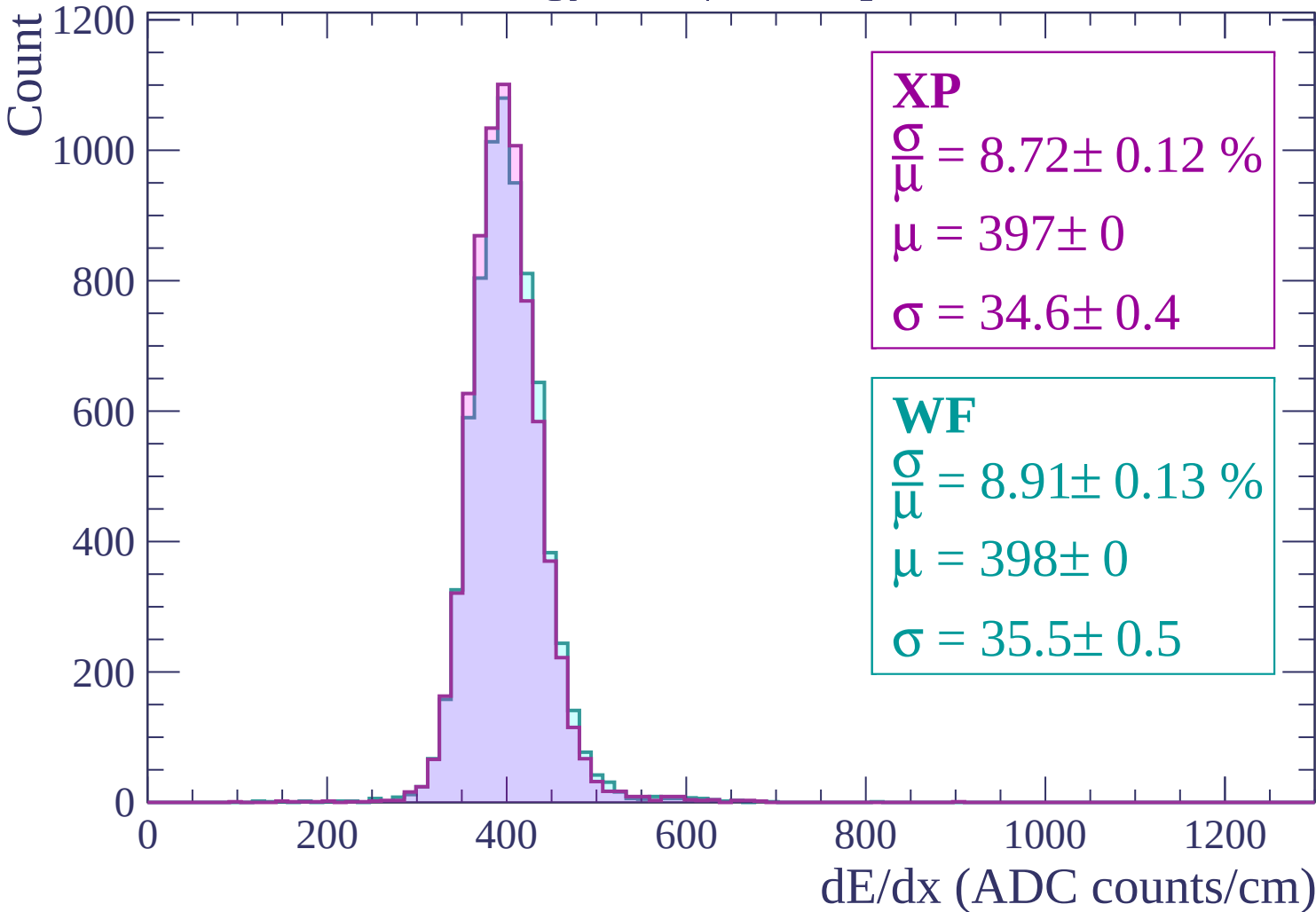
200

0



dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$



Energy loss | $420 < p < 480$

Count

1200

1000

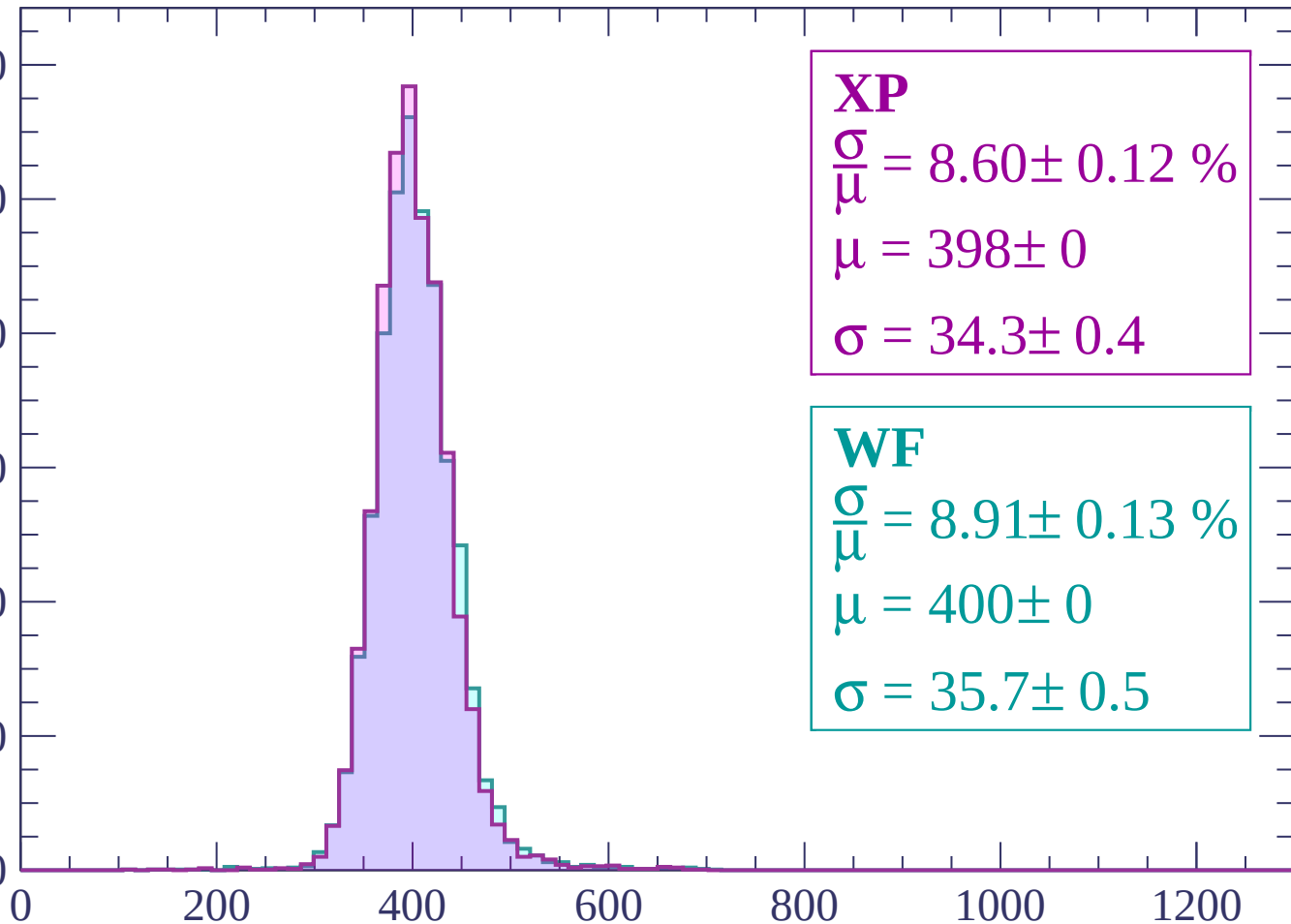
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 8.60 \pm 0.12 \%$$

$$\mu = 398 \pm 0$$

$$\sigma = 34.3 \pm 0.4$$

WF

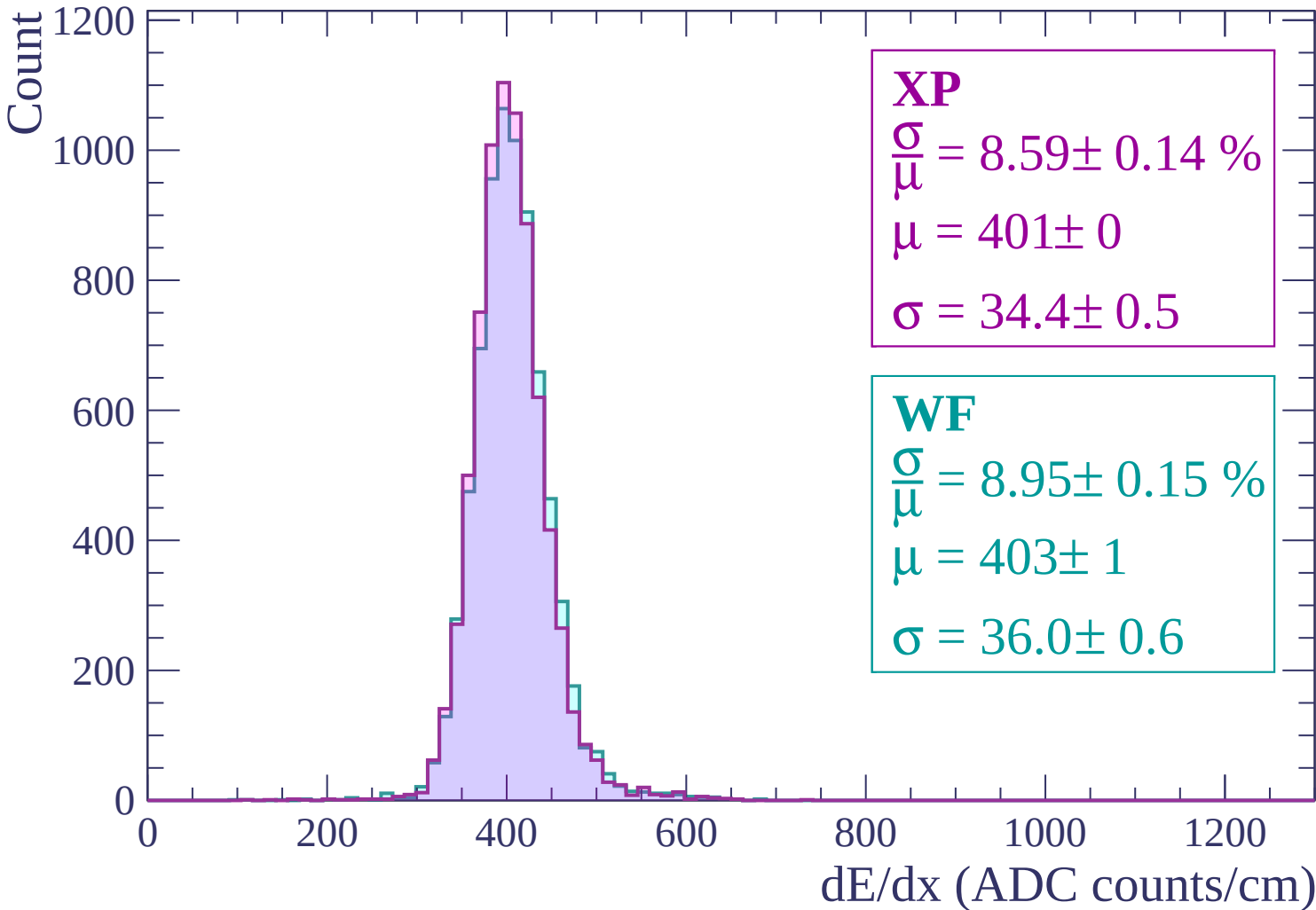
$$\frac{\sigma}{\mu} = 8.91 \pm 0.13 \%$$

$$\mu = 400 \pm 0$$

$$\sigma = 35.7 \pm 0.5$$

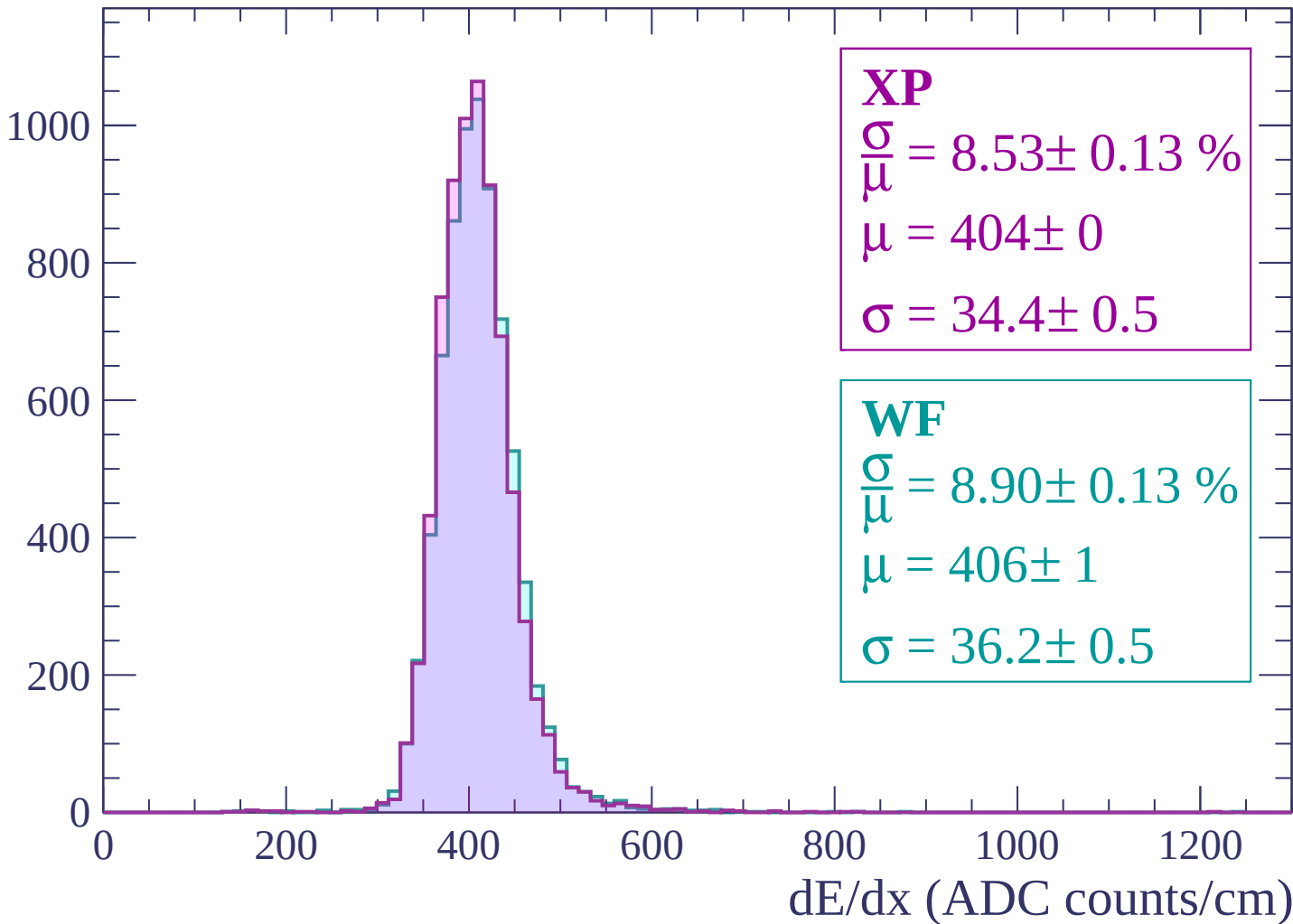
dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count



Energy loss | $600 < p < 660$

Count

1000

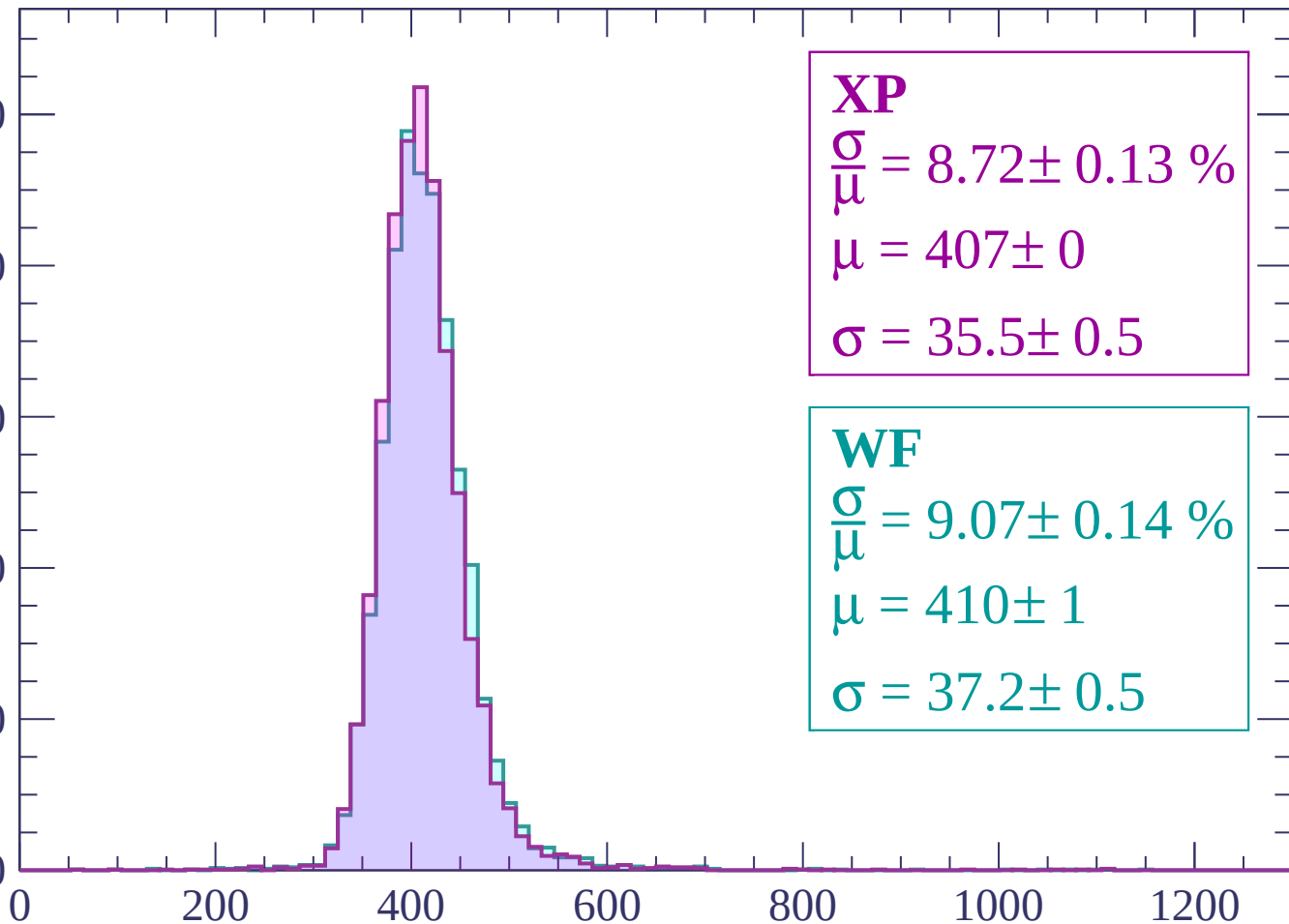
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.13 \%$$

$$\mu = 407 \pm 0$$

$$\sigma = 35.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.07 \pm 0.14 \%$$

$$\mu = 410 \pm 1$$

$$\sigma = 37.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count

1000

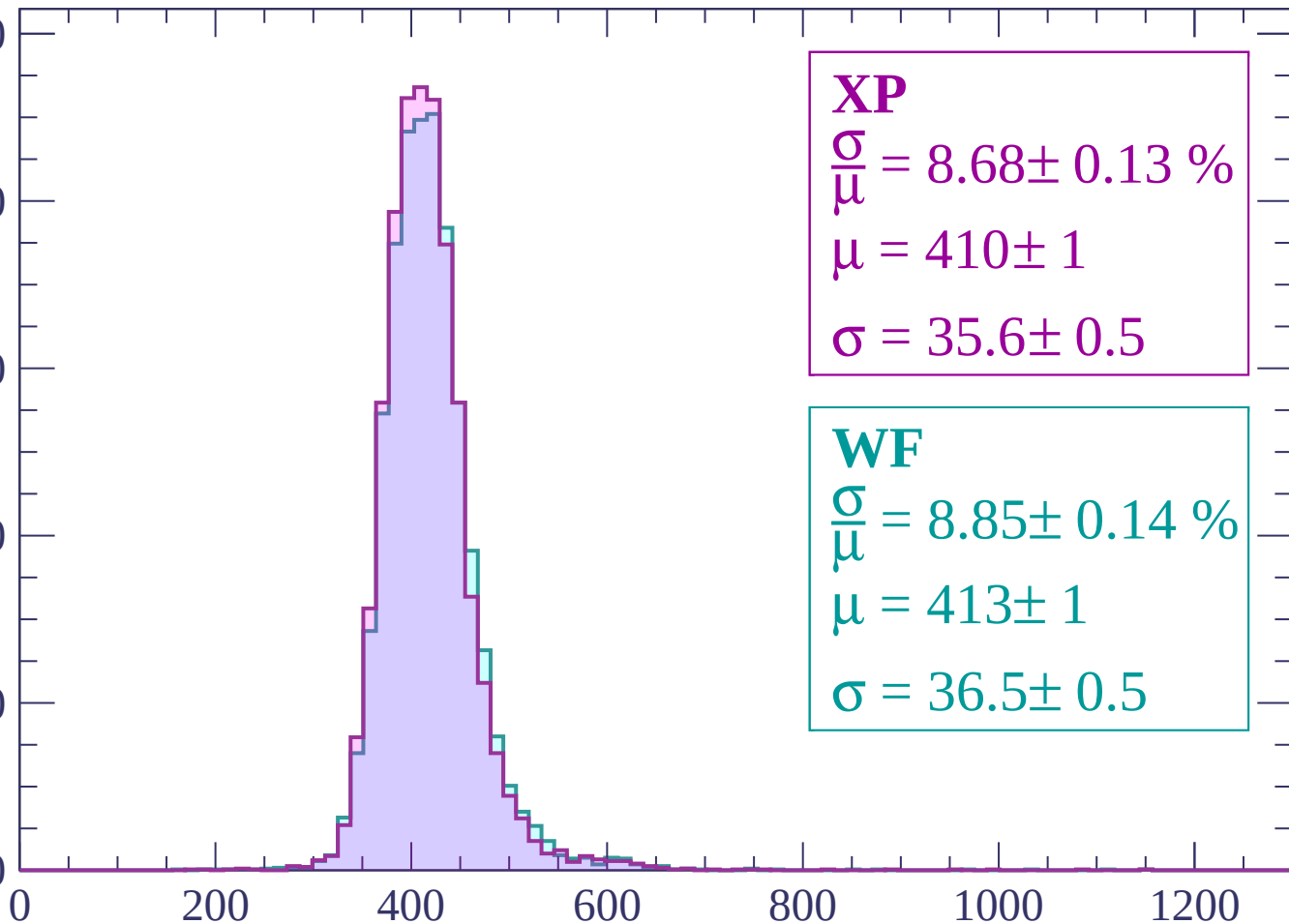
800

600

400

200

0



dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\bar{\mu}} = 8.63 \pm 0.13 \%$$

$$\mu = 414 \pm 1$$

$$\sigma = 35.7 \pm 0.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.00 \pm 0.13 \%$$

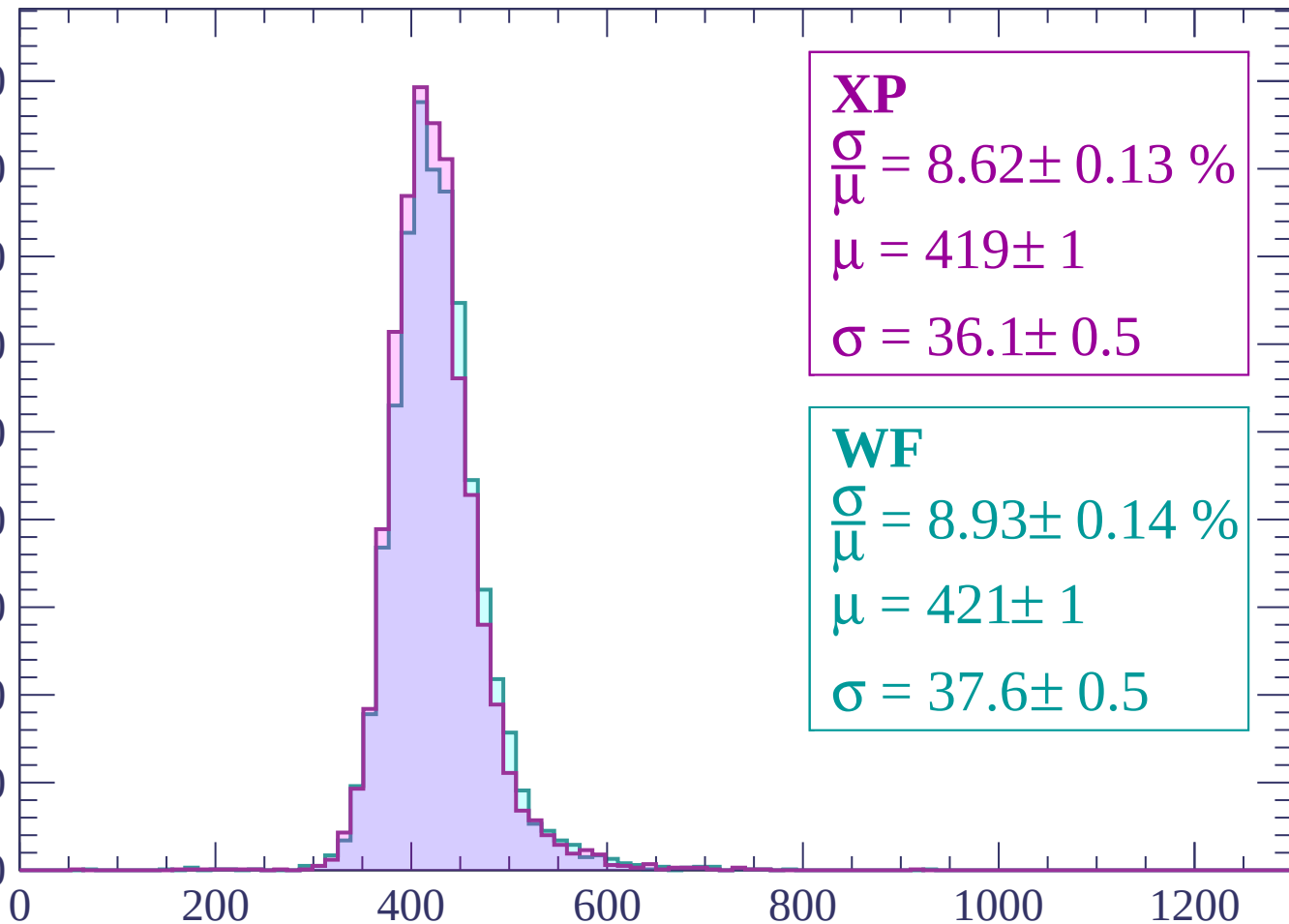
$$\mu = 417 \pm 1$$

$$\sigma = 37.5 \pm 0.5$$

Energy loss | $780 < p < 840$

Count

900
800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 8.62 \pm 0.13 \%$$

$$\mu = 419 \pm 1$$

$$\sigma = 36.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.14 \%$$

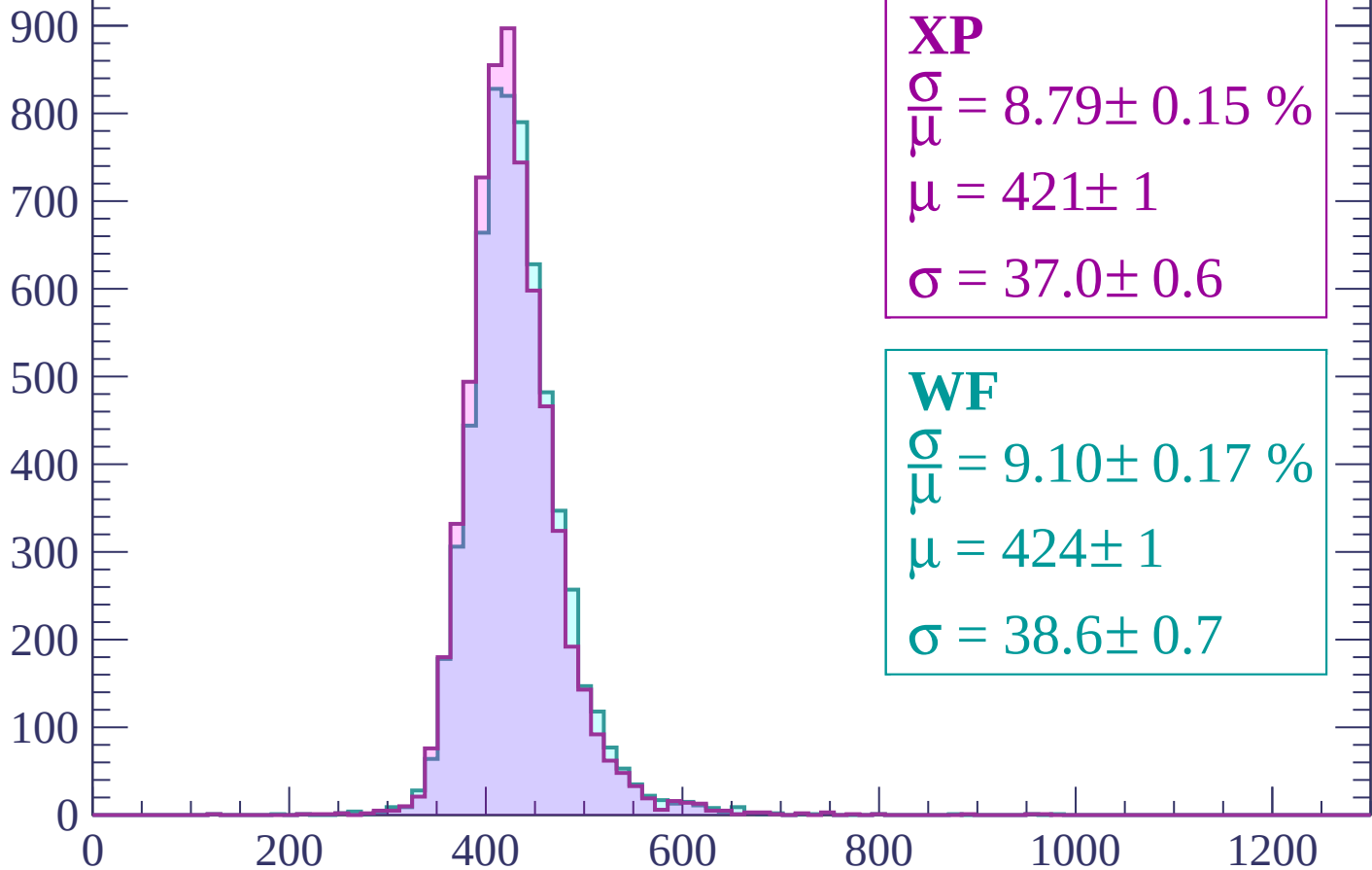
$$\mu = 421 \pm 1$$

$$\sigma = 37.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.15 \%$$

$$\mu = 421 \pm 1$$

$$\sigma = 37.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.17 \%$$

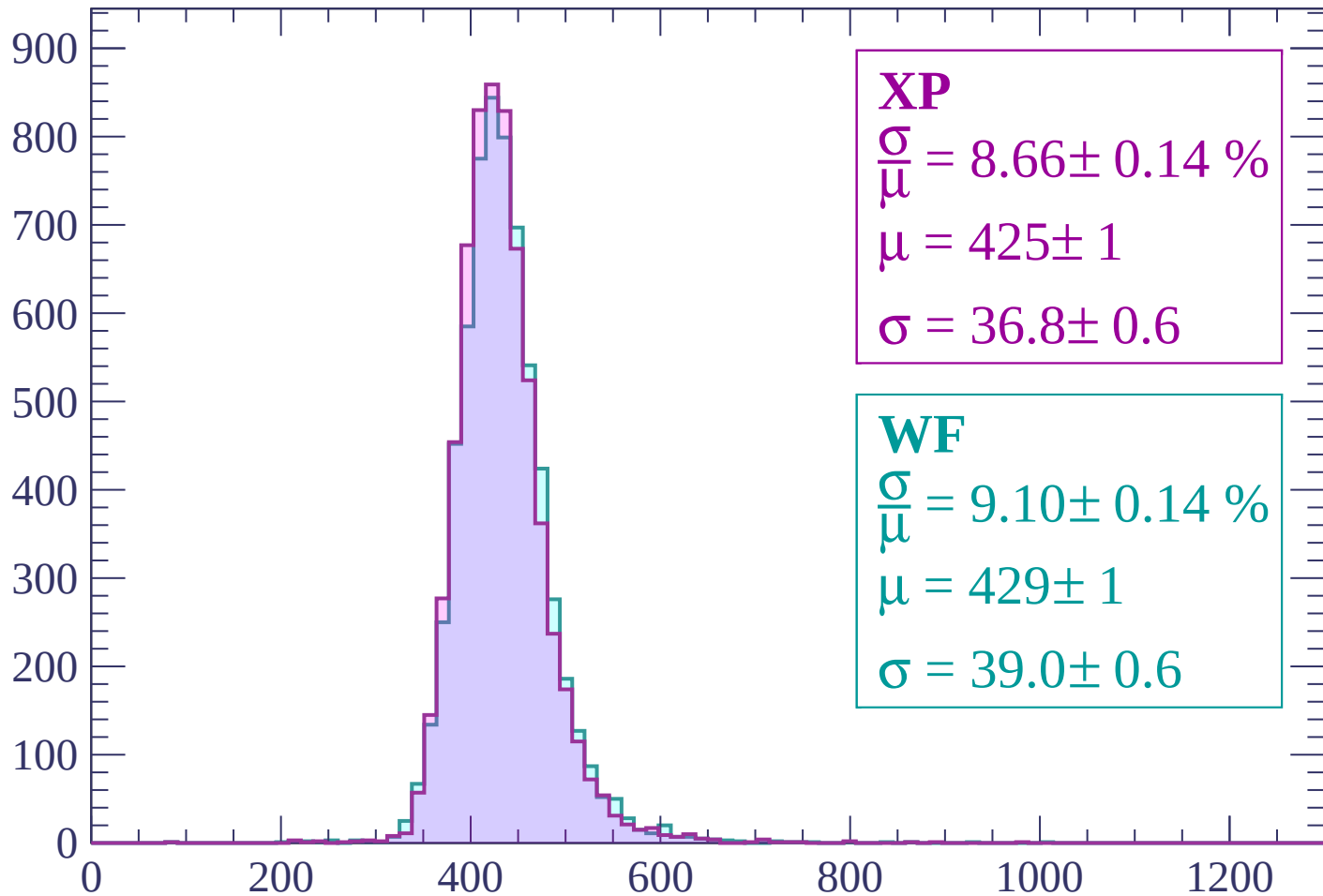
$$\mu = 424 \pm 1$$

$$\sigma = 38.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.14 \%$$

$$\mu = 425 \pm 1$$

$$\sigma = 36.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.14 \%$$

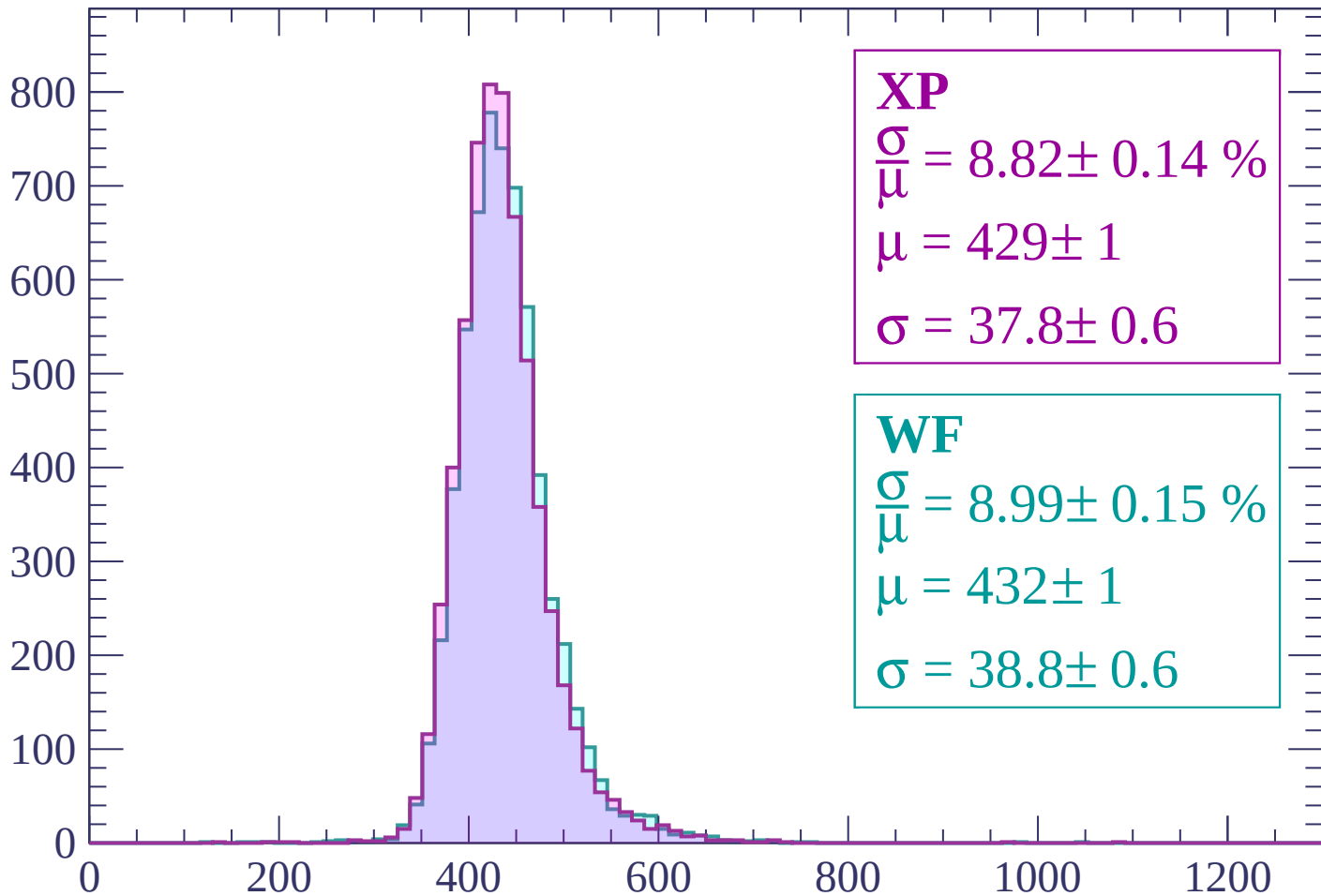
$$\mu = 429 \pm 1$$

$$\sigma = 39.0 \pm 0.6$$

dE/dx (ADC counts/cm)

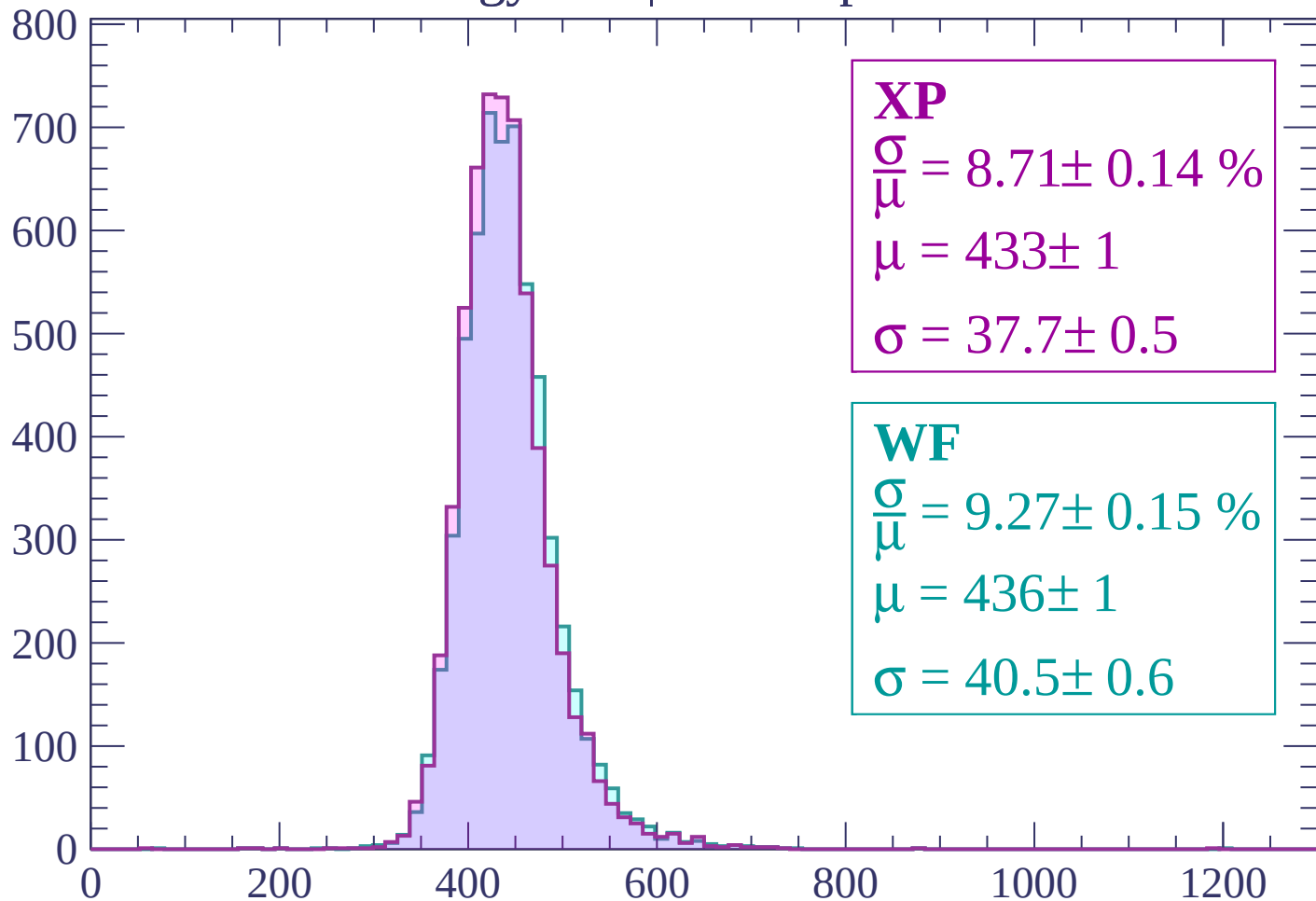
Energy loss | $960 < p < 1020$

Count



Energy loss | $1020 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 8.71 \pm 0.14 \%$$

$$\mu = 433 \pm 1$$

$$\sigma = 37.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.27 \pm 0.15 \%$$

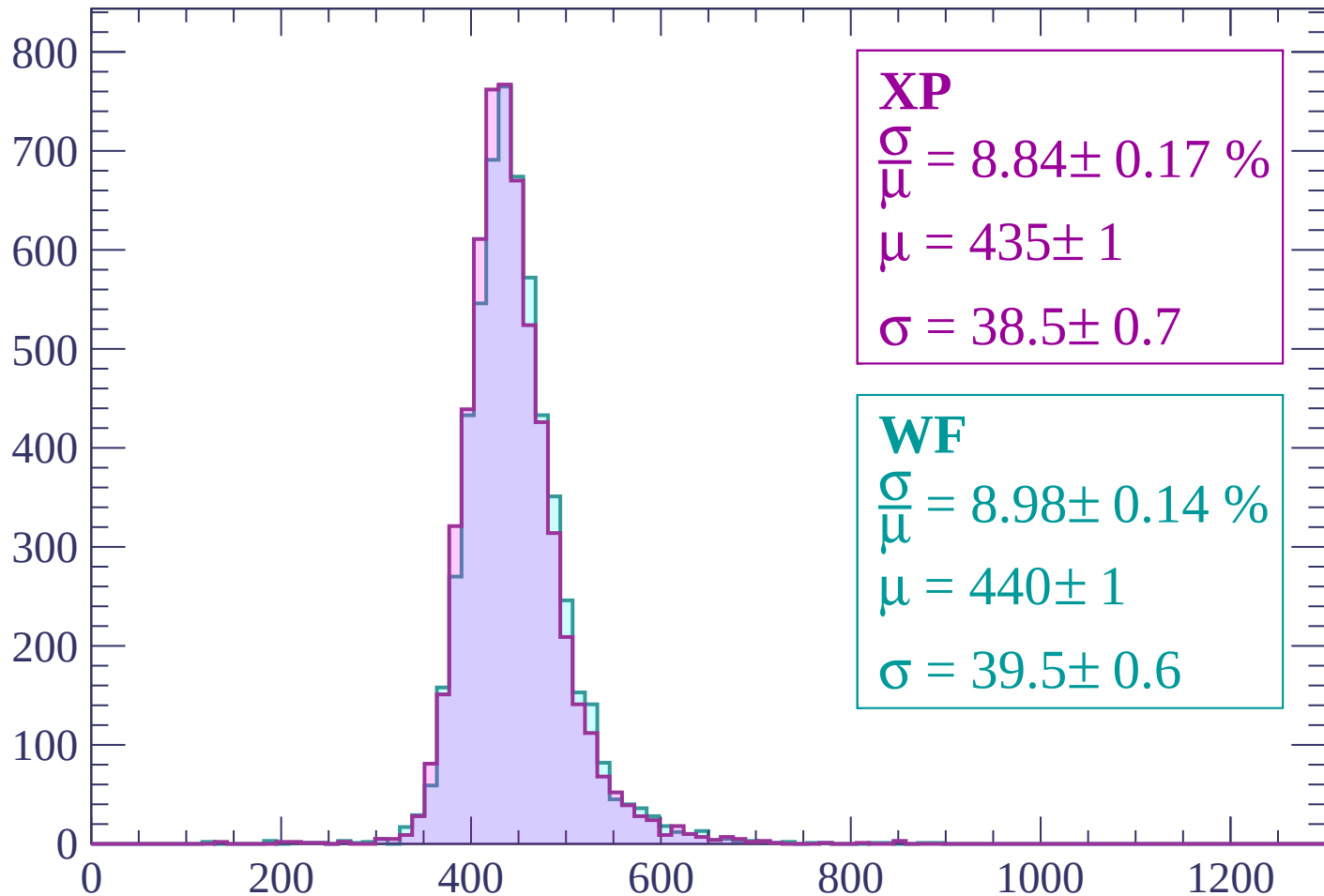
$$\mu = 436 \pm 1$$

$$\sigma = 40.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.17 \%$$

$$\mu = 435 \pm 1$$

$$\sigma = 38.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.14 \%$$

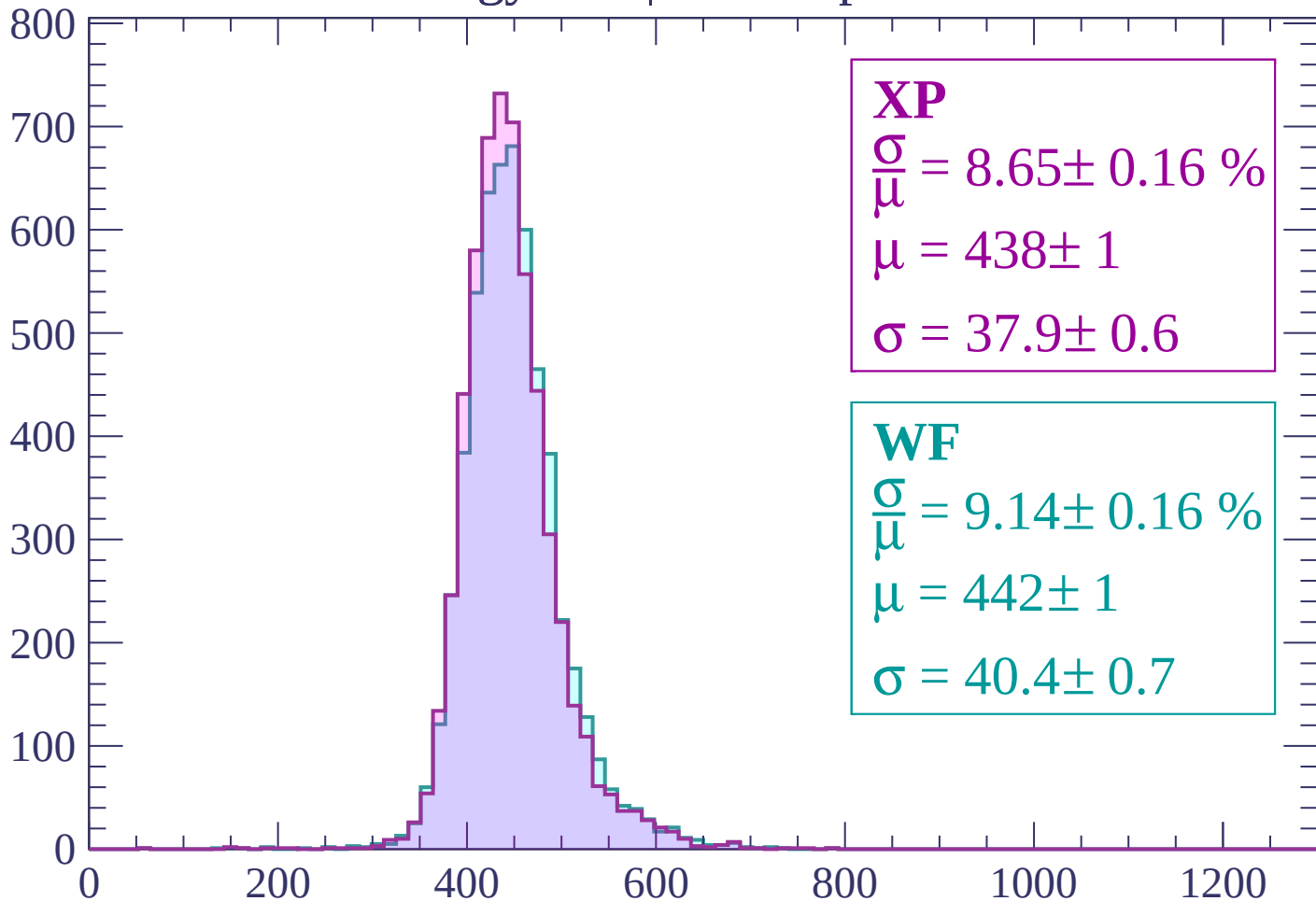
$$\mu = 440 \pm 1$$

$$\sigma = 39.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.65 \pm 0.16 \%$$

$$\mu = 438 \pm 1$$

$$\sigma = 37.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.14 \pm 0.16 \%$$

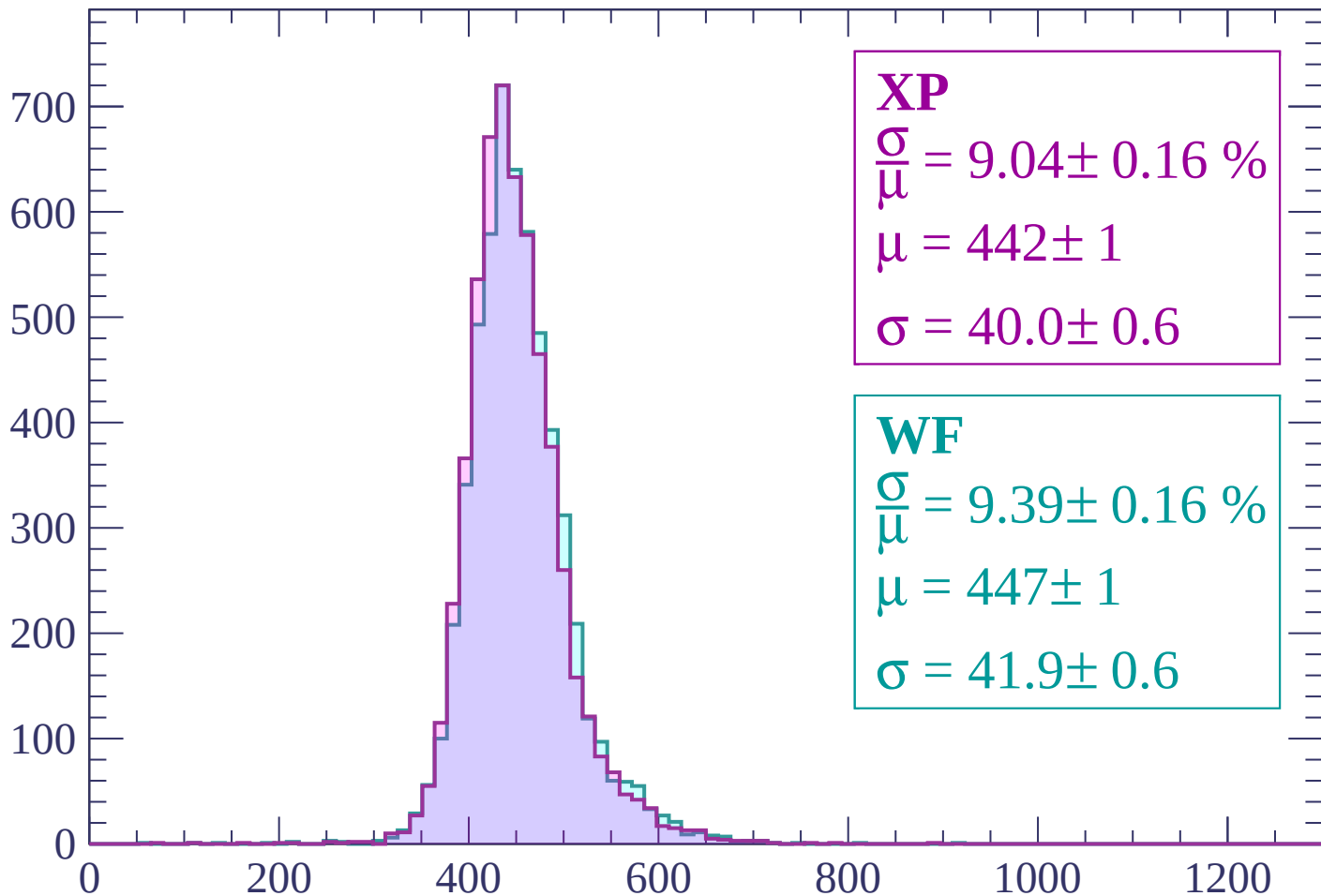
$$\mu = 442 \pm 1$$

$$\sigma = 40.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.16 \%$$

$$\mu = 442 \pm 1$$

$$\sigma = 40.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.39 \pm 0.16 \%$$

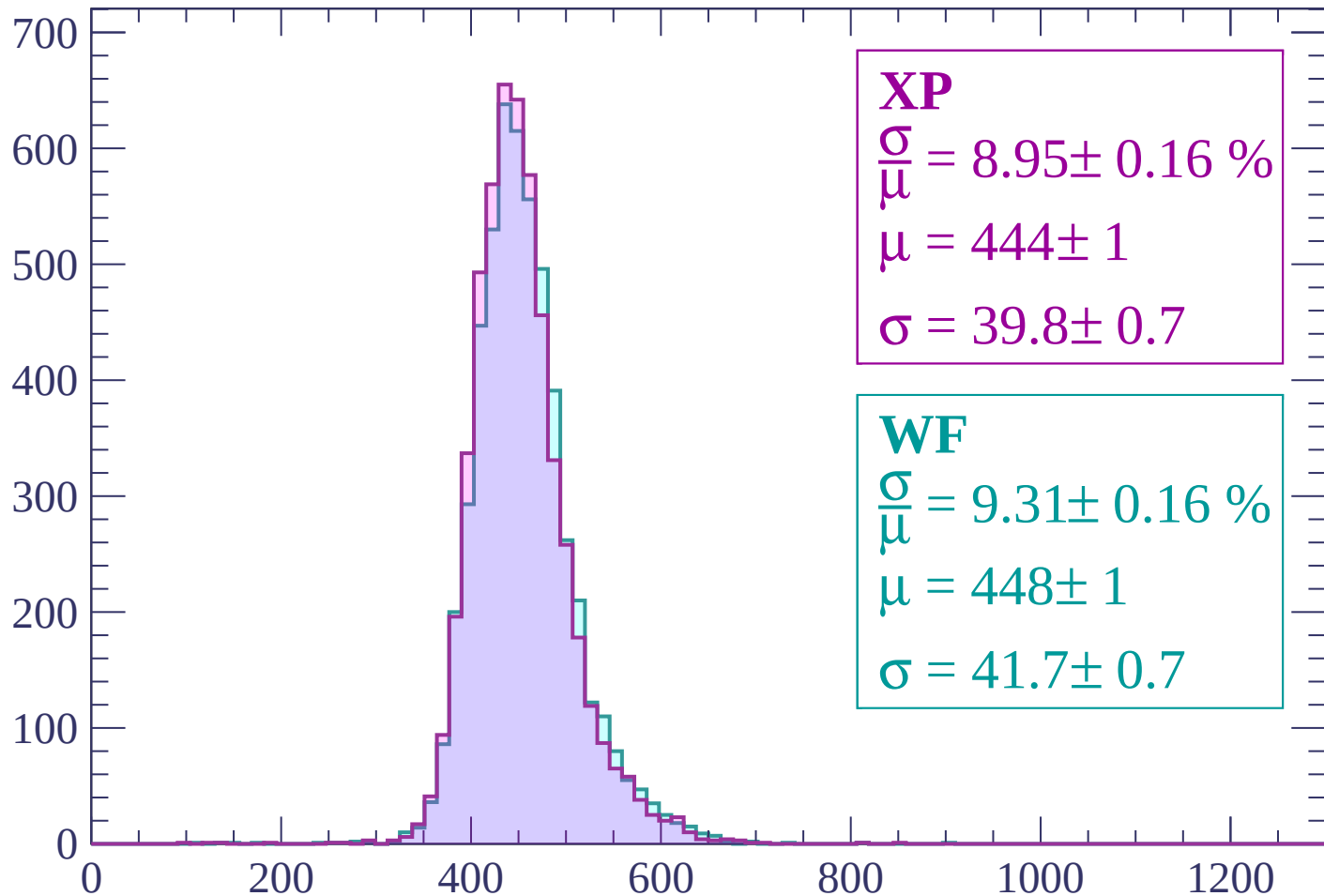
$$\mu = 447 \pm 1$$

$$\sigma = 41.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.16 \%$$

$$\mu = 444 \pm 1$$

$$\sigma = 39.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.31 \pm 0.16 \%$$

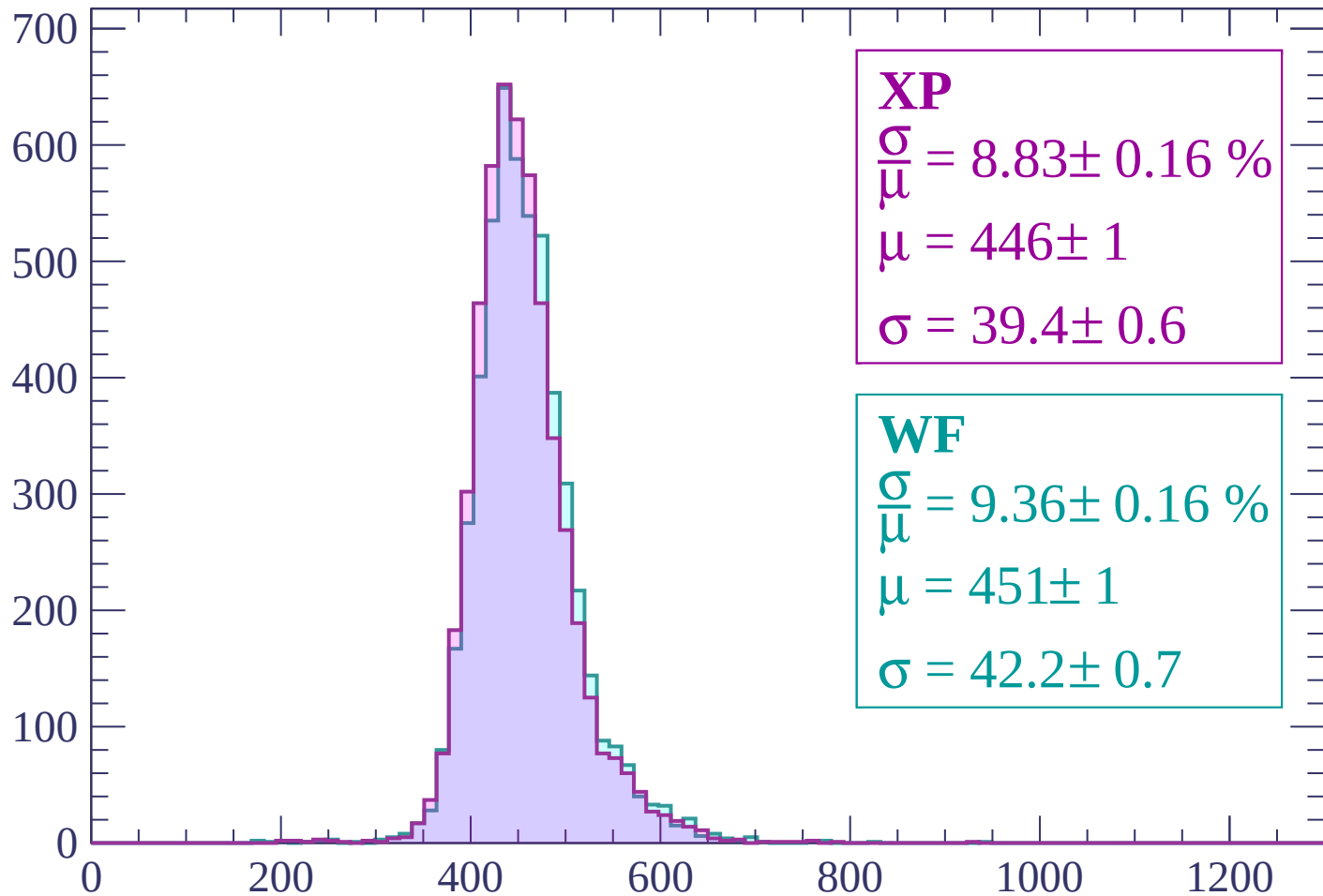
$$\mu = 448 \pm 1$$

$$\sigma = 41.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$$

$$\mu = 446 \pm 1$$

$$\sigma = 39.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.36 \pm 0.16 \%$$

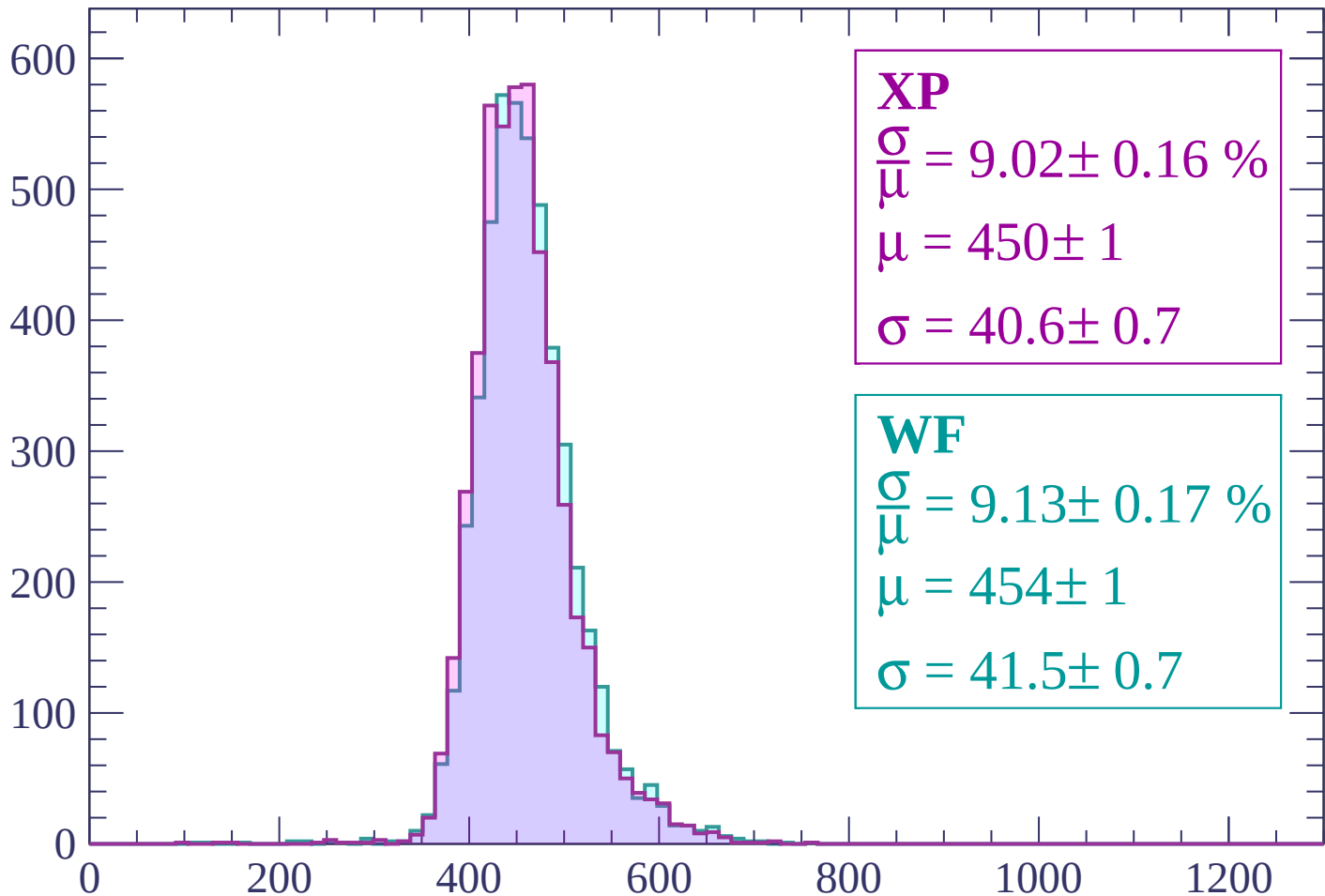
$$\mu = 451 \pm 1$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$$

$$\mu = 450 \pm 1$$

$$\sigma = 40.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.13 \pm 0.17 \%$$

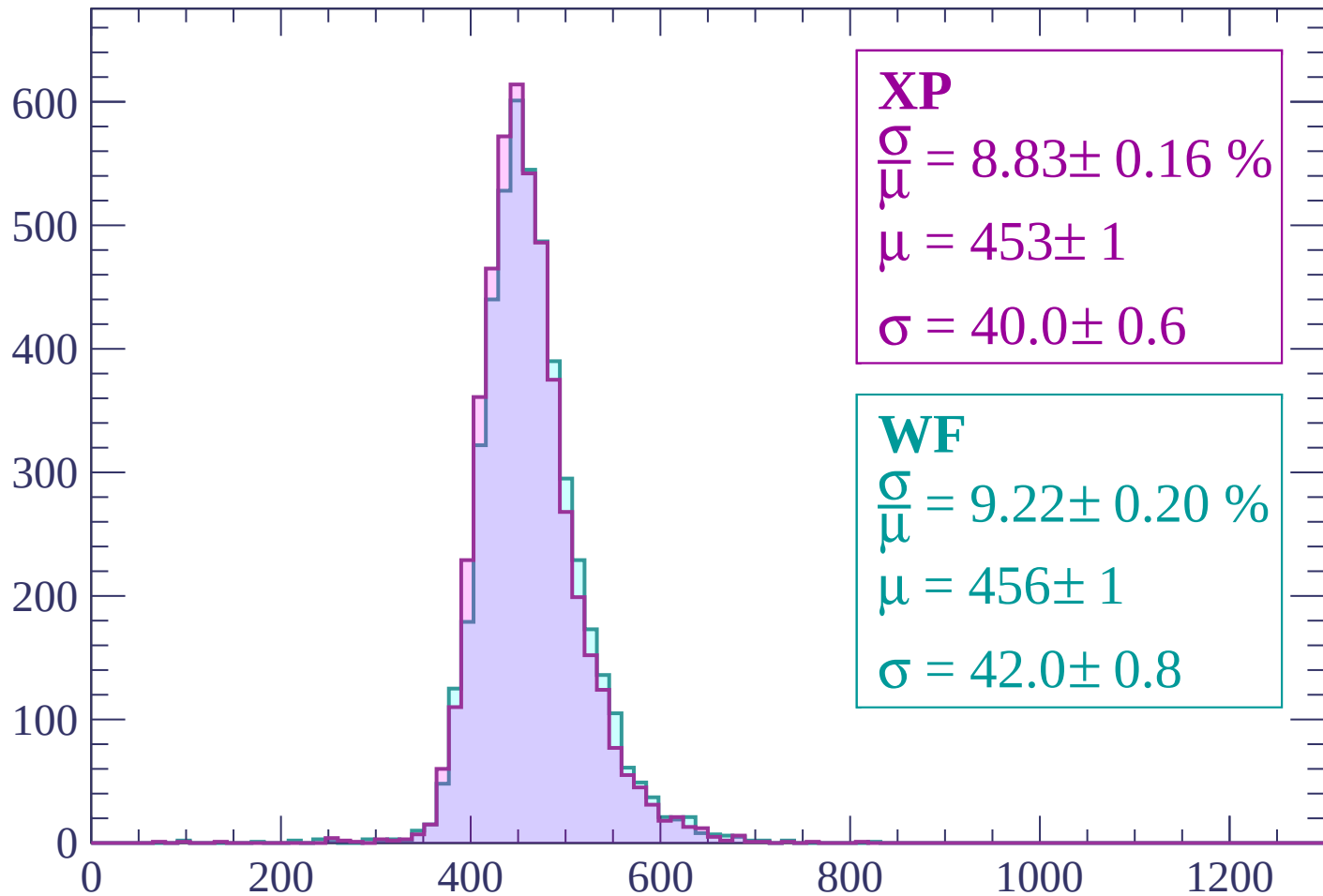
$$\mu = 454 \pm 1$$

$$\sigma = 41.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$$

$$\mu = 453 \pm 1$$

$$\sigma = 40.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.20 \%$$

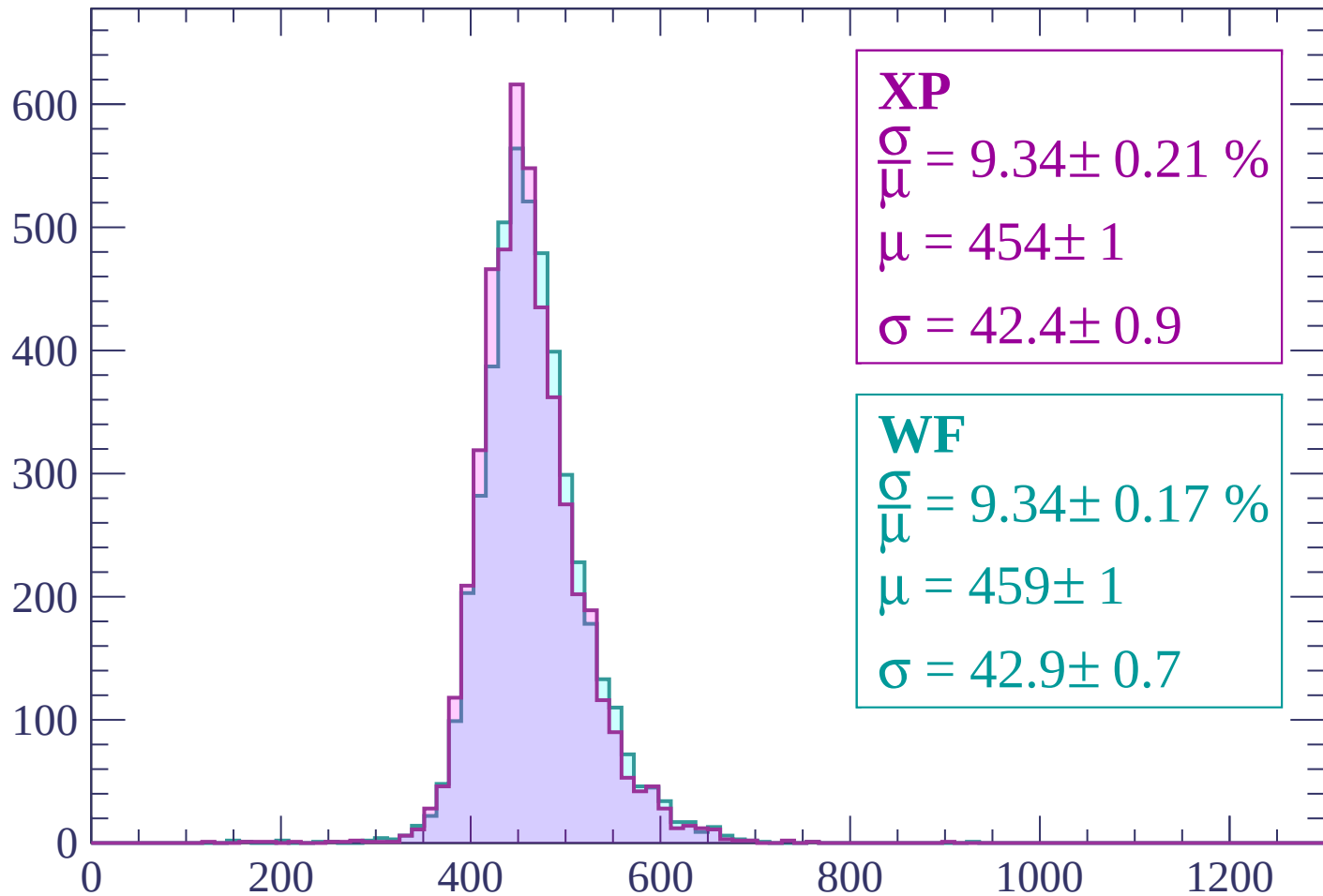
$$\mu = 456 \pm 1$$

$$\sigma = 42.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.34 \pm 0.21 \%$$

$$\mu = 454 \pm 1$$

$$\sigma = 42.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.34 \pm 0.17 \%$$

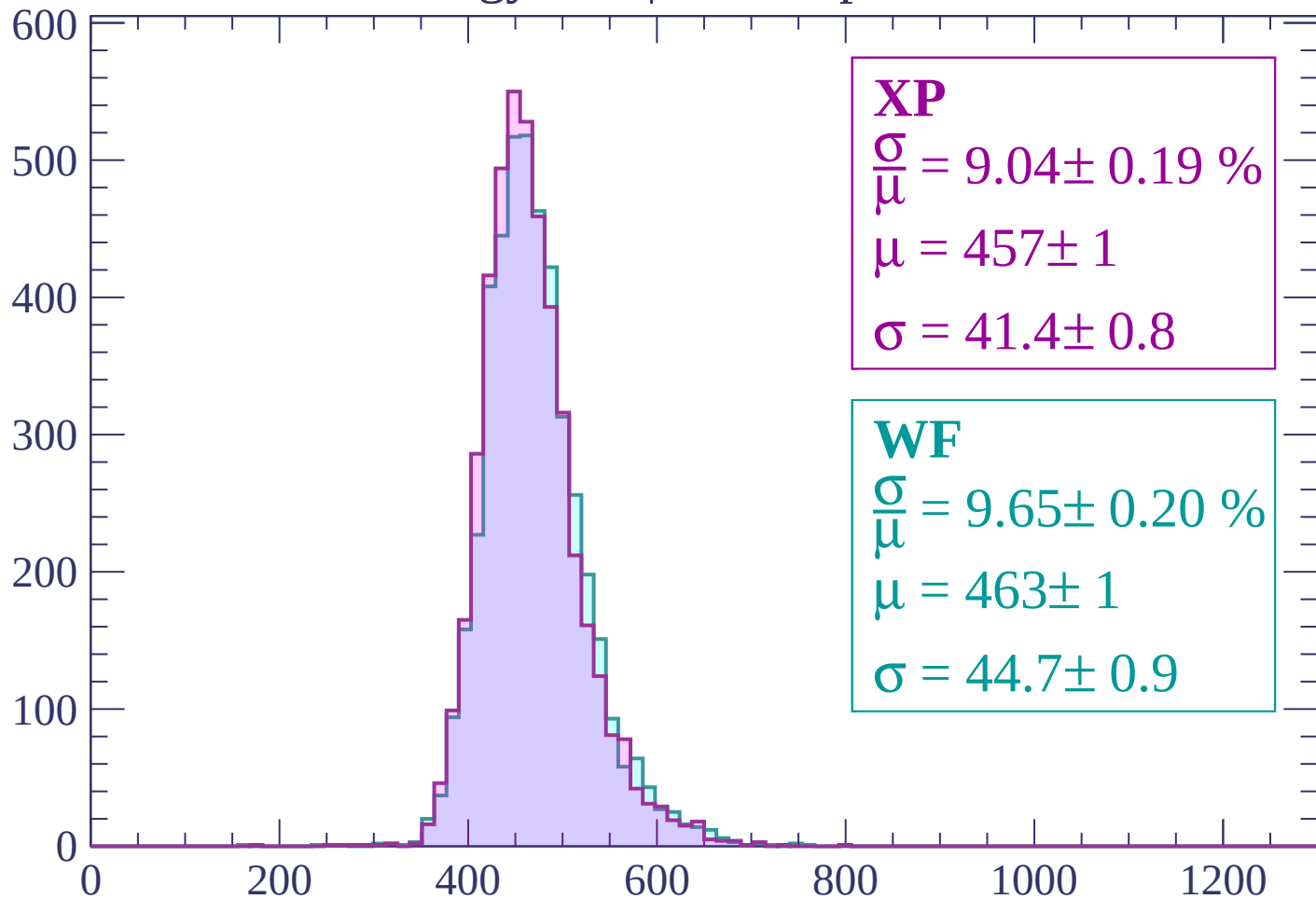
$$\mu = 459 \pm 1$$

$$\sigma = 42.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.19 \%$$

$$\mu = 457 \pm 1$$

$$\sigma = 41.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.65 \pm 0.20 \%$$

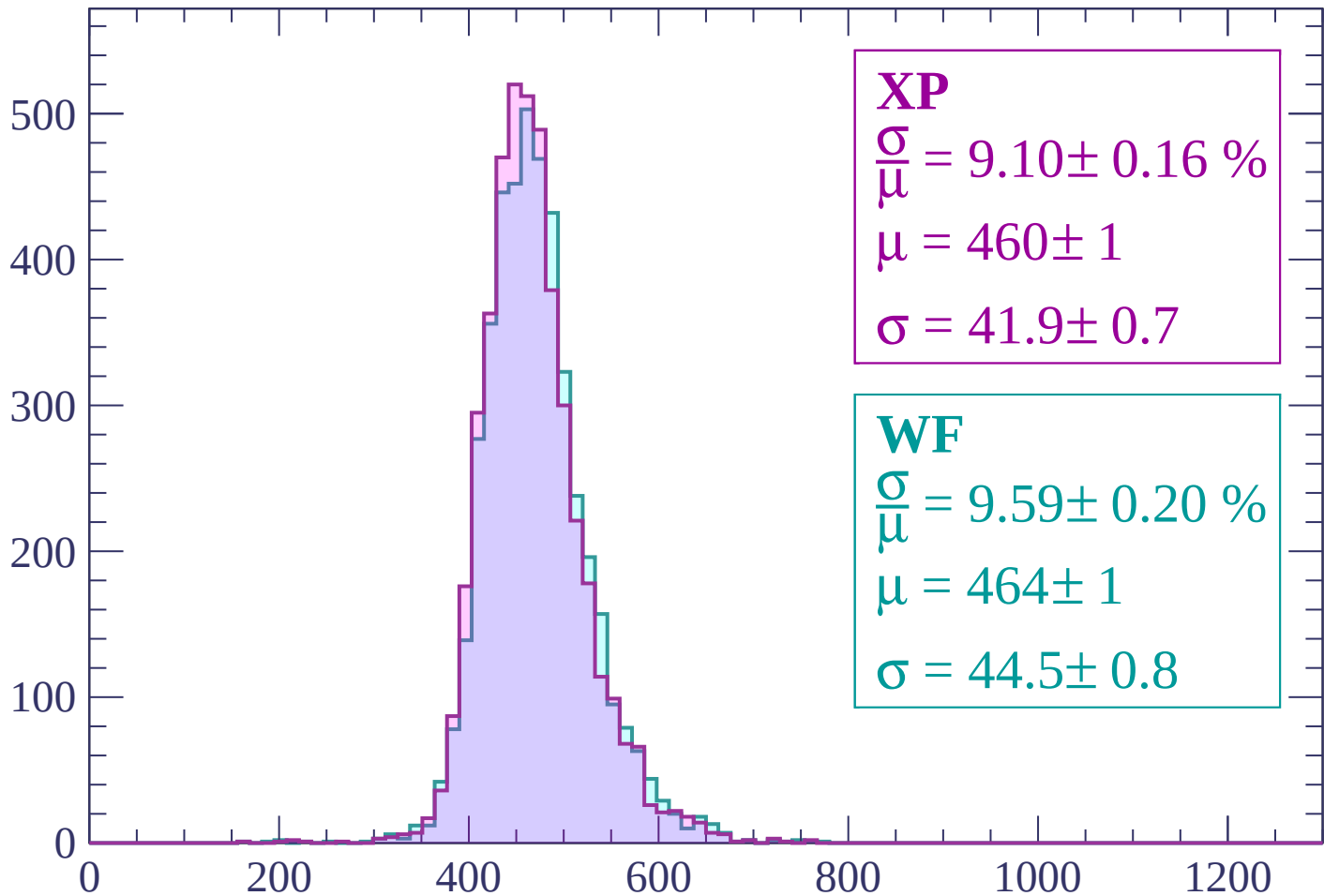
$$\mu = 463 \pm 1$$

$$\sigma = 44.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.16 \%$$

$$\mu = 460 \pm 1$$

$$\sigma = 41.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.20 \%$$

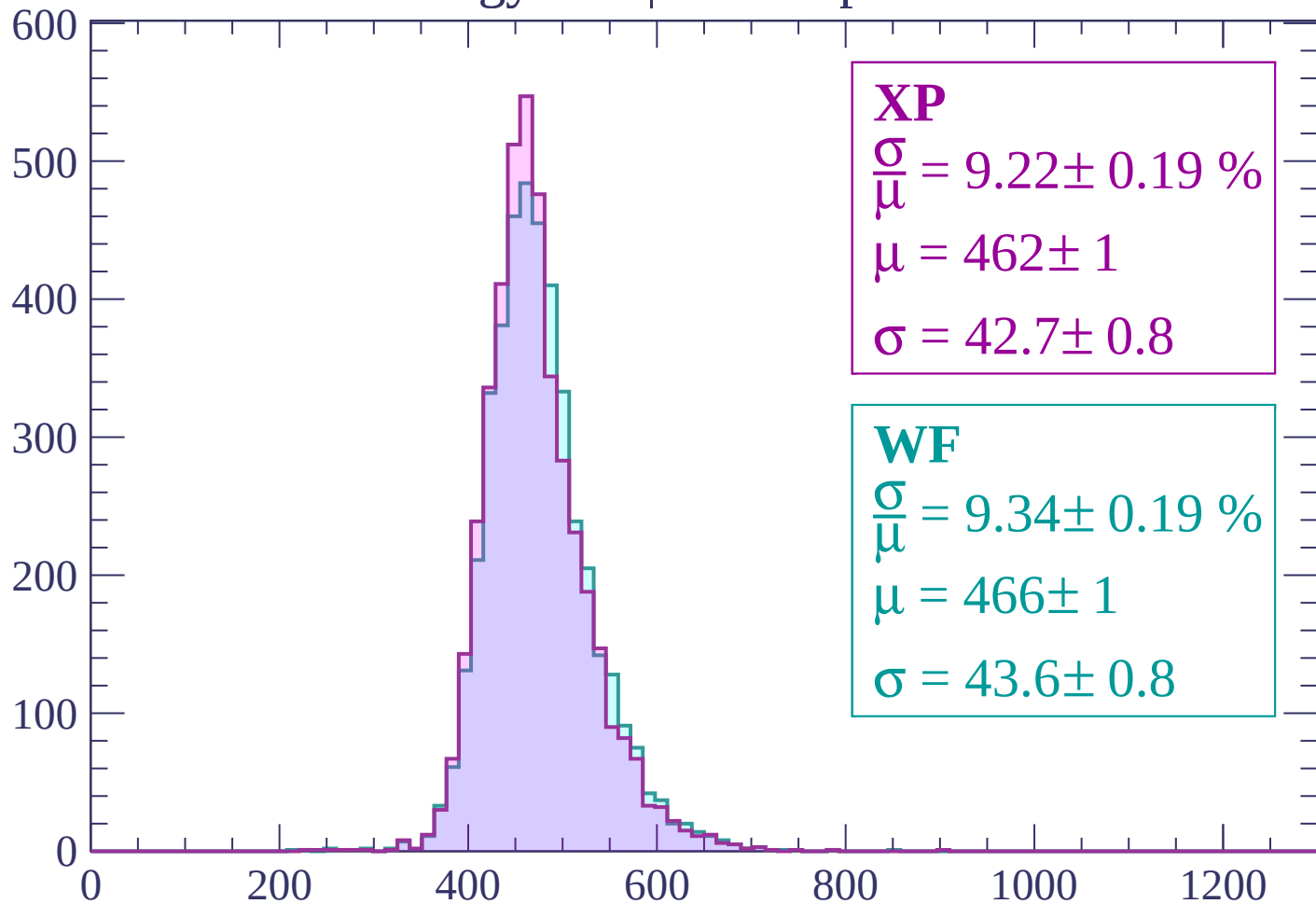
$$\mu = 464 \pm 1$$

$$\sigma = 44.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.19 \%$$

$$\mu = 462 \pm 1$$

$$\sigma = 42.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.34 \pm 0.19 \%$$

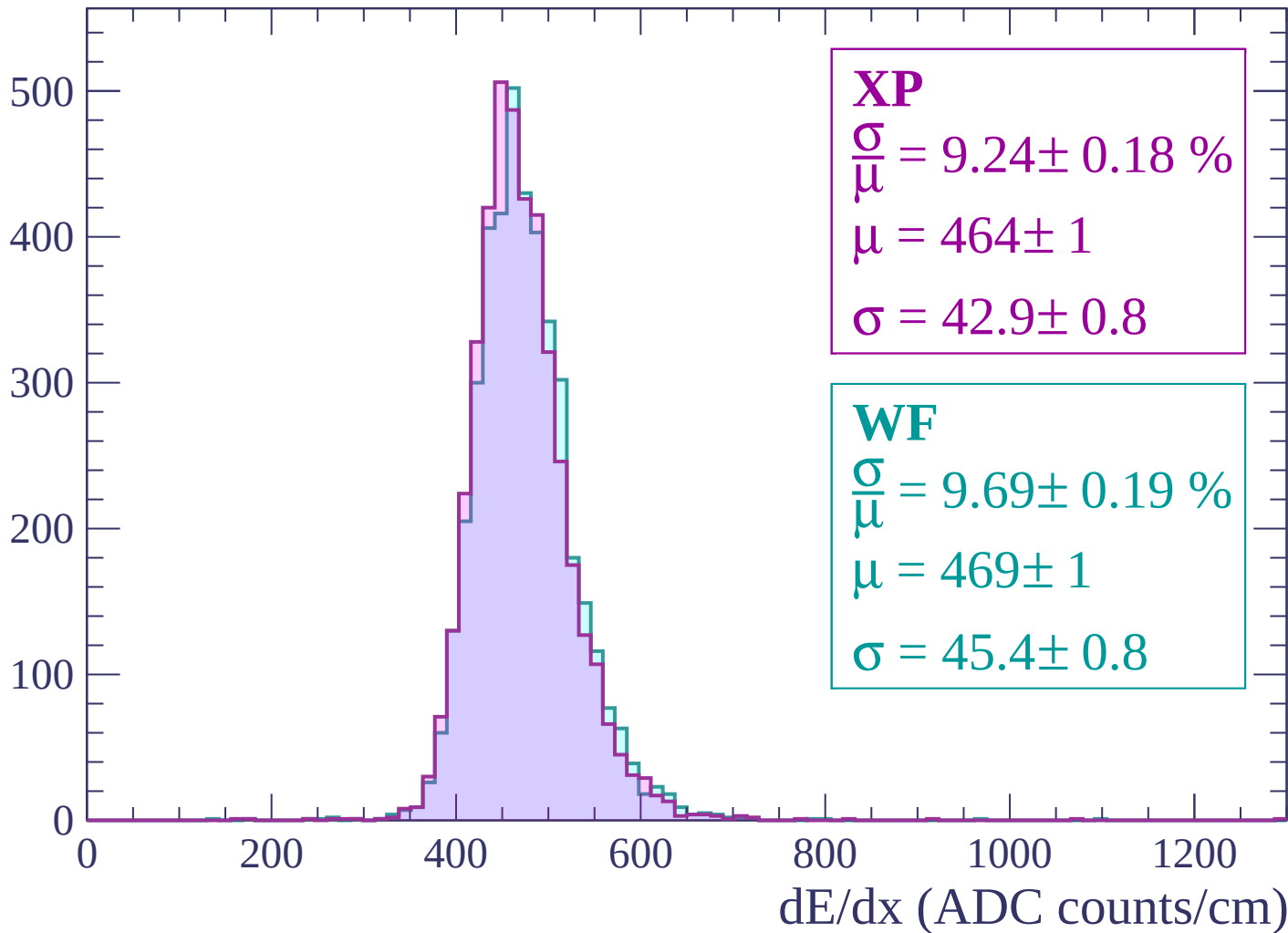
$$\mu = 466 \pm 1$$

$$\sigma = 43.6 \pm 0.8$$

dE/dx (ADC counts/cm)

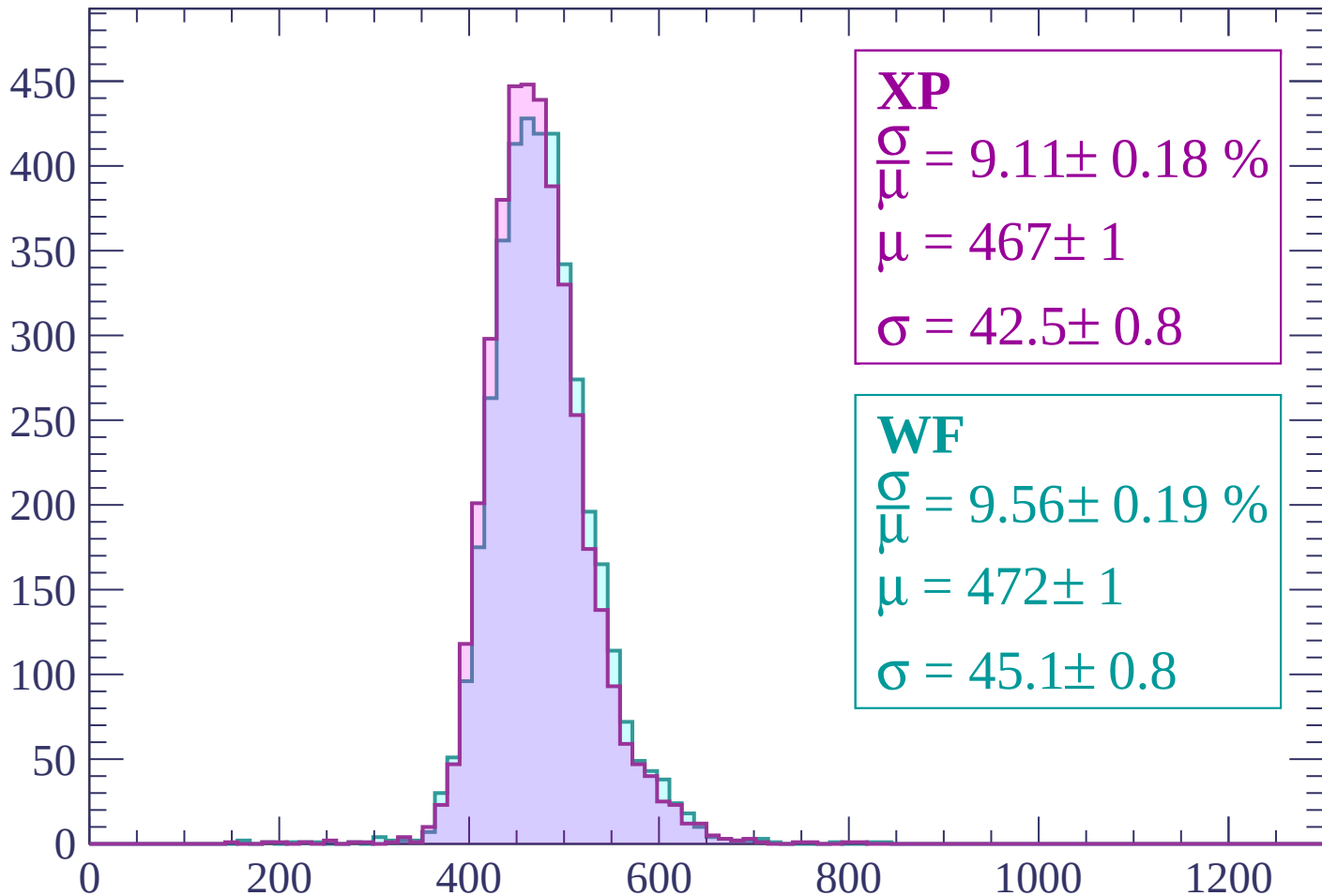
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.18 \%$$

$$\mu = 467 \pm 1$$

$$\sigma = 42.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.19 \%$$

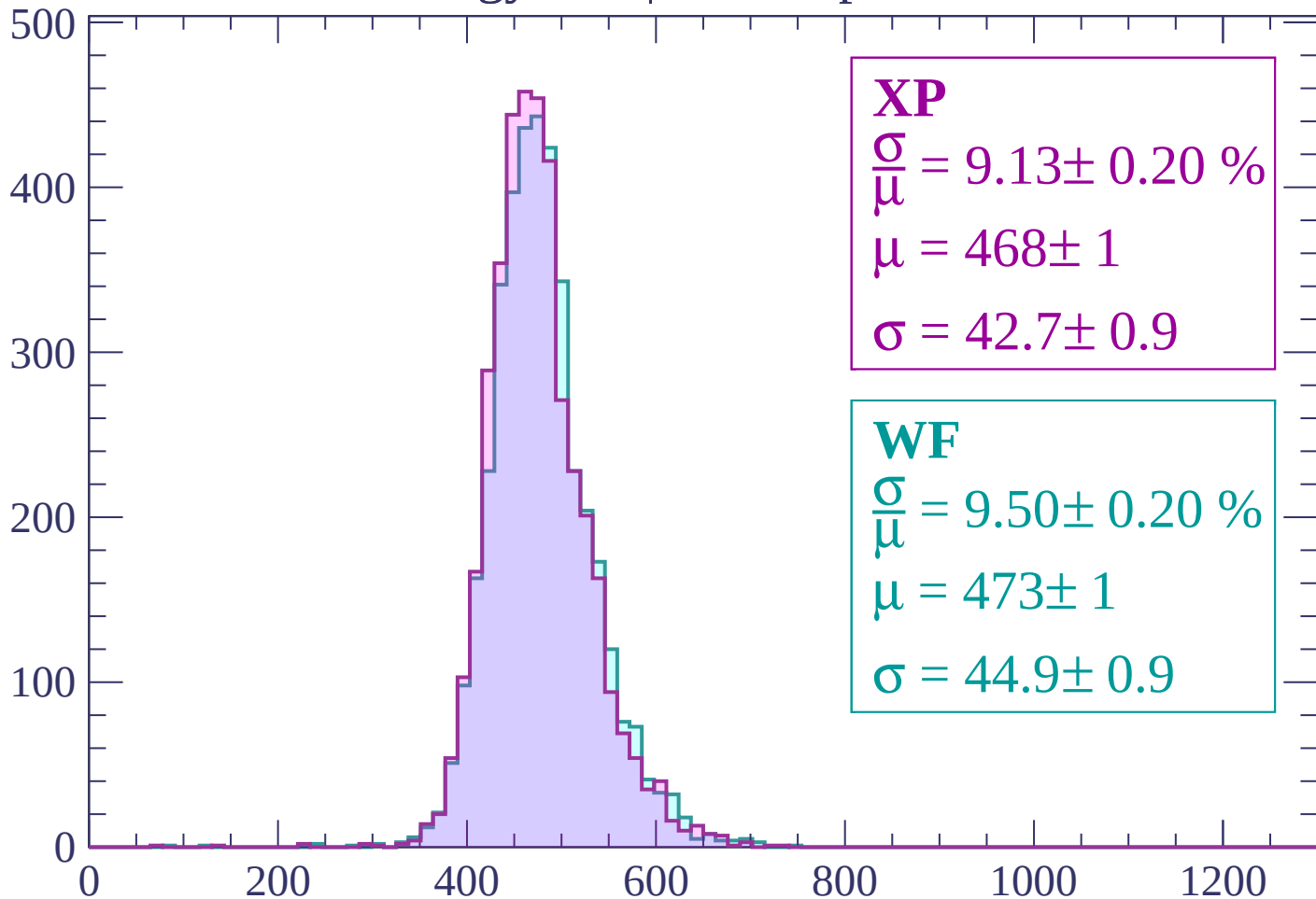
$$\mu = 472 \pm 1$$

$$\sigma = 45.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.20 \%$$

$$\mu = 468 \pm 1$$

$$\sigma = 42.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.20 \%$$

$$\mu = 473 \pm 1$$

$$\sigma = 44.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1980$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.52 \pm 0.20 \%$$

$$\mu = 471 \pm 1$$

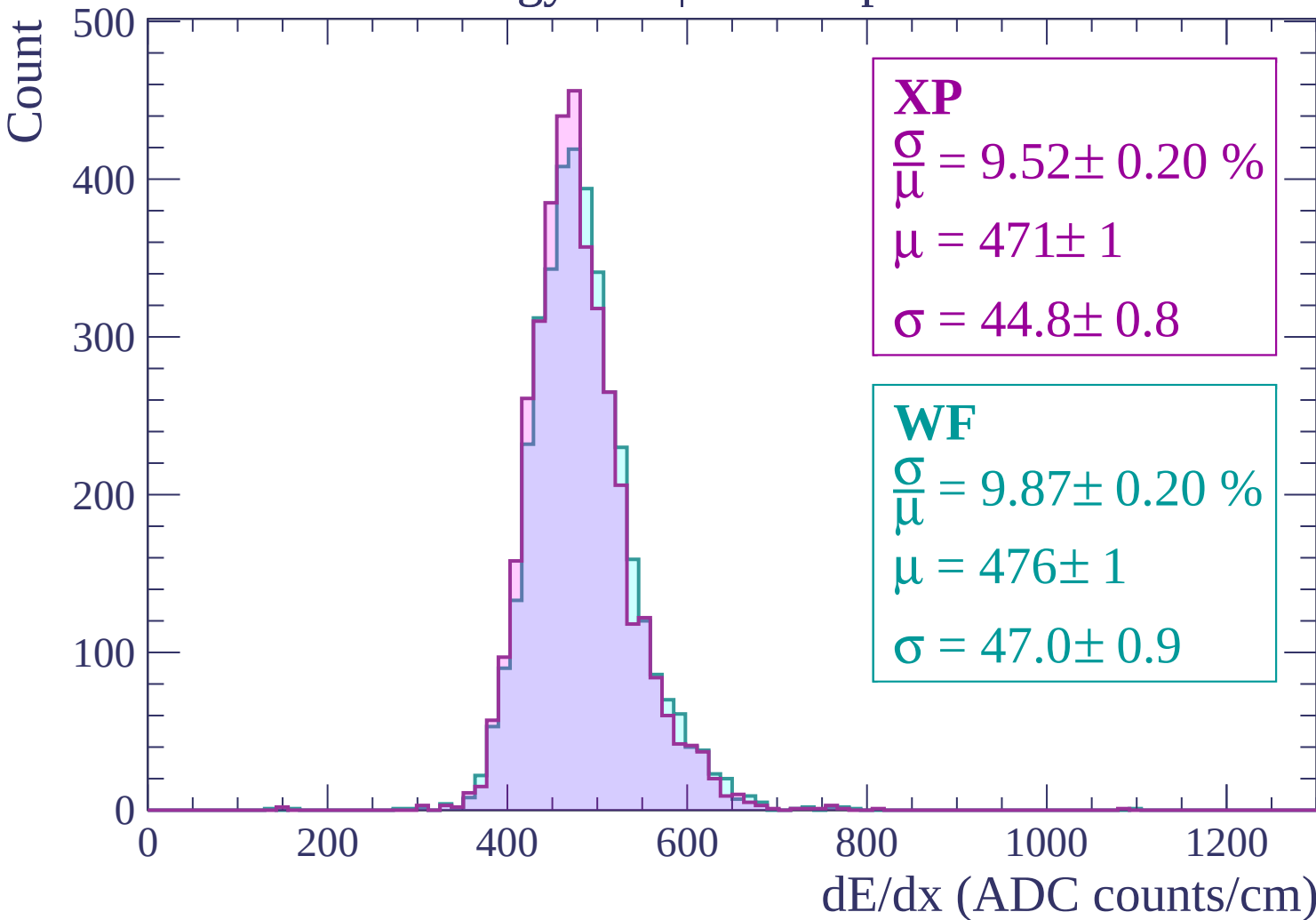
$$\sigma = 44.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.87 \pm 0.20 \%$$

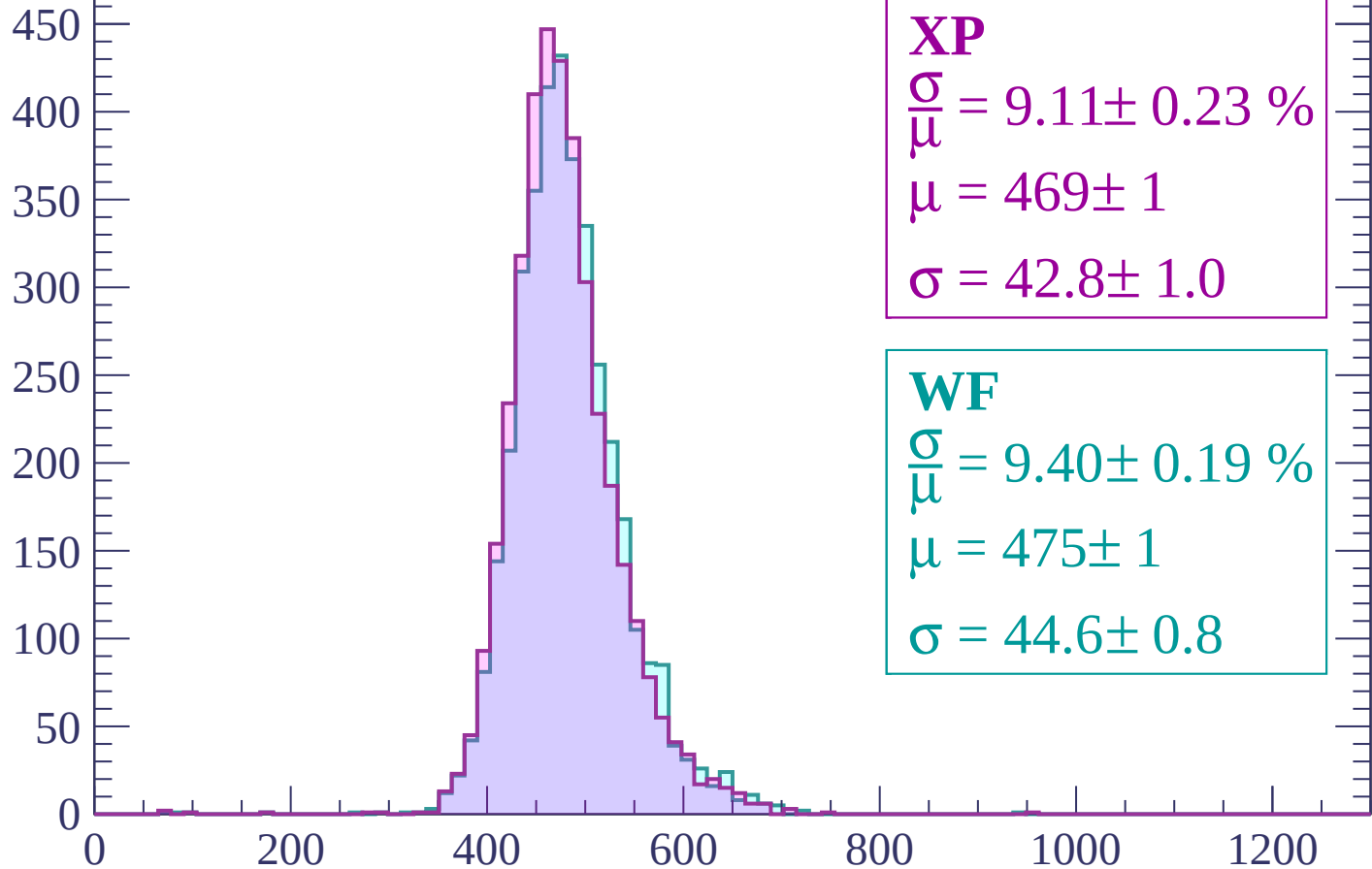
$$\mu = 476 \pm 1$$

$$\sigma = 47.0 \pm 0.9$$



Energy loss | $1980 < p < 2040$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.23 \%$$

$$\mu = 469 \pm 1$$

$$\sigma = 42.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.19 \%$$

$$\mu = 475 \pm 1$$

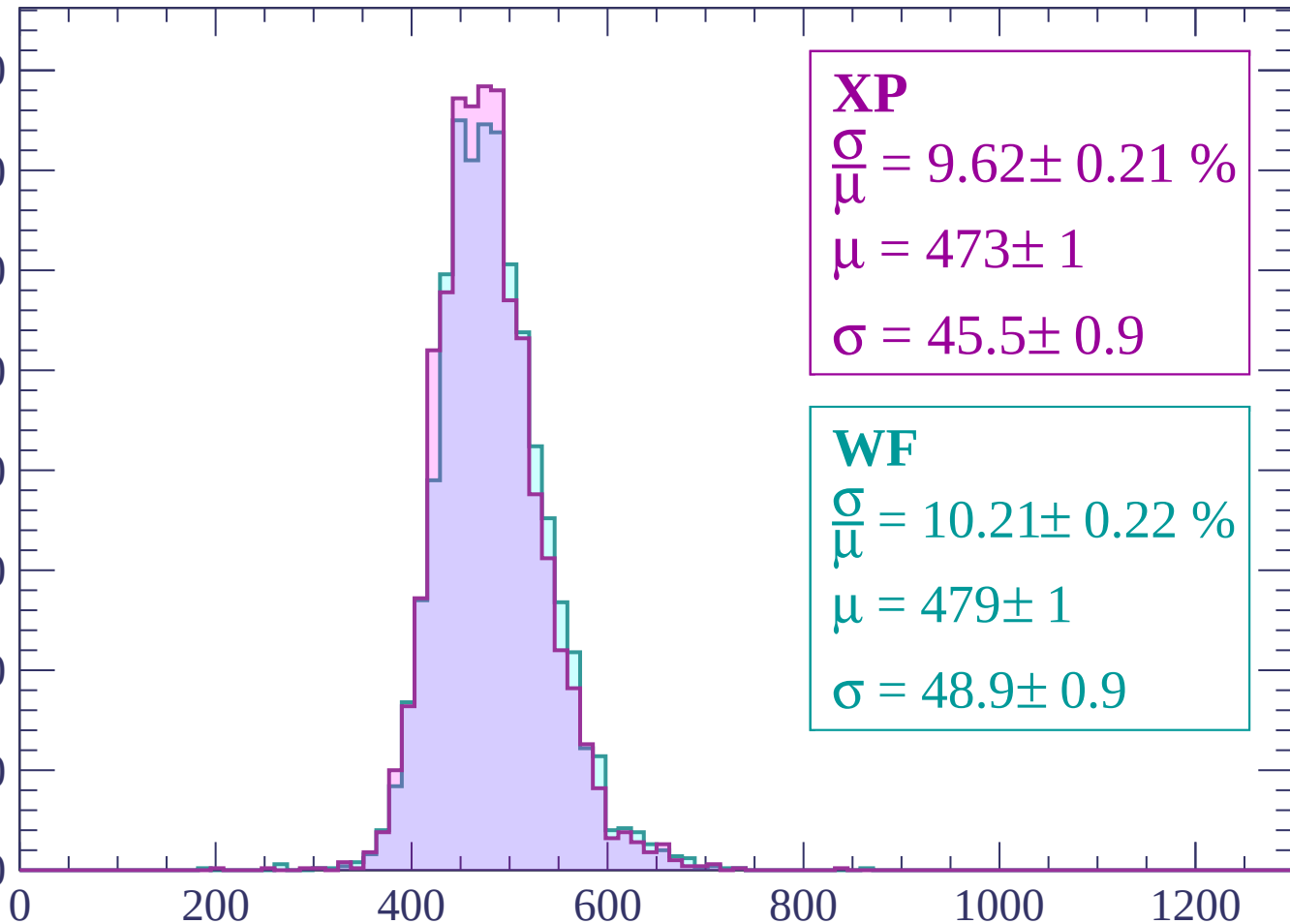
$$\sigma = 44.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.62 \pm 0.21 \%$$

$$\mu = 473 \pm 1$$

$$\sigma = 45.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$$

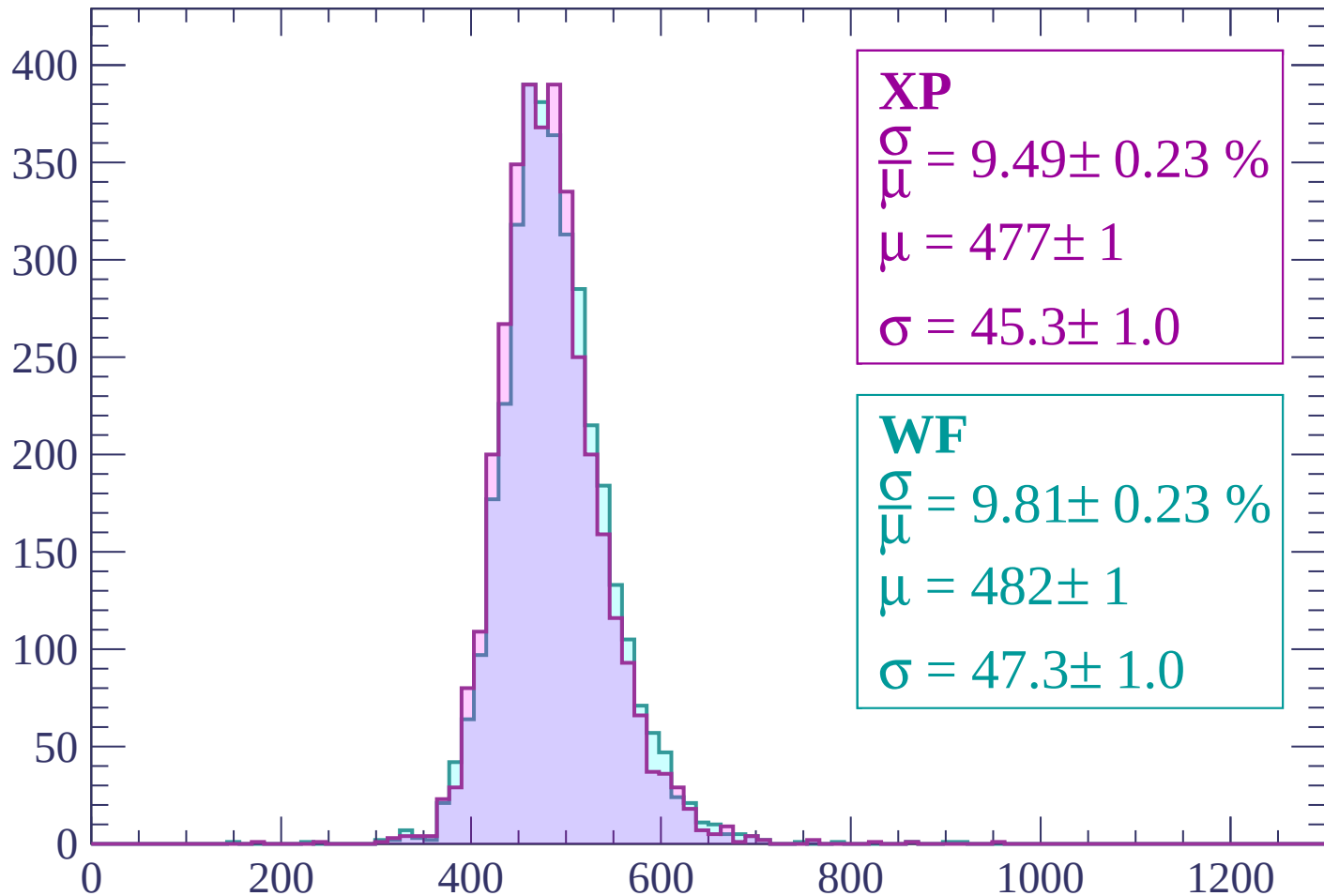
$$\mu = 479 \pm 1$$

$$\sigma = 48.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.23 \%$$

$$\mu = 477 \pm 1$$

$$\sigma = 45.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.81 \pm 0.23 \%$$

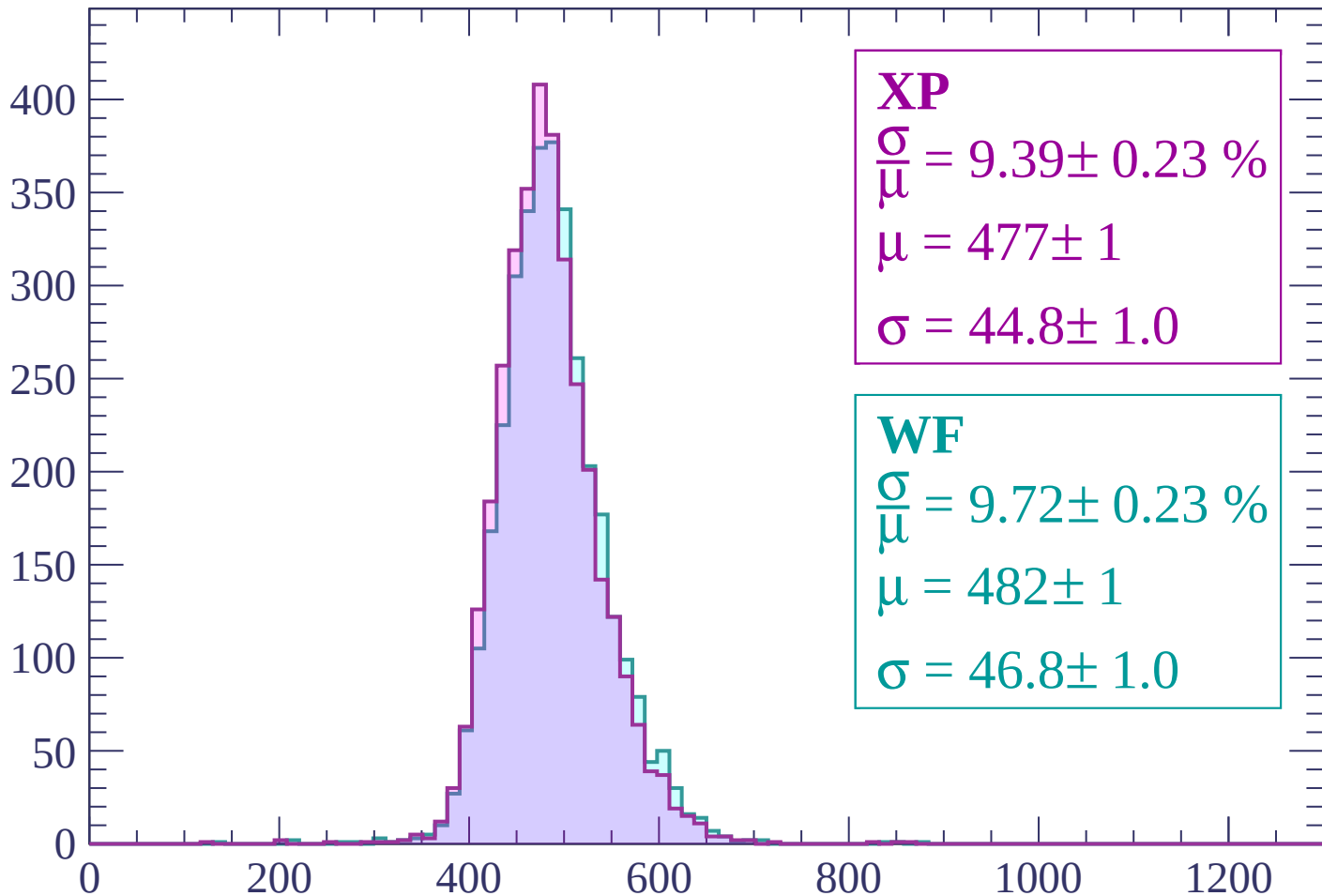
$$\mu = 482 \pm 1$$

$$\sigma = 47.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.23 \%$$

$$\mu = 477 \pm 1$$

$$\sigma = 44.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.72 \pm 0.23 \%$$

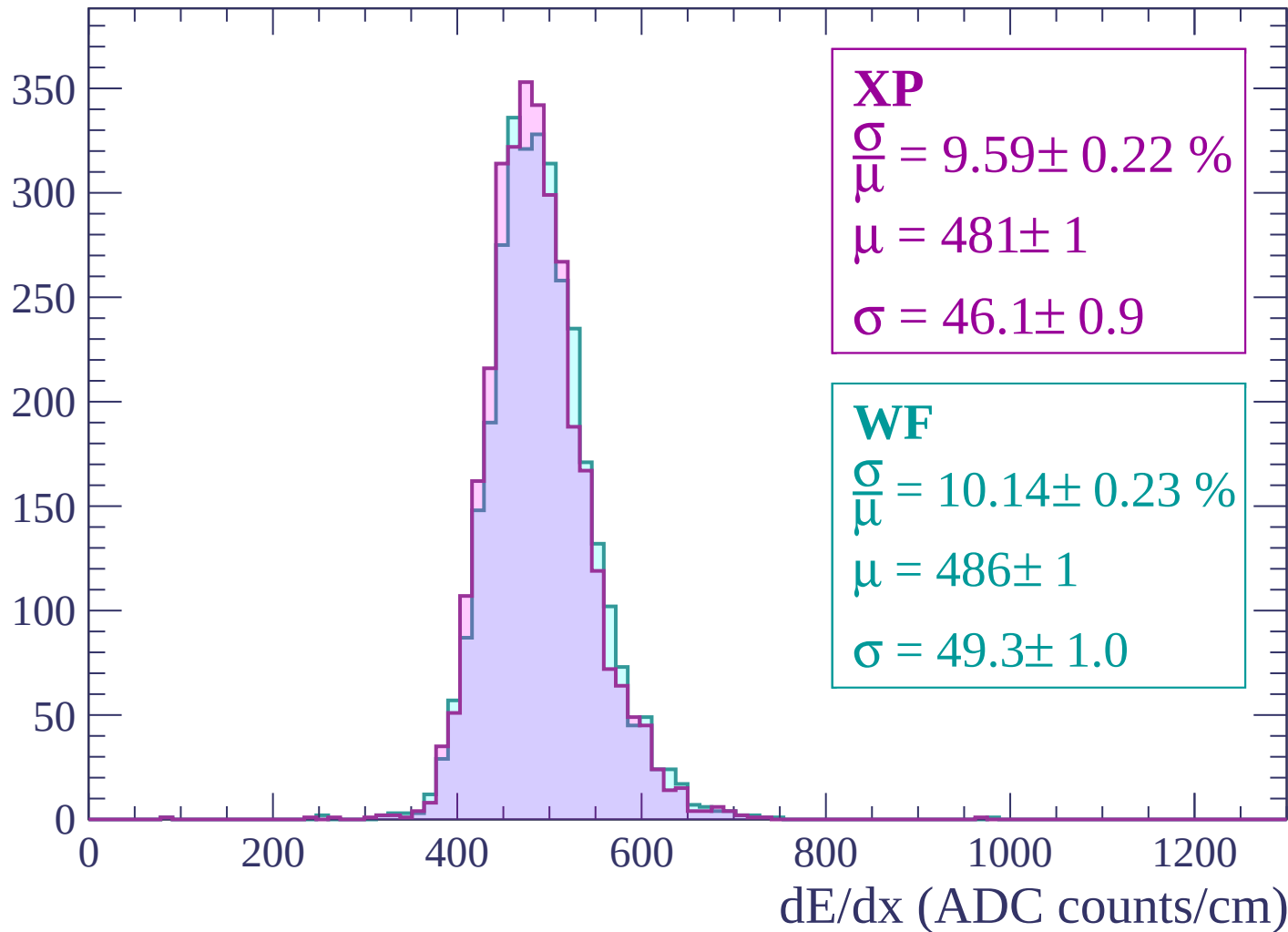
$$\mu = 482 \pm 1$$

$$\sigma = 46.8 \pm 1.0$$

dE/dx (ADC counts/cm)

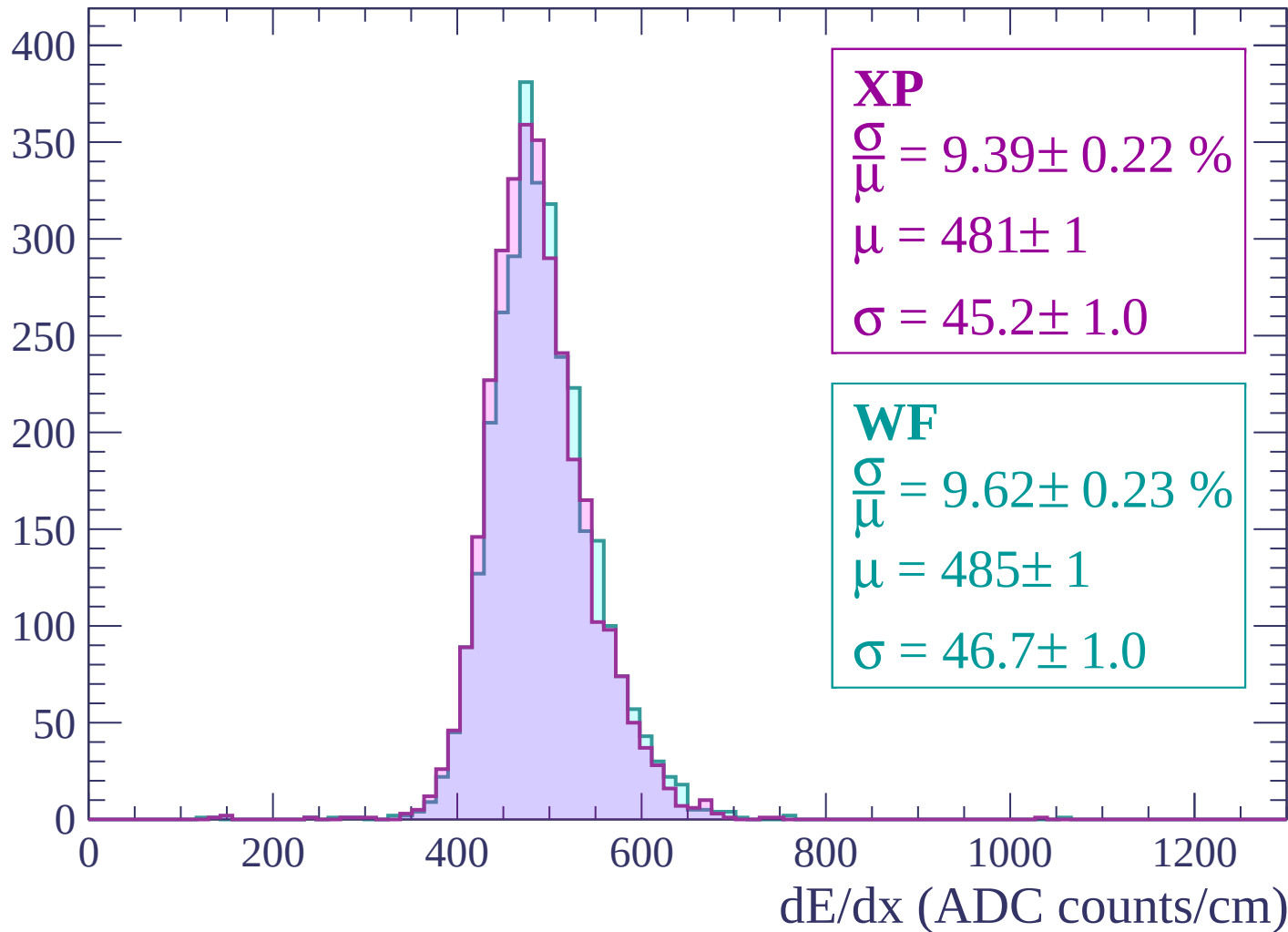
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

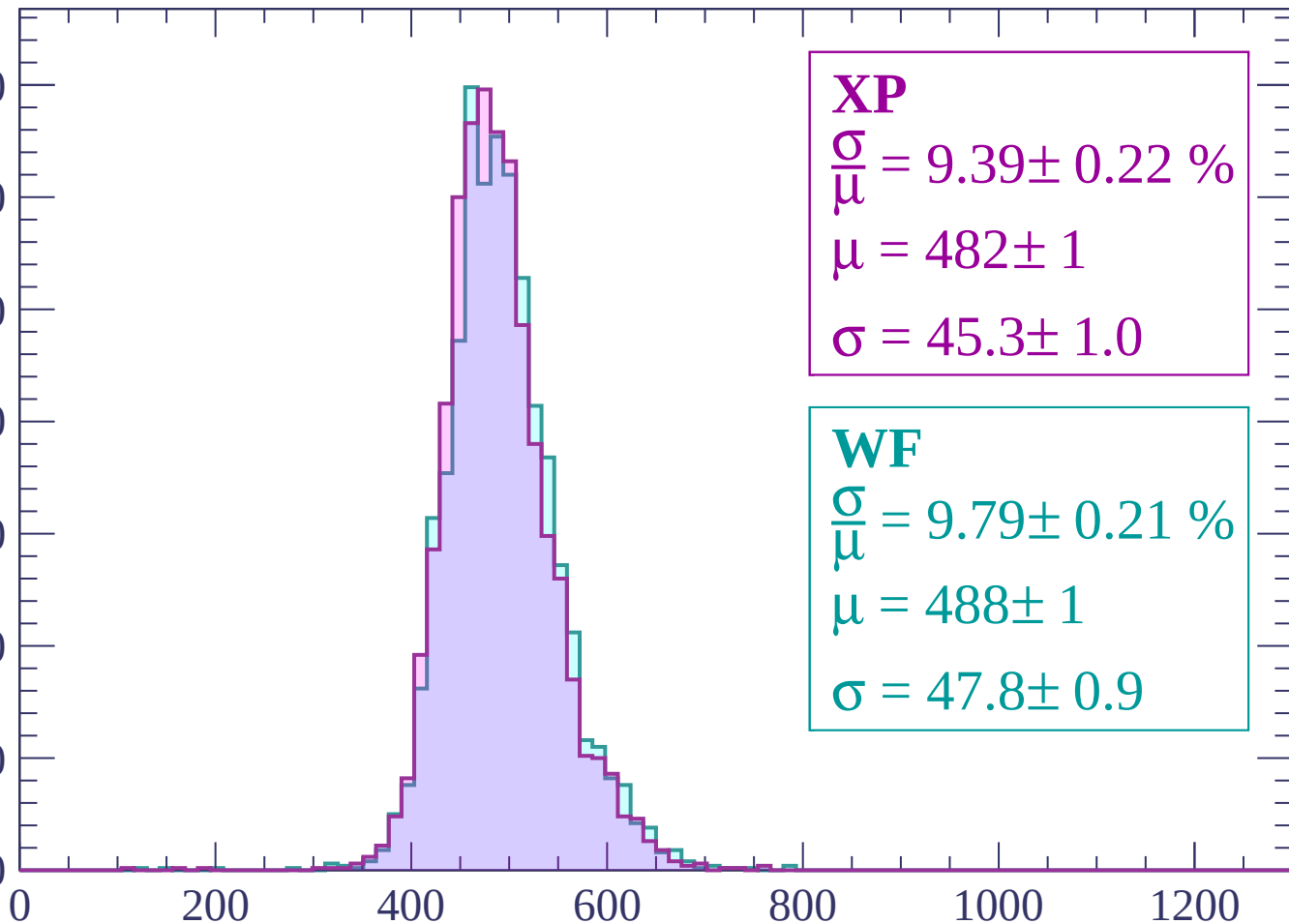
Count



Energy loss | $2340 < p < 2400$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.22 \%$$

$$\mu = 482 \pm 1$$

$$\sigma = 45.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.21 \%$$

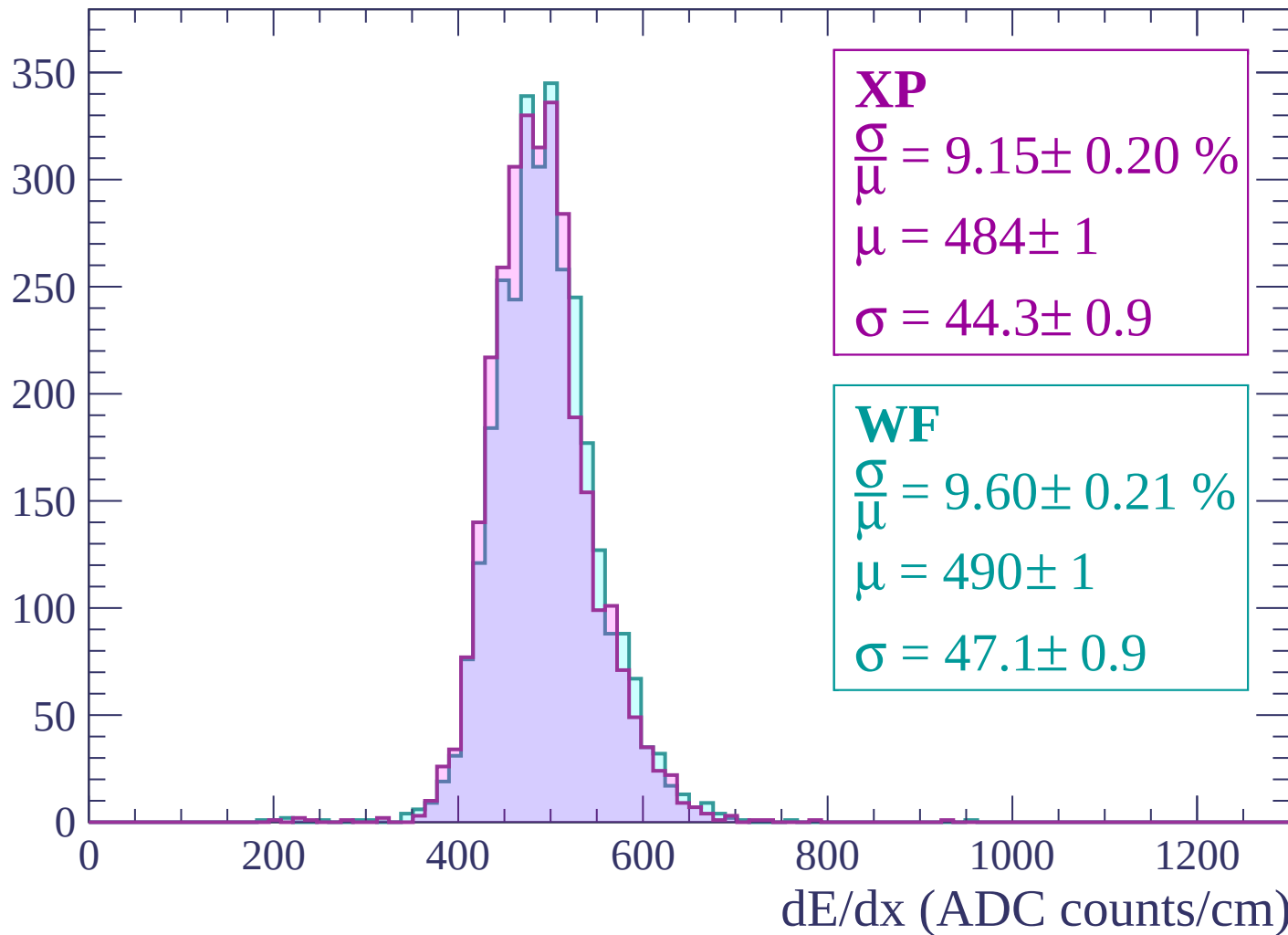
$$\mu = 488 \pm 1$$

$$\sigma = 47.8 \pm 0.9$$

dE/dx (ADC counts/cm)

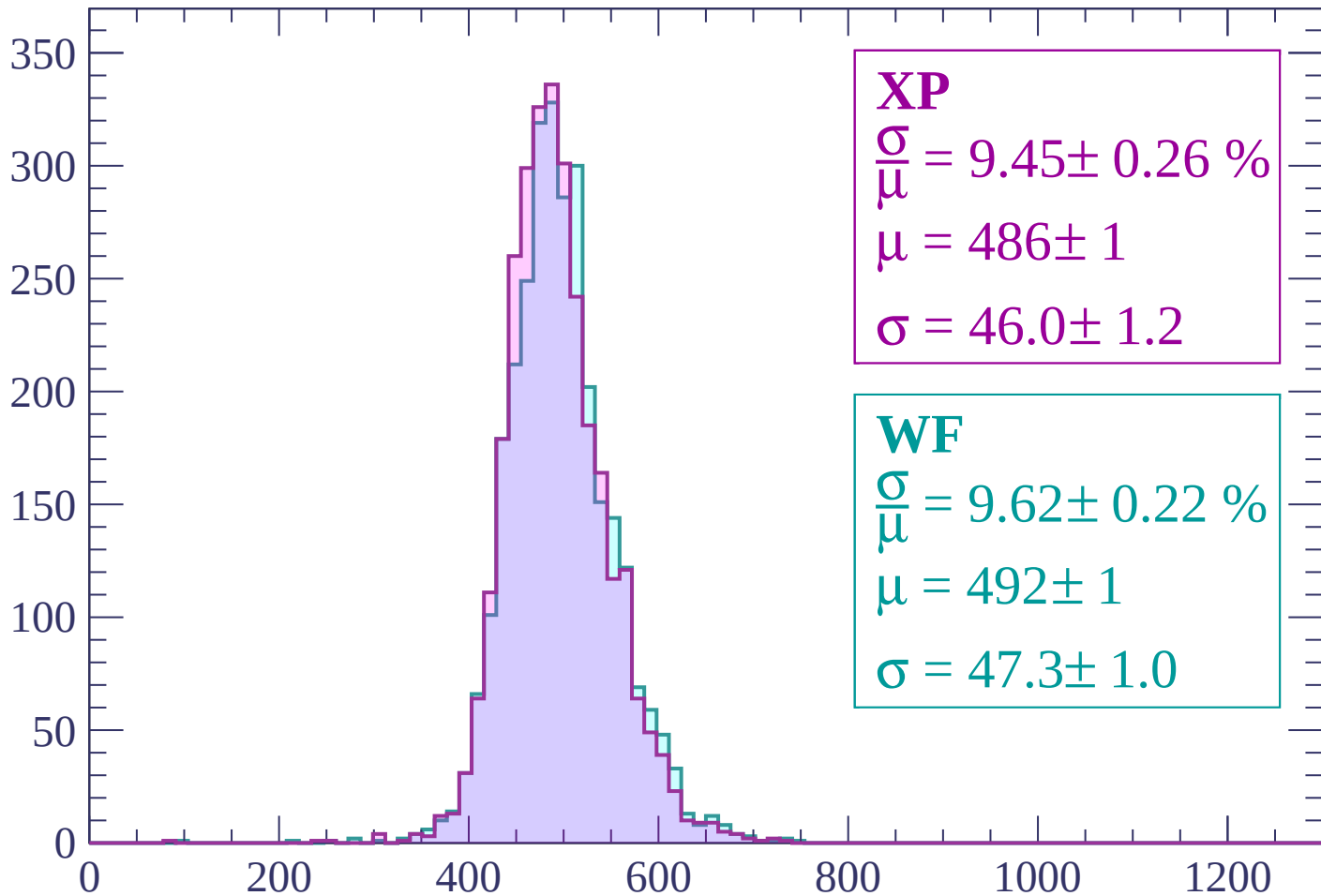
Energy loss | $2400 < p < 2460$

Count



Energy loss | $2460 < p < 2520$

Count



XP

$$\frac{\sigma}{\mu} = 9.45 \pm 0.26 \%$$

$$\mu = 486 \pm 1$$

$$\sigma = 46.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.62 \pm 0.22 \%$$

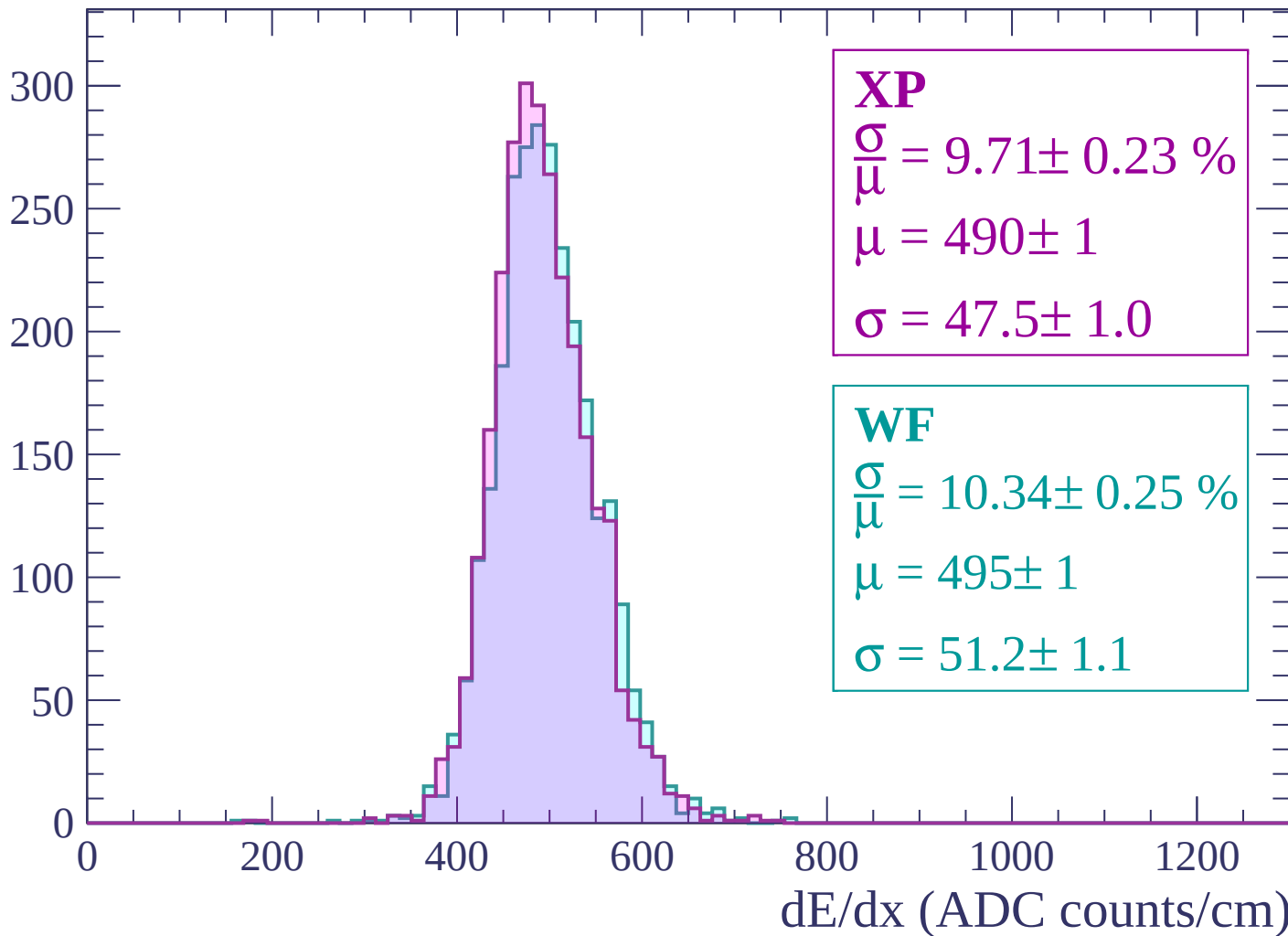
$$\mu = 492 \pm 1$$

$$\sigma = 47.3 \pm 1.0$$

dE/dx (ADC counts/cm)

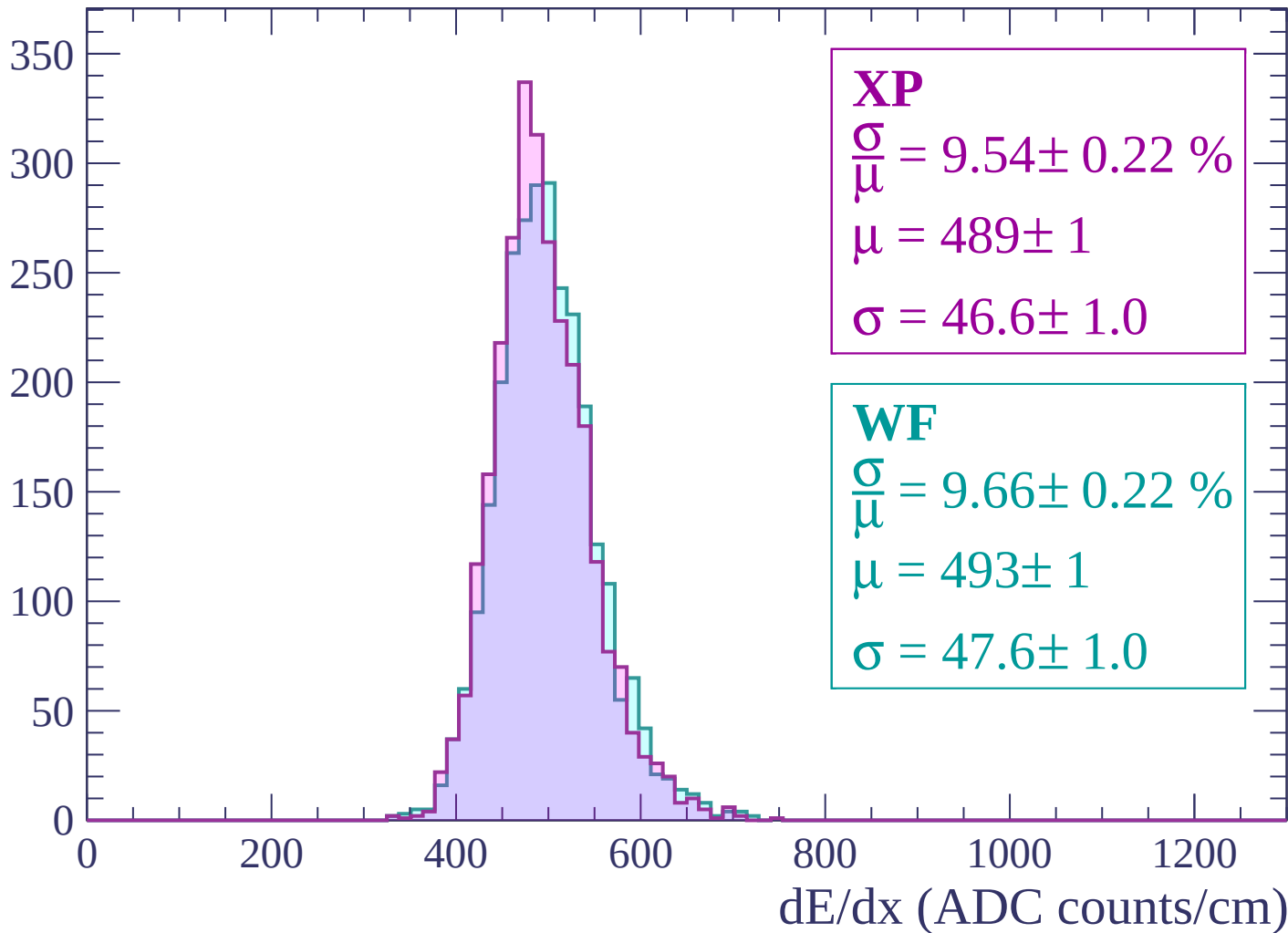
Energy loss | $2520 < p < 2580$

Count



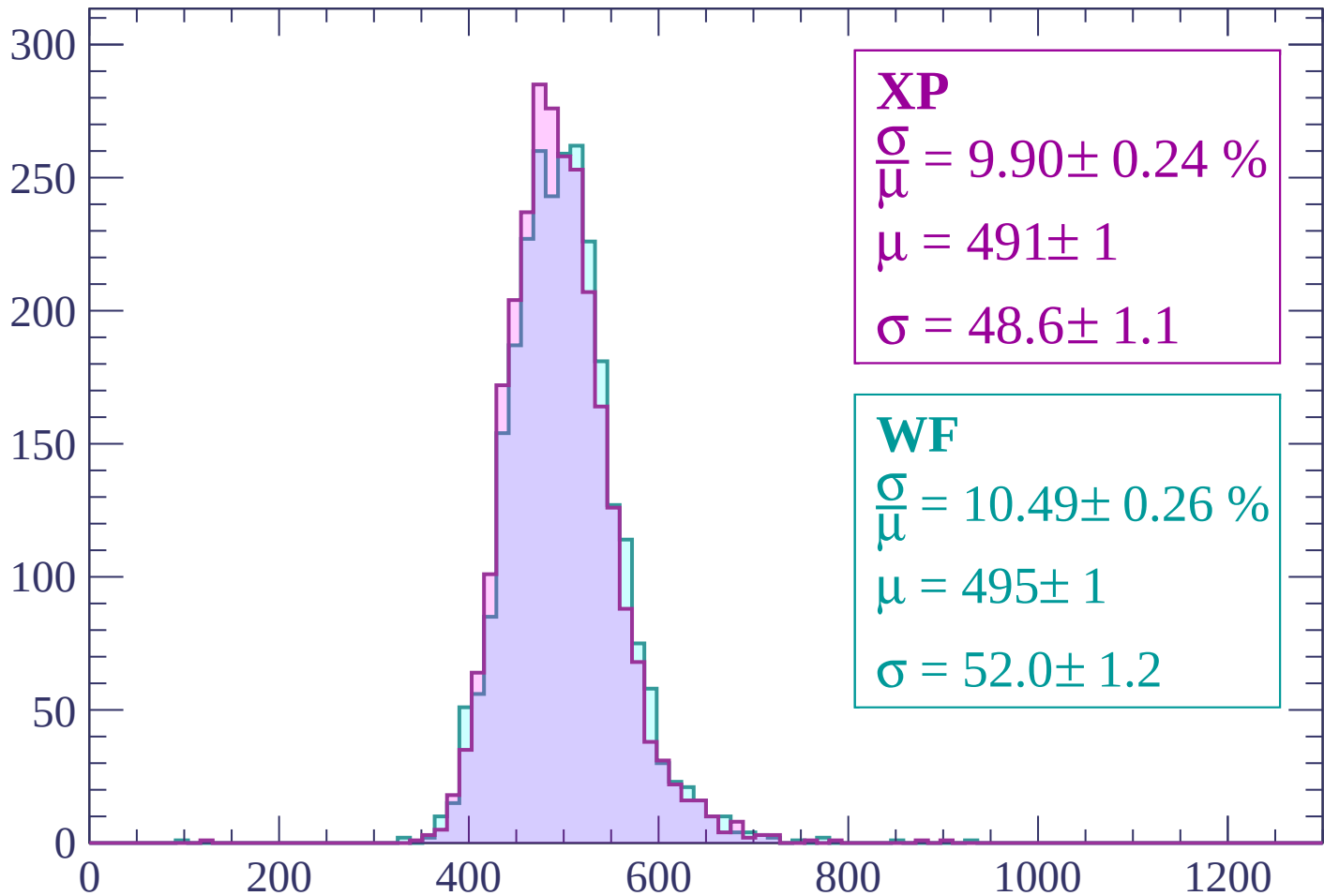
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP

$$\frac{\sigma}{\mu} = 9.90 \pm 0.24 \%$$

$$\mu = 491 \pm 1$$

$$\sigma = 48.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.49 \pm 0.26 \%$$

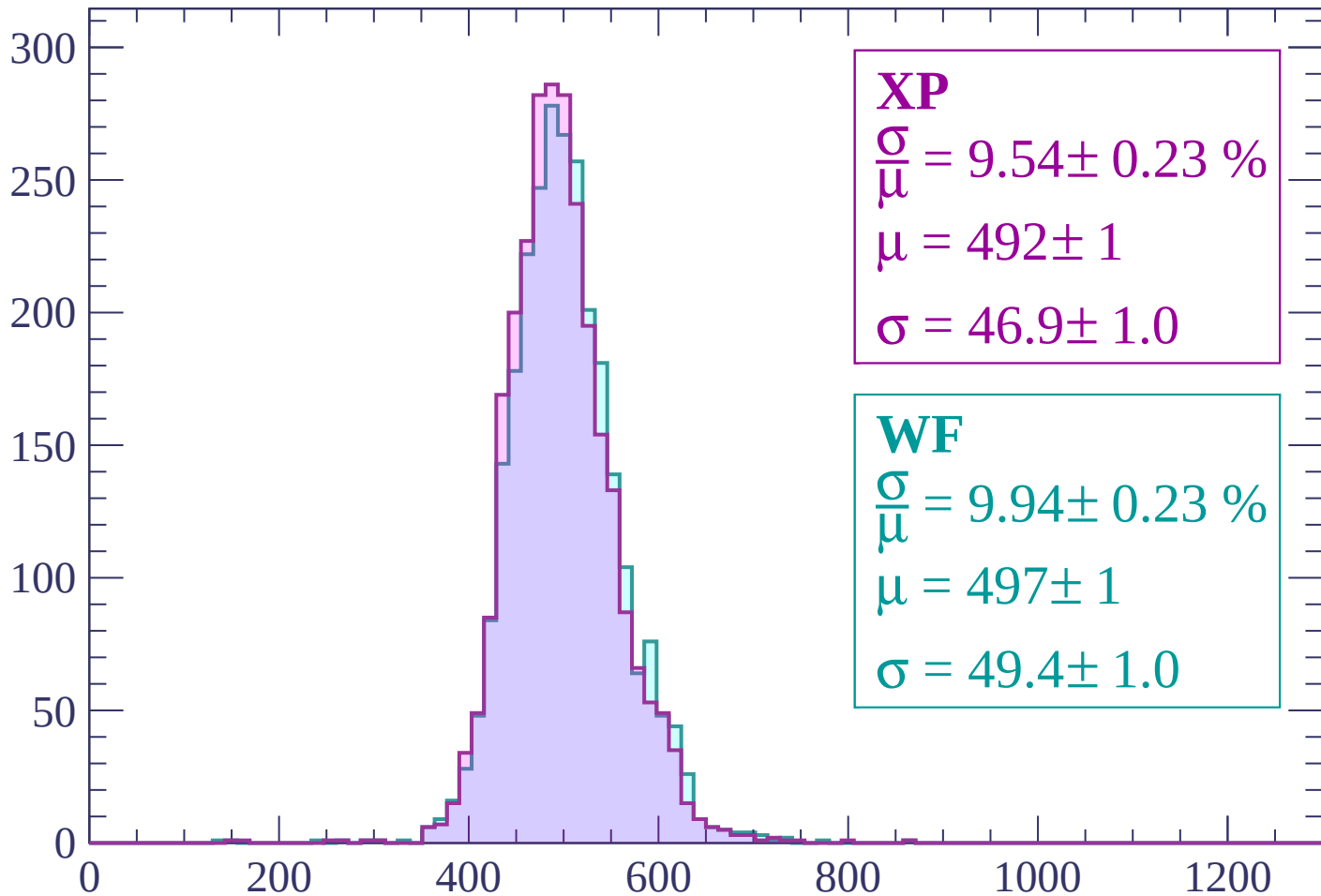
$$\mu = 495 \pm 1$$

$$\sigma = 52.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP

$$\frac{\sigma}{\mu} = 9.54 \pm 0.23 \%$$

$$\mu = 492 \pm 1$$

$$\sigma = 46.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.94 \pm 0.23 \%$$

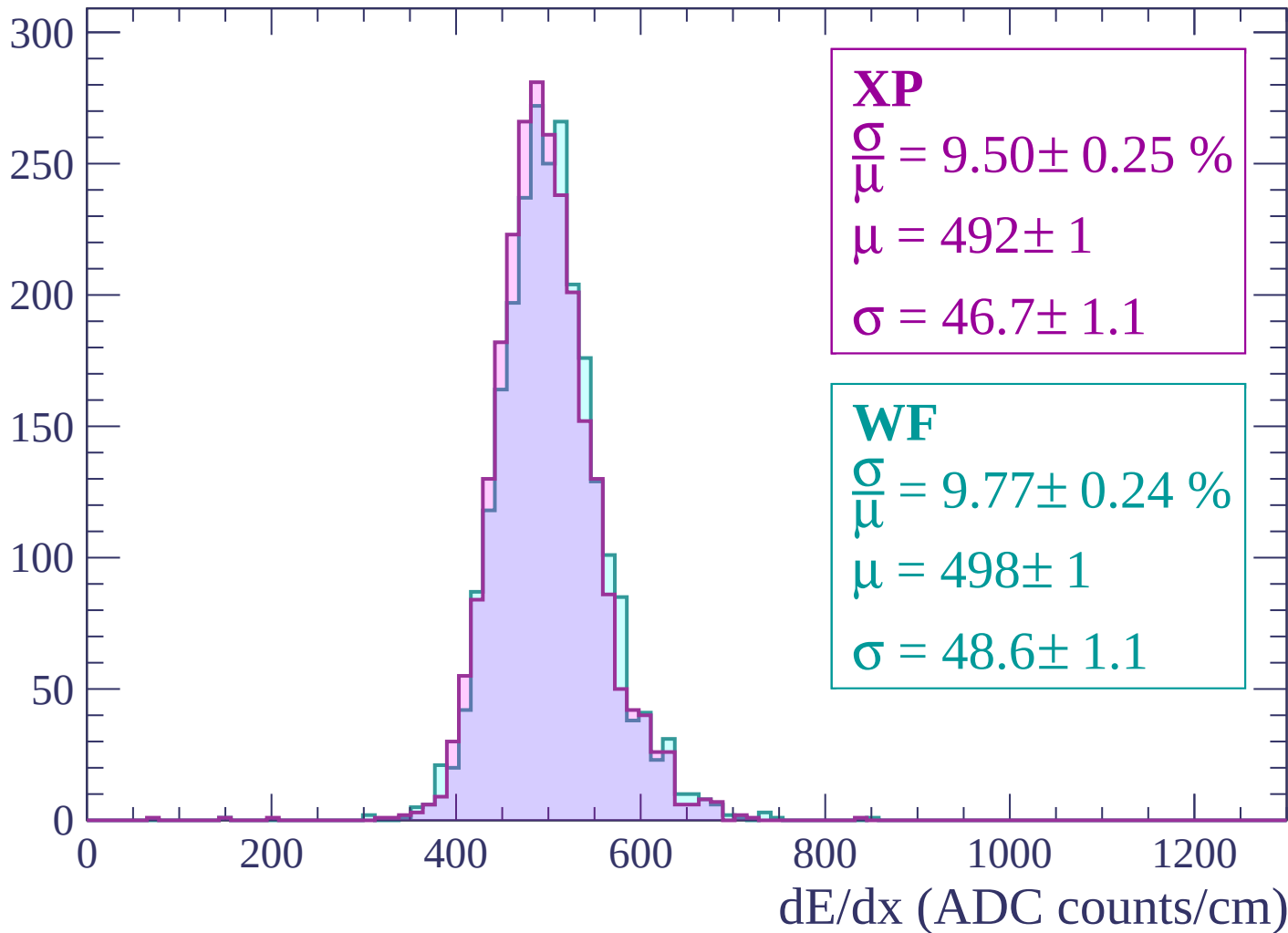
$$\mu = 497 \pm 1$$

$$\sigma = 49.4 \pm 1.0$$

dE/dx (ADC counts/cm)

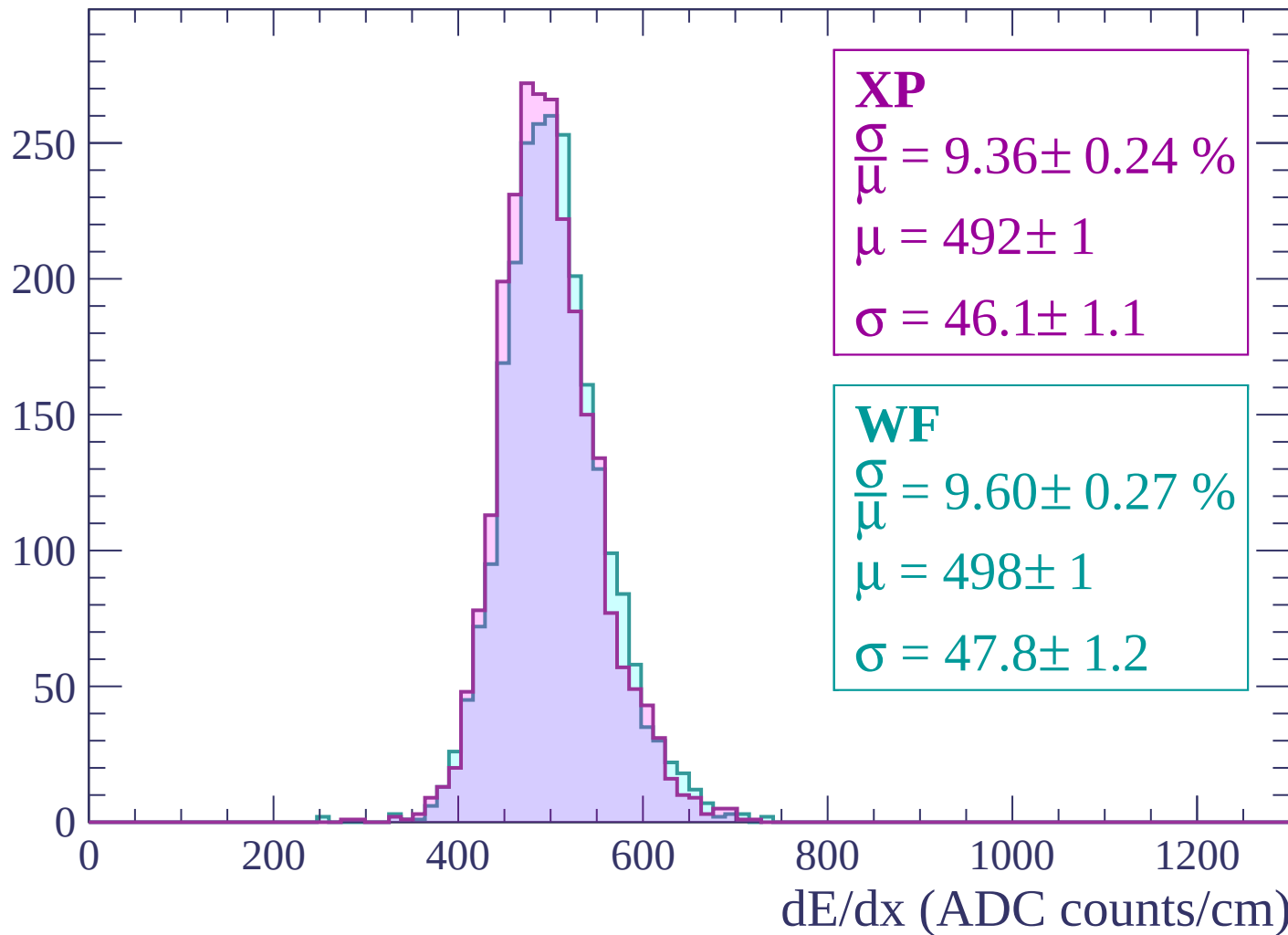
Energy loss | $2760 < p < 2820$

Count



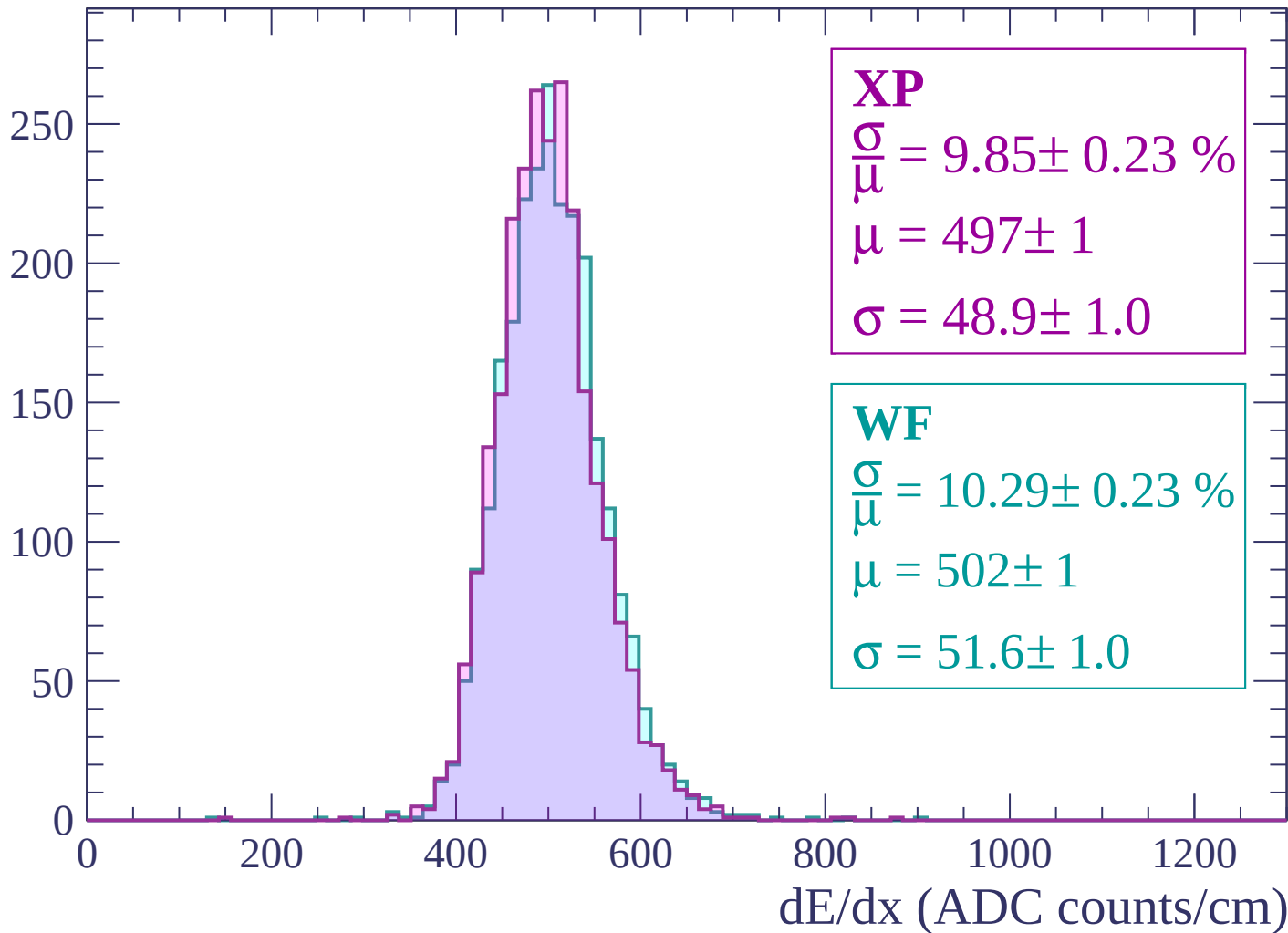
Energy loss | $2820 < p < 2880$

Count



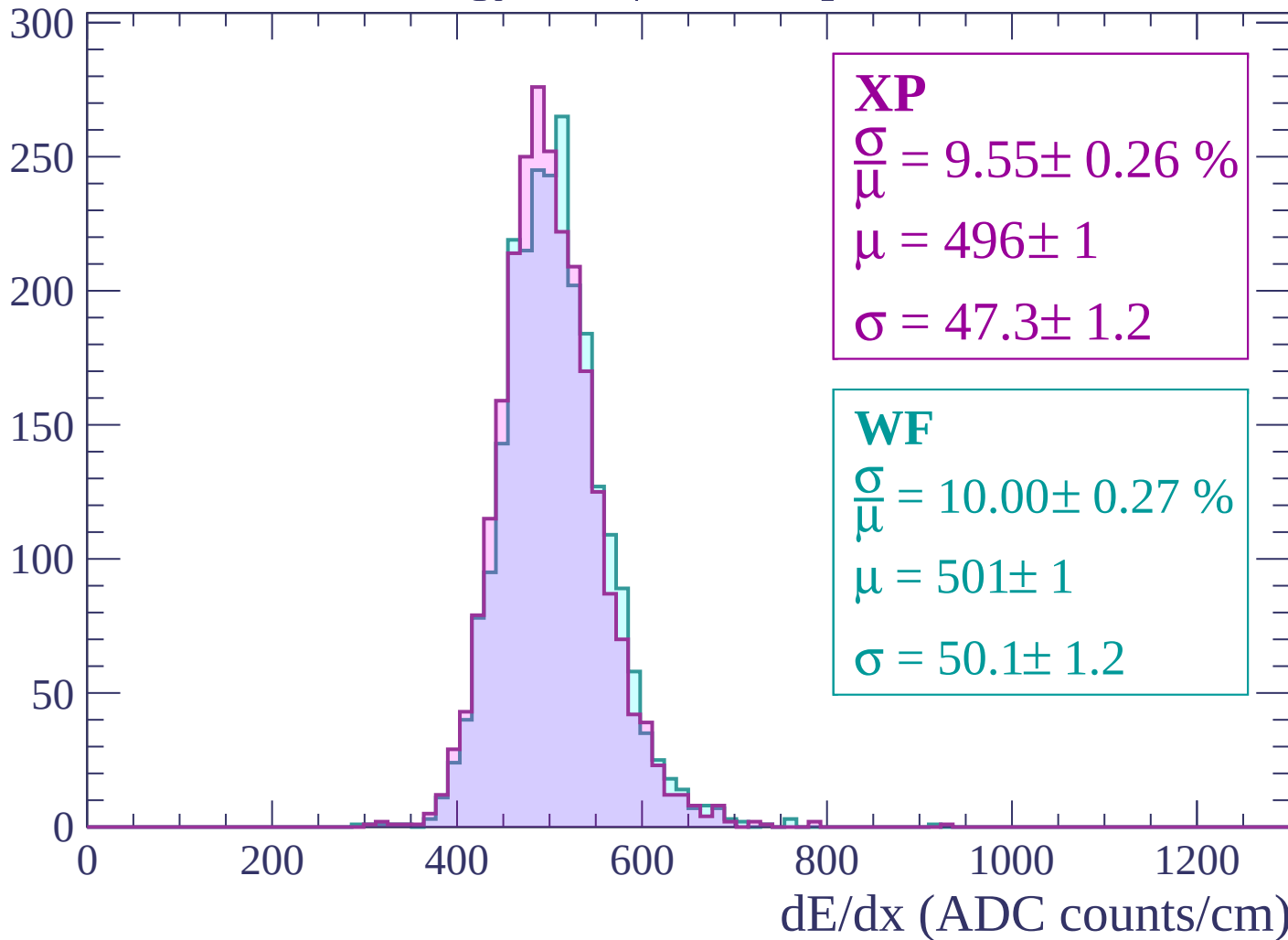
Energy loss | $2880 < p < 2940$

Count



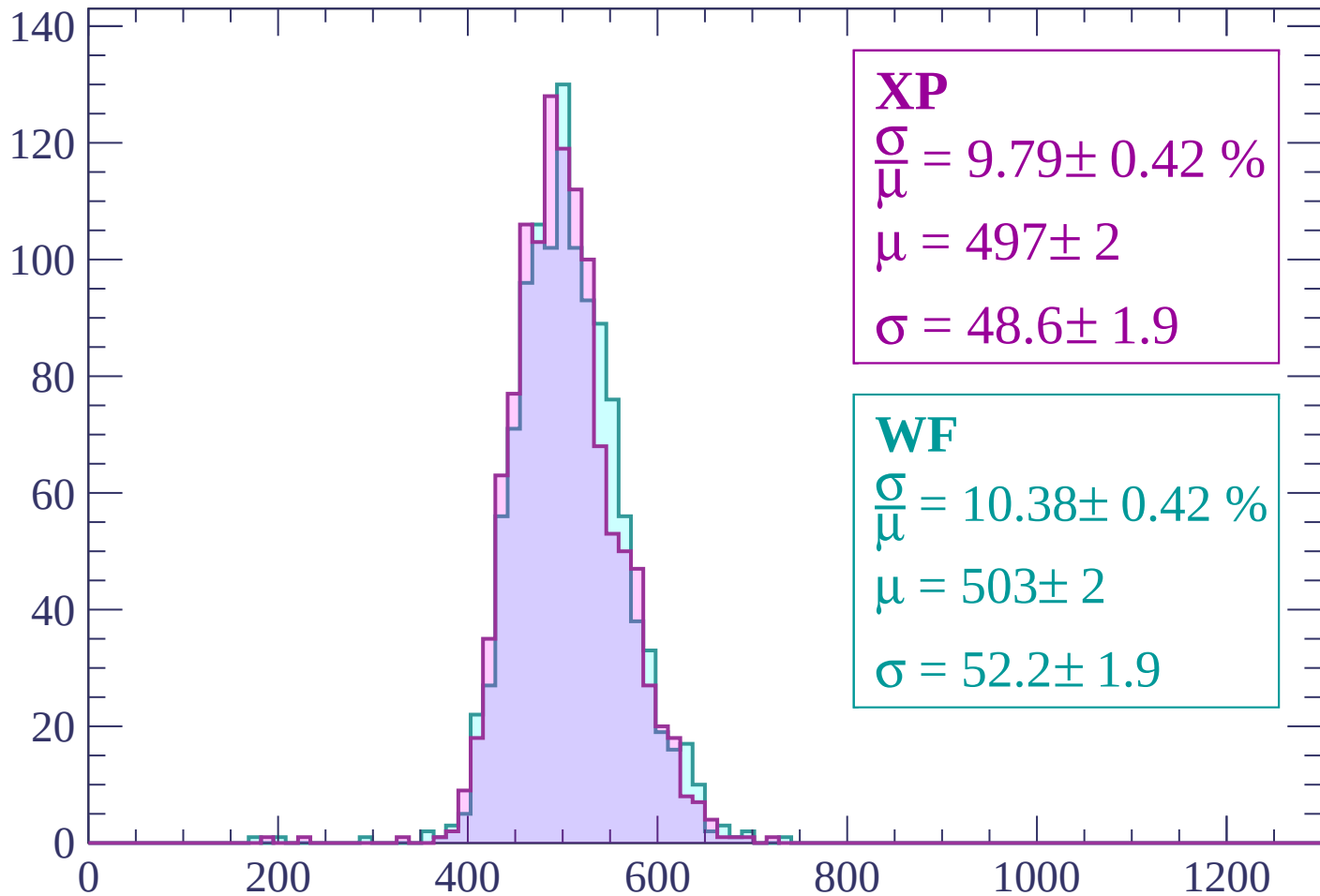
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP

$$\frac{\sigma}{\mu} = 9.79 \pm 0.42 \%$$

$$\mu = 497 \pm 2$$

$$\sigma = 48.6 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.38 \pm 0.42 \%$$

$$\mu = 503 \pm 2$$

$$\sigma = 52.2 \pm 1.9$$

dE/dx (ADC counts/cm)

